Advanced Combustion Microgravity Experiment (ACME)

Engineering Requirements Document (ERD)

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ZIN Technologies, Inc.

Advanced Combustion via Microgravity Experiments

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PREFACE

The Advanced Combustion via Microgravity Experiments (ACME) is a next generation mini-facility designed to support Principal Investigators (PI) interested in studying various aspects of advanced gaseous combustion in a microgravity environment. ACME will be integrated into the Fluids and Combustion Facility's (FCF) Combustion Integrated Rack (CIR) onboard the International Space Station in the US Lab Module.

The ACME mission is to perform microgravity experiments to study fire suppression/flame extinguishment investigations using the Combustion Integrated Rack. The ACME will allow investigation using pure fuels and mixtures of fuels at various chamber atmosphere pressures, compositions, and oxygen concentrations. The microgravity environment allows for measurement and observation of flame formation and extinguishment processes that cannot be made in sustained gravity. The modular design of the ACME allows for flexibility in configuring for specific experiments and easy maintainability of the hardware. This hardware will provide the apparatus for dispensing, deploying and igniting fuel. When new experiments are proposed the additional experiment specific hardware will be provided. This requirements document reflects such a case, including only the new requirements of the ACME investigation.

DOCUMENT HISTORY LOG

Revision	Date	Description	Approval
-	12/08	Baseline Release	
A	06/09	ACME-ECO-002: Incorporated updates to Science Requirement Doc draft dated 2009-04-03	BB 06/24/09 CG 6/24/09 LF 06/25/09
B	05/2010	ACME-ECO-004: Incorporated changes from PI, for RDR RID Review	LF 05/18/10 BB 05/19/10
C	06/2010	ACME-ECO-006: Incorporated changes from RDR Panel initial comments and decomposition activities interim to receiving RDR Final Review Package	LF 06/30/10 BB 06/30/10
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J	11/2013	ACME-ECO-240: 1) Updates figures to reflect design evolution. Updates requirements language.	LF 11/18/13 BB: 11/18/13
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TABLE OF CONTENTS

1.0 INTRODUCTION 1
1.1 Purpose 1
1.2 Scope 1
2.0 APPLICABLE DOCUMENTS 2
2.1 Order of Precedence for Documents 2
2.2 Applicable Documents 2
2.3 Reference Documents 3
3.0 REQUIREMENTS MANAGEMENT 4
3.1 Preparation for Requirements 4
3.1.1 Stakeholder Analysis 4
3.1.1.1 Identify Stakeholders 4
3.1.1.2 Identify Level 0 Stakeholder Requirements 5
3.2 Obtaining Requirements 5
3.2.1 Identifying All Stakeholder Requirements 5
3.3 Analyzing Requirements 6
3.3.1 Requirement Quality 7
3.3.2 Requirement Validation 7
3.4 Approving Requirements 7
3.5 Maintaining Requirements 8
4.0 ACME REQUIREMENT SOURCES 9
4.1 Evaluation of Requirement Sources 9
5.0 ACME HARDWARE PRODUCT DESCRIPTION 10
5.1 ACME Chamber Insert Assembly (CIA) System 11
5.2 Experiment Mounting Structure (EMS) System 11
5.3 Cooling Plate System 12
5.4 Gas Flow System (FLOW) 12
5.5 Gas Burner System (Burner) 12
5.6 Ignition System (IGN) 12
5.7 Analog Camera System (OpsCam) 12
5.8 Illumination System (OpsCam) 12
5.9 Diagnostic System (DIAG) 13
5.10 Radiometer System 13
5.11 Photomultiplier Tube (PMT) System 13
5.12 Thin Fiber Pyrometry (TFP) Array System 13
5.13 Measurement System (MEAS) 13
5.14 E-Field System (E-FIELD) 13
5.15 ACME Avionics Package System (AVP) 14
5.16 Avionics Package (AVP) 14
5.16.1 Main Computer System (MCS) – Control and Data Management 14
5.16.2 Color Camera Data Store (CDS) System 15
5.16.3 Power Distribution System (PDS) 15
5.17 ACME Imaging Package System (IMAG) 15
5.18 Data Color Camera Imaging System (DataCam) 15

5.19 CIR Imaging Subsystems 15
6.0 INTERFACES 16
6.1 Internal Interfaces 17
6.2 External Interfaces 17
6.2.1 Launch Configuration Interfaces 17
6.2.2 On-orbit External Interfaces 17
7.0 ACME ENGINEERING REQUIREMENTS 18
8.0 ACME OPERATIONAL REQUIREMENTS 72
9.0 CARRIER REQUIRMENTS 85
10.0 DERIVED REQUIREMENTS 86
11.0 SOFTWARE REQUIREMENTS 87
12.0 SAFETY REQUIREMENTS 115
APPENDIX A ACRONYMS AND ABBREVIATIONS <APPENDIX> 116
A.1 Scope 116

TABLE OF FIGURES

Figure 3-1—Example Project Requirements Source Diagram 6
FIGURE 1. ACME Requirements Flow 9
FIGURE 2. ACME Product Breakdown Structure (PBS) 10
FIGURE 3. ACME Chamber Insert Assembly (CIA) Design 11
FIGURE 4. ACME Avionic Package (AVP) Design 14
FIGURE 5. ACME Data Color Camera Package (DataCam) Design 16

TABLE OF TABLES

Table 2-1—Applicable Documents 2
Table 2-2—Reference Documents 3
Table 5-1—Engineering Requirements 18
Table 6-1—Operational Requirements 72
Table 8-1—Derived Requirements 86
Table 9-1—Software Requirements 87
Table A-1—Acronyms 116

1.0 INTRODUCTION

1.1 Purpose

This document will define the engineering requirements necessary to successfully complete science and safety requirements. This document will be used by the ACME design engineers to develop the systems necessary for ACME. Interface and carrier requirements levied on ACME will be documented in the *ACME ICD*, FCF-ICD-CIR-ACME. ACME will use the existing Multi-User Droplet Combustion Apparatus (MDCA) hardware and software platform design already developed as a baseline for meeting the requirements found in this document where applicable. Due to the reuse of the MDCA platform design, ACME operational safety concerns will be addressed in existing MDCA documents where applicable. Requirements in this document levied on the ACME carrier, the Combustion Integrated Rack (CIR), will be communicated to the ACME-CIR Interface / Sustaining Engineering Team

1.2 Scope

This Engineering Requirements Document assesses the system requirements and lessons learned of the existing Multi-user Droplet Combustion Apparatus (MDCA) and Flame Extinguishment Experiment requirements needed to successfully build the hardware and develop avionics software and how these requirements are implemented. The Engineering Requirements Document (ERD) will provide information regarding the ACME experiment and ensure that requirements are quantifiable, verifiable and specific. The ACME ERD consists of six sets of requirements: Science Requirements, Operational Requirements, FCF/CIR Integration Requirements, Derived Requirements, Software Requirements and Safety Requirements.

2.0 APPLICABLE DOCUMENTS

This section lists the NASA/Government and non-NASA/Government specifications, standards, guidelines, handbooks, or other special publications applicable to this document.

2.1 Order of Precedence for Documents

In the event of a conflict between this document and other documents specified herein, the requirements of this document shall apply. In the event of a conflict between this document and higher level documents, the higher level documents shall take precedence over this document.

Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

2.2 Applicable Documents

Applicable documents are those documents that form a part of this document. These documents carry the same weight as if they were stated within the body of this document.

Table 2-1—Applicable Documents

Document Number	Document Title
FCF-IA-MV-ACME	ACME Integration Agreement
FCF-ICD-0076	FCF Software Interface Control Document
FCF-ICD-CIR-ACME	ACME Interface Control Document for the FCF
FCF-IDD-CIR_Rev.C	CIR Payload Interface Definition Document
CIR-DOC-4064	FCF CIR Payloads Accommodations Handbook
CIR-DOC-3850A	CIR Phase III Flight Safety Data Package
ACME-SRD-001	ACME Science Requirements Document
ACME-PLN-002	ACME Systems Engineering Management Plan (SEMP)
ACME-PLN-003	ACME Science and Engineering Verification Plan (SEVP)
P40020	SpaceDOC II Safety and Mission Assurance Plan
P04005	Requirements Management
P40030	Software Safety Plan

2.3 Reference Documents

Reference documents are those documents that, though not a part of this document, serve to clarify the intent and contents of this document.

Table 2-2—Reference Documents

Document Number	Document Title
KHB 1700.7	Space Shuttle Payload Ground Safety Handbook
MSFC-HDBK-527/ JSC 09604	Materials Selection List for Space Hardware Systems
MSFC-SPEC-522	Design Criteria for Stress Corrosion Cracking
NSTS 13830	Implementation Procedure for NSTS Payloads System Safety Requirements
NSTS 14046	Payload Verification Requirements
NSTS 1700.7	Safety Policy and Requirements for Payloads Using the Space Transportation System
NSTS 1700.7 ISS Addendum	Safety Policy and Requirements for Payloads Using the International Space Station
NSTS 18798A MA2-95-048	Interpretation of NSTS Payload Safety Requirements
SSP 50005	International Space Station Flight Crew Integration Standard
SSP 52005	International Space Station Payload Flight Equipment Requirements and Guidelines for Safety-Critical Structures

3.0 REQUIREMENTS MANAGEMENT

3.1 Preparation for Requirements

The preparation for developing proper project requirements is an important first step. Preparing in this sense means establishing procedures for identifying, approving, allocating and managing changes to requirements. This document along with the ZIN Requirements Management Procedure (P04005) is meant to facilitate this process. Other project documentation supports this effort and their completion is also required. A project shall use the following documents and procedures during the requirements development process:

- ZIN Requirements Management Procedure P04005
- SpaceDOC II Systems Engineering Management Plan (SEMP) or a Project SEMP as necessary T04001
- ZIN Requirements Database T04003
- Project Concept of Operations (ConOps) T04002
- Project Engineering Requirements Document (ERD)

3.1.1 Stakeholder Analysis

An understanding of a customer's needs and goals for a project is paramount to the requirements development process. Ultimately, it is the customer who will be using the hardware built to the specifications laid out by the requirements a project develops. A customer's satisfaction with the performance of the developed hardware and the benefits the hardware's operation provides to stakeholders is the ultimate measurement of a projects success. Therefore, a project shall perform a stakeholder analysis in the early phases (Pre-phase A or Phase A) of the project life cycle.

3.1.1.1 Identify Stakeholders

Project requirements are elicited from Project Stakeholders. A stakeholder is any entity with a legitimate interest in the project. They are any person or organization that is actively involved in a project, or whose interests may be positively or negatively affected by execution or completion of the project. Some stakes an entity may have in your project are: Interest, rights, ownership, knowledge, impact or influence, contribution, etc. Examples of stakeholders can include:

- ZIN Technologies (further broken into Project Manager, QA Manager, Resource Manager, etc.)
- Principal Investigators and Project Scientists
- Customer
- NASA (further broken into GRC, JSC, ISS, etc.)
- Product users
- Product integrator

Stakeholders shall be identified and documented in a Project SEMP using the Project SEMP template T04001.

3.1.1.2 Identify Level 0 Stakeholder Requirements

To initiate the requirements elicitation process a project should discuss and develop a short list of the main goals and products of the project. Ideally there would be less than 10 and around 4 or 5 main Level 0 requirements. This step should immediately follow the identification of the Project Stakeholders.

3.2 Obtaining Requirements

Once the preparations for developing requirements have been made early in the life cycle of a project, the engineering team can begin to develop a set of requirements that will guide the design and development of the systems necessary to fulfill the Project Stakeholder's needs. These "build-to" or engineering requirements begin to take shape typically in Phase A of a project life cycle and continue to be developed throughout the design phase of the project.

3.2.1 Identifying All Stakeholder Requirements

After the initial, Level 0, stakeholder requirements have formed a foundation for discussion, a project can initiate the collection of all stakeholder requirements. Inputs for identifying more detailed stakeholder requirements will include: Source documents, stakeholder needs and project constraints such as cost, schedule and hardware/software solution restrictions. Stakeholder requirements can also be found in interface documents, concept documents developed by the project, ZIN organizational and industry standards, test results and design maturation discussions, stakeholder meetings, etc. At this point in the project trade studies are ongoing and specific hardware and software solutions are beginning to be formulated.

Requirements shall be properly documented with traceability and captured in a Requirements Development Matrix Database using the ZIN template T04003. Traceability shall include (as appropriate):

- The source document identity
- The paragraph title
- The sentence number

The requirements will then further be documented using this template as the project's Engineering Requirements Document, which is generated from the database.

Figure 1 shows the various sources of project requirements in a block flow diagram. This section should also make apparent what requirements are housed and maintained by the project engineering team and which are external requirements that are not included in this document but must also be considered during the design process. Note the color coded differentiation between internal project documents and external documents.

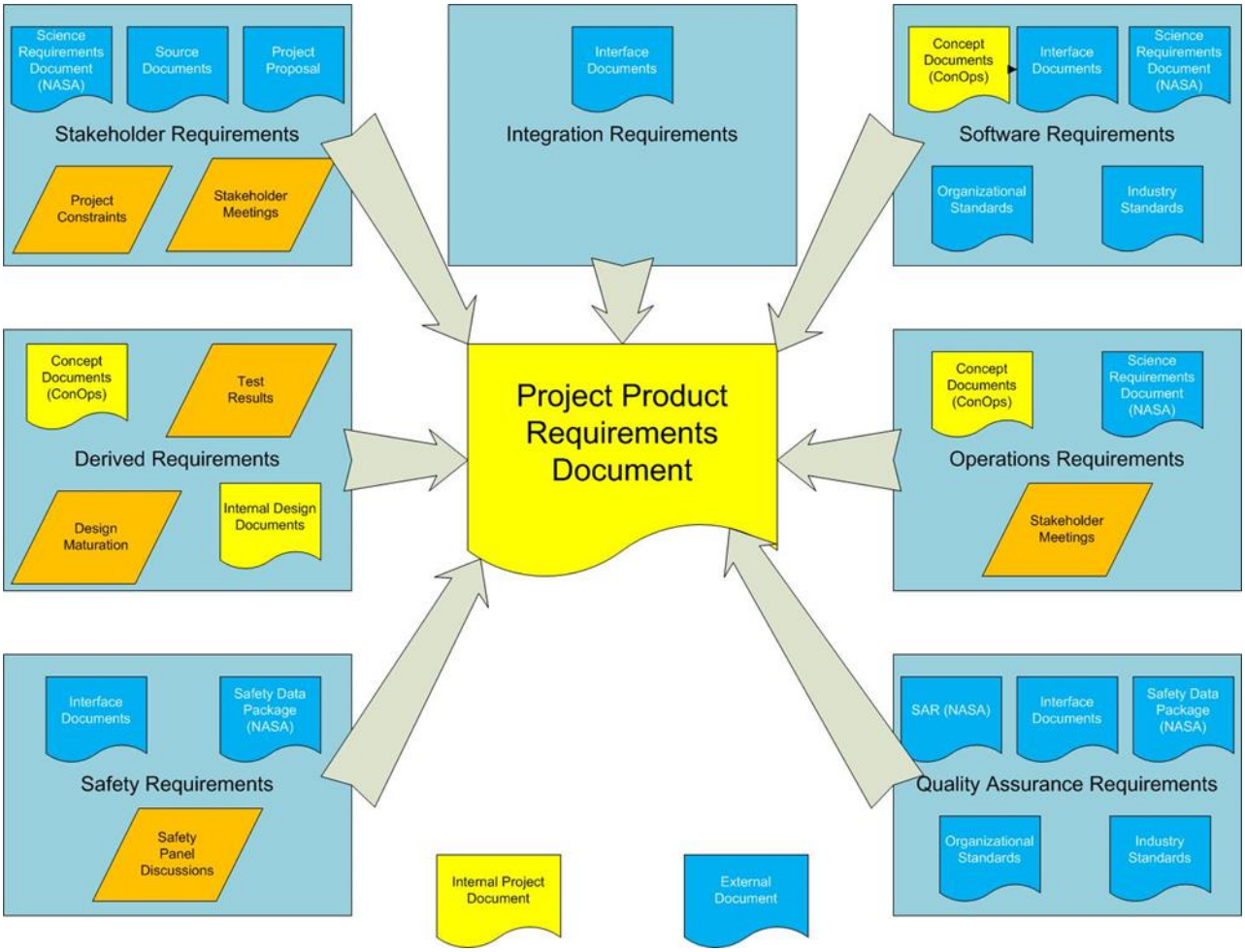


Figure 3-1—Example Project Requirements Source Diagram

3.3 Analyzing Requirements

Once all stakeholder requirements have been collected and organized it is essential that these requirements be analyzed. Ensuring that the requirement statements meet the necessary standard of quality for communication and implementation is part of this process. Projects shall review their requirement statements for quality and validity.

3.3.1 Requirement Quality

Requirement statements shall be clear and concise and decomposed to such a level that there is only one requirement in each requirement statement. Requirements shall use the word "shall" to clearly identify the demand upon the project team. Some tips to identify poor requirement statements are:

- Use of superlatives such as best, most
- Subjective language such as "user friendly", "easy to use"
- Vague pronouns
- Ambiguous adverbs and adjectives such as "almost always", "significant", "real-time", "various", "few", "many", "appropriately"
- Open-ended, non-verifiable terms such as "provide support", "as a minimum", "but not limited to"
- Comparative phrases like "better than" or "higher quality"
- Loopholes such as "if possible", "as appropriate"
- Other indefinites such as "etc.", "to be determined (TBD)"

At the time a requirement is identified, a verification method shall also be identified and documented in the Requirements Database. Verification methods include test, inspection, demonstration and analysis. This will encourage the writers of the requirement to evaluate whether they have written a verifiable requirement and provide an instant check of the requirement's value. This step shall be done early in the project, typically in Phase B of the project life cycle.

3.3.2 Requirement Validation

As mentioned above, understanding stakeholders' needs and goals is an important step in forming requirements. Ensuring that your requirements are written in a way that fulfills these expectations is the next step. Along with understanding the manner in which the requirement will be verified it is also important to analyze whether or not the requirements and the solutions they offer align themselves with stakeholders initial operational and programmatic goals.

Requirement validation requires forethought as to whether or not the hardware that the engineering requirements will produce can perform in a manner that fulfills the stakeholders goals. Simplified, validation becomes the question: Are you building the right hardware? For example, have your requirements properly captured the precision and accuracy needs of a particular measurement device? All project requirements shall be validated against stakeholder's requirements. This includes any transition or derived requirements.

3.4 Approving Requirements

The approval of the ERD serves as an official method of communication between the engineering team and the NASA Project Manager. Communication between the engineering team and NASA stakeholders will have invariably been occurring throughout the requirements formulation process. However, the ERD approval is where these requirements are formally agreed upon and baselined.

3.5 Maintaining Requirements

Maintenance of requirements is one of the most essential steps to a project's requirements development effort. Even after the ERD has been baselined there could be many instances when it would need to be updated. A new stakeholder could be identified, maturation of the product design could bring about new derived requirements from within the engineering team, and of course frequent review of the requirements could result in finding flaws in the requirement statements that can be fixed.

Following the logic of this process it is easy to see how many of the project's fundamental documents are affected by the requirements. Therefore, when a requirement is modified or added, each document discussed in this process is also subject to change.

This step shall happen regularly in the Project Life-Cycle to constantly ensure the integrity of the end product. Requirements documents, the Requirements Development Matrix and associated project documents shall be updated prior to major project gateway reviews at a minimum.

4.0 ACME REQUIREMENT SOURCES

4.1 Evaluation of Requirement Sources

Figure 1 represents the requirements flow for all of ACME requirements. Science requirements originate from the ACME Science Requirements Document (SRD). Operational requirements originate from the Science Operations Requirements Document (SORD). The carrier requirements are found in the CIR – ACME Interface Control Document (FCF-ICD-CIR-ACME). Derived Requirements develop from engineering requirements and are created by the ACME Project Team. Software Requirements are derived by the ACME engineering team with the purpose of command and control of hardware as well as telemetry reporting. Safety requirements are created from the ACME safety review hazard reports. These requirements are all incorporated into this the *ACME Engineering Requirements Document* (ACME-REQ-0001).

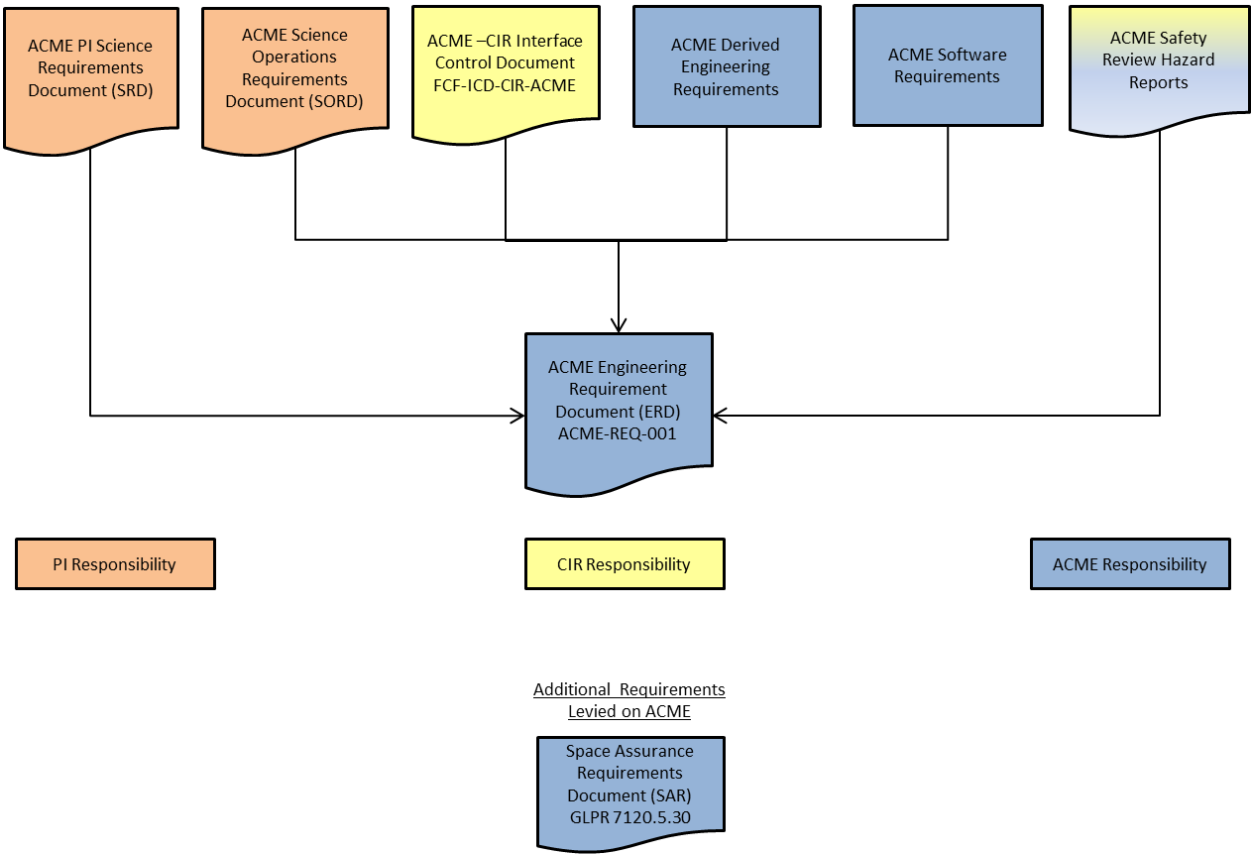


FIGURE 1. ACME Requirements Flow

5.0 ACME HARDWARE PRODUCT DESCRIPTION

The ACME investigation will be using the MDCA hardware and software concepts and design when appropriate. The ACME hardware consists of three hardware system domains, the Chamber Insert Assembly (CIA) System, the Avionics Package (AVP) System and the ACME Imaging Packages (IMAG) System.

The CIA provides the mechanism for the combustion event. The ACME CIA Flow subsystem, which interfaces with the CIR Fuel/Oxidizer Management Assembly (FOMA), provides gas mixing and flow control. The AVP is the center of control and power distribution for ACME. And, the IMAG provides imaging of the combustion event including that provided by the carrier (CIR).

The ACME PBS numbering convention for the systems in Figure 2 corresponds to the ACME Work Breakdown Structure (WBS) numbering for traceability across the ACME Project Lifecycle. The number convention utilizes a SSC (System/Subsystem Classification) followed by an underscore and the uniquely allocated four digit requirement identification number.

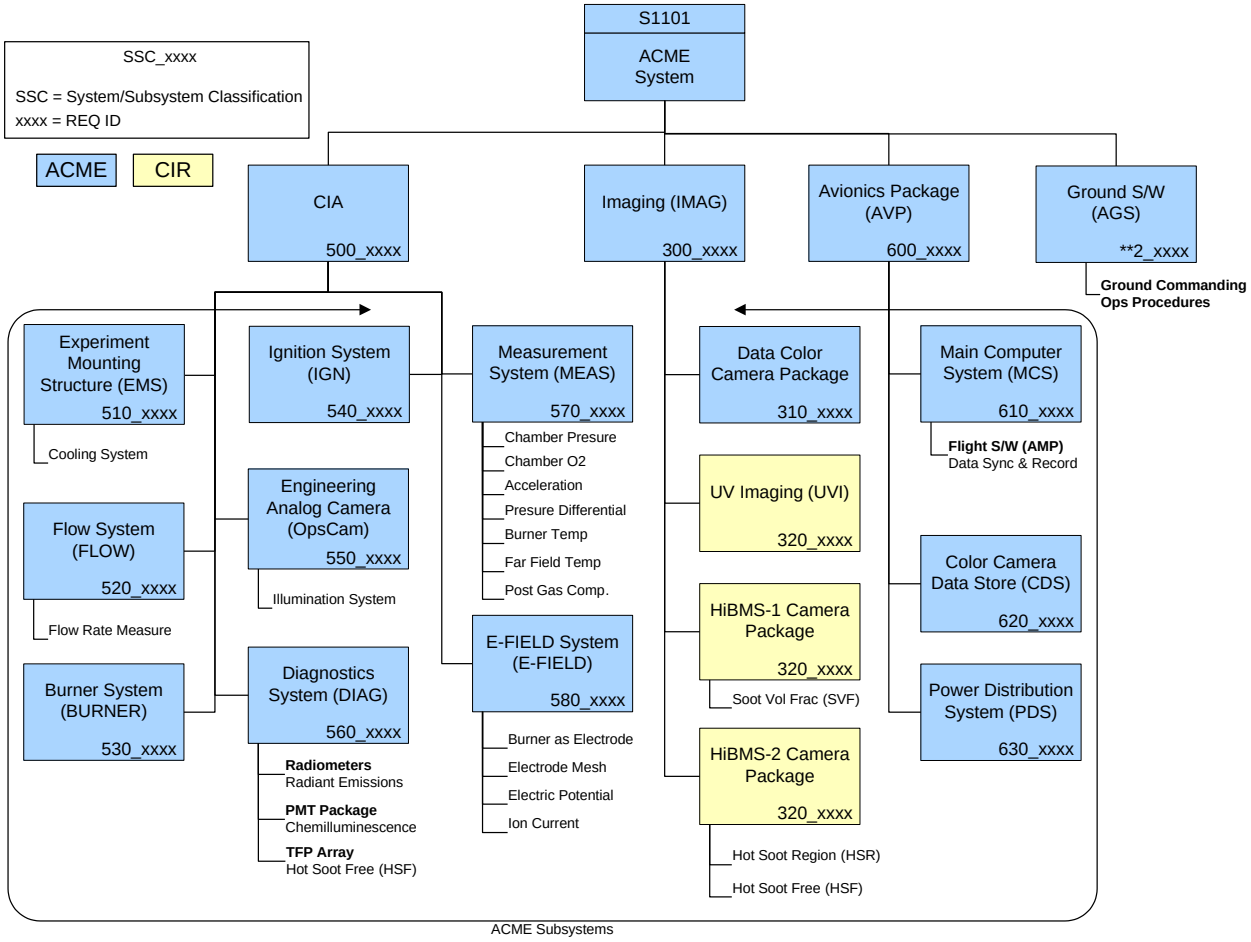


FIGURE 2. ACME Product Breakdown Structure (PBS)

5.1 ACME Chamber Insert Assembly (CIA) System

The ACME CIA System provides all the necessary hardware and interfaces to perform science on one platform. The available interfaces are the CIR Water Thermal Cooling System (WTCS), and the CIR Fuel/Oxidizer Management Assembly (FOMA). Most experimental functions occur within the CIA structure and many associated subsystems are found on the CIA. The CIA consists of the Experiment Mounting Structure (EMS) which is the functional hardware and upon which the various subsystems are structurally based. Other subsystems found on and around the CIA include the Gas Flow System (FLOW), Exchangeable Gas Burner System (Burner), Ignition System (IGN), Engineering Analog Camera System (OpsCam), Diagnostics System (DIAG), Measurement System (MEAS), E-Field System (E-FIELD).

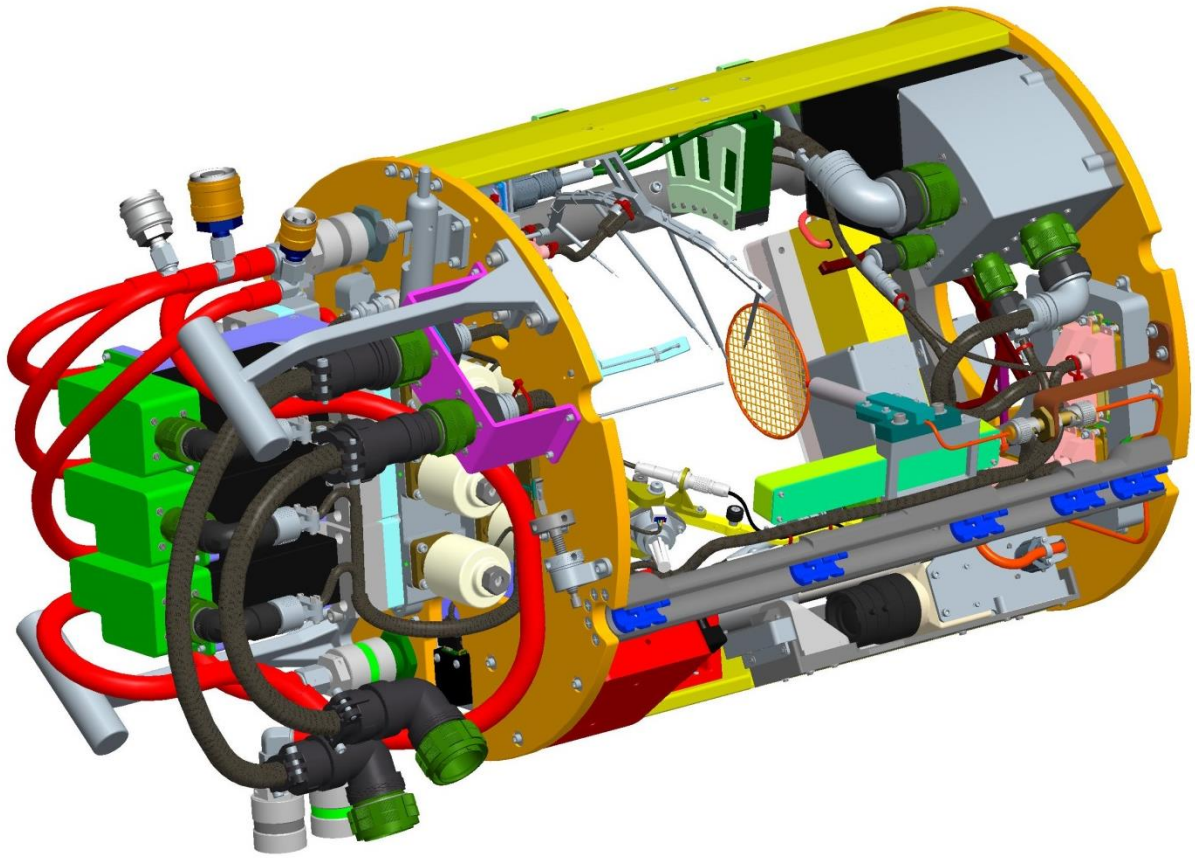


FIGURE 3. ACME Chamber Insert Assembly (CIA) Design

5.2 Experiment Mounting Structure (EMS) System

The EMS provides the main structure for the CIA and acts as the primary mechanical interface with the CIR combustion chamber. It consists of two endplates, support structure, and extraction handles.

5.3 Cooling Plate System

ACME will use two cold plates mounted to the Experiment Mounting Structure interfacing with the CIR Water Thermal Control System (WTCS). One cooling plate can be attached to each end plate. The hardware to accommodate this is identical to that used on MDCA System. The cooling plates are plumbed together using rigid tubing, and attached to the IRR WTCS interfaces with flexible hoses and quick disconnects.

5.4 Gas Flow System (FLOW)

There is a reconfigurable flow system on the CIA that is auxiliary to the CIR FOMA. Gases are received from FOMA via quick disconnects on the Interface Resource Ring (IRR). The gases are directed through interchangeable Mass Flow Controllers, which are replaced as experiments require. Gases are then directed through a series of switching and isolation valves to the burner or directly vented into the chamber for chamber fills.

5.5 Gas Burner System (Burner)

The burner system consists of three different types of exchangeable burners made from varying inner and outer diameter stainless steel tubing supported by an electrically insulating ceramic base. There are 4 types of burners: Spherical (3), Gas Jet (7), Co-flow (1) and Burning Rate Emulator (2). Each burner tip is smooth and without burs or sharp edges to aid in the gas deployment. The burners are mounted with a coupling that allows exchangeability of the burners and maintains the precise alignment of the system. Spherical and Gas-Jet burners also include burner temperature thermocouples.

5.6 Ignition System (IGN)

The Ignition System will reuse the design of the MDCA System igniter, with a different deployment scheme.

The ignition system consists of a hot wire igniter subassembly mounted to a deployment arm. A Haydon-Kerk linear actuator motor and driver board controls the position of the deployment arm. The igniter tip is mounted to an insulating ceramic structure attached to the end of the arm. The entire top assembly is removed from the arm and replaced on an experiment basis to maintain the precise geometry necessary for igniter functionality and to reposition the igniter tip based on the burner tip's position.

5.7 Analog Camera System (OpsCam)

The Analog Camera System will be provided, internal to the combustion chamber, primarily as a near real-time visual aid to engineering for hardware operation monitoring and troubleshooting. In addition, this view will provide a near real-time "look" into the chamber for science throughout the flame combustion event.

5.8 Illumination System (OpsCam)

The Illumination System consists of an array of high intensity, white light LEDs that receive power from the relay board hardware and are controlled by the Data Acquisition Unit. The

illumination system intensity will be variable to avoid saturation of diagnostics. This illumination system is provided to support the Analog Camera System.

5.9 Diagnostic System (DIAG)

Two of the diagnostic systems the Radiometers, the Photomultiplier Tube Modules (PMTs), and the Thin Fiber Pyrometry (TFP) Array are located on the Insert. This hardware supports Diagnostic Analysis of the Flames.

5.10 Radiometer System

The Radiometers will provide relative measure of flame radiation and emissions, over the wavelength range of 0.2 μm – 11 μm, throughout the combustion event. The Radiometer System will be mounted on the CIA providing a direct view of the combustion event. Data from the Radiometers will be handled by the Data Acquisition Unit mounted inside the Chamber. Power via the Relay Board to the Radiometer System is provided by ACME AVP System (section 3.4.2).

5.11 Photomultiplier Tube (PMT) System

The Photomultiplier Tube Package will provide high-sensitivity relative measurements of flame radiation and chemiluminescent emissions of electromagnetic energy over a wide range of frequencies and intensity levels throughout the combustion event as well as flame detection and extinction determination. The PMT System consists of three Hamamatsu PMT modules, mounted on the CIA providing a direct view of the combustion event. Additional filters will be required for the 2 PMT modules used for narrowband emissions. Data from the PMT modules will be handled by the Data Acquisition Unit mounted inside the Chamber. Power via the Relay Board to the PMT System is provided by ACME AVP System (section 3.4.2).

5.12 Thin Fiber Pyrometry (TFP) Array System

A Thin Fiber Pyrometry (TFP) Array arm mounted to a translating motor to move the array into and out of the flame. The translating motor will be an Ultramotion brand linear stepper motor and encoder, similar to that used by MDCA System for its igniter. The Array will be stowed out of the combustion area during E-Field testing, or when the Array is not required, but will not be removed from the chamber. The array can be aligned with either the HiBMs-2 or ACME data color camera.

5.13 Measurement System (MEAS)

The ACME measurement system encompasses the measurement subsystems involved in detection and measurement of the experiments with the exception of diagnostic systems mentioned above in section 3.4.1.6 and CIR provided diagnostics. The following sensors are a part of the Measurement System: Far Field Thermocouple Array, Pressure Transducers for differential pressure measurement, and heat flux sensors for the BRE burners.

5.14 E-Field System (E-FIELD)

The E-Field system consists of two High Voltage (HV) sources (one negative and one positive polarity), a passive network, an insulated mesh electrode and an insulated burner electrically connected to the returns of the HV supplies.

5.15 ACME Avionics Package System (AVP)

The Avionics Package System (AVP) provides power conditioning and power distribution, system control, data acquisition and CIR and ground interface for the ACME insert. The AVP will house the main electronics for the ACME hardware and will be mounted on the CIR Optics Bench.

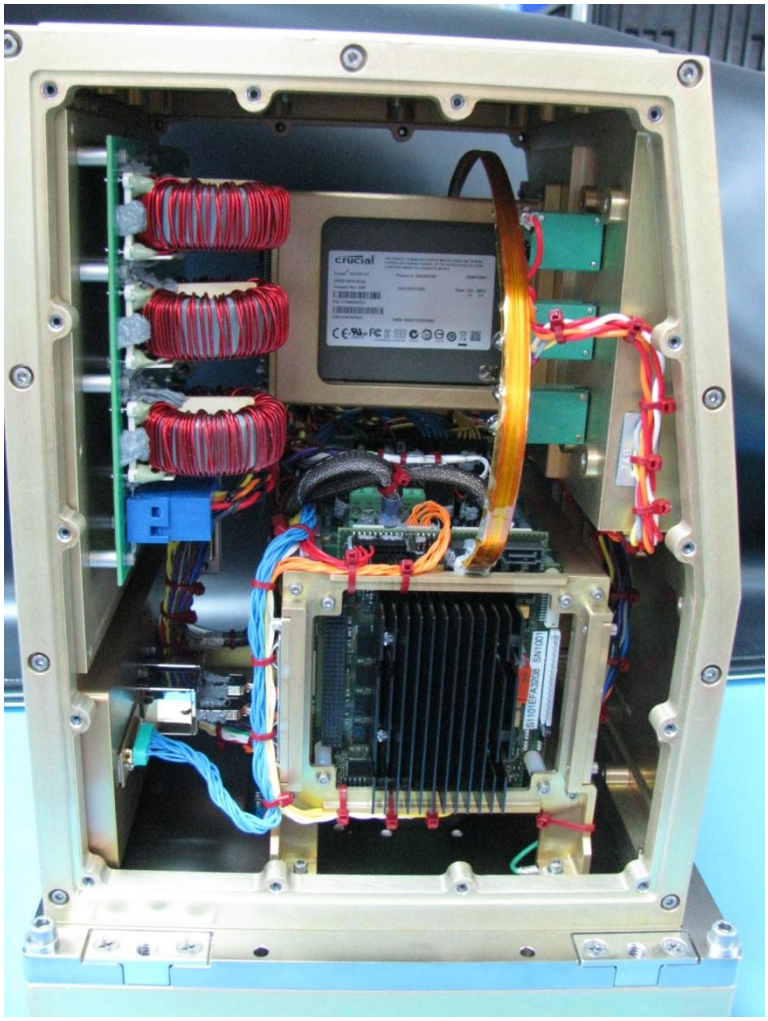


FIGURE 4. ACME Avionic Package (AVP) Design

5.16 Avionics Package (AVP)

The Avionics Package will house the main electronics and data storage for the ACME hardware and will be mounted at the CIR PIL (PI location), using the standard Optics Bench interfaces.

5.16.1 Main Computer System (MCS) – Control and Data Management

A main computer processor is responsible for controlling CIA hardware, imaging hardware to execute test points as well as acquiring, storing, and downlinking experiment data. Experiment control and sequencing will be uplinked from the ACME Ground System (AGS) software, at the TSC, to the ACME Main Processor (AMP) System software that is resident on the main processor in a PC/104 stack configuration. The main processor communicates with the CIR

Input/Output Processor (IOP). The interface to the CIR IOP is necessary for transferring commands, downlinked data and images, through the International Space Station (ISS) communications system.

5.16.2 Color Camera Data Store (CDS) System

The ACME Color Camera data is stored on a hard drive in the AVP. The hard drive is removable and replaceable on orbit.

5.16.3 Power Distribution System (PDS)

The AVP Power & Distribution System provides power to the equipment in within the AVP and the CIA via four 28VDC, 8 amp circuits from the CIR's Electrical Power Control Unit (EPCU). In the AVP, the power is converted to the necessary voltages, regulated and filtered. Power distribution and control within the Chamber is handled by a data acquisition unit and a pair of custom designed relay boards. These devices handle all of the communications with the AVP as well as the activation of various components of the CIA.

5.17 ACME Imaging Package System (IMAG)

The ACME Imaging (IMAG) System provides all optical imaging and non-optical imaging information for acquisition and storage concerning each of the ACME experiments.

5.18 Data Color Camera Imaging System (DataCam)

The ACME Data Color Camera Package (see Figure 4) will provide a view of the combustion event. The current Color Camera Package concept consists of an Allied Vision (formerly Prosilica) GC1380CH (30fps) camera. The camera is external to the chamber mounted at a UML Location on the Optics Bench. Power to the Data Color Camera Package is supplied via the UML location.

5.19 CIR Imaging Subsystems

ACME will utilize the CIR LLL-UV, HiBMS-1, and HiBMS-2 Packages.

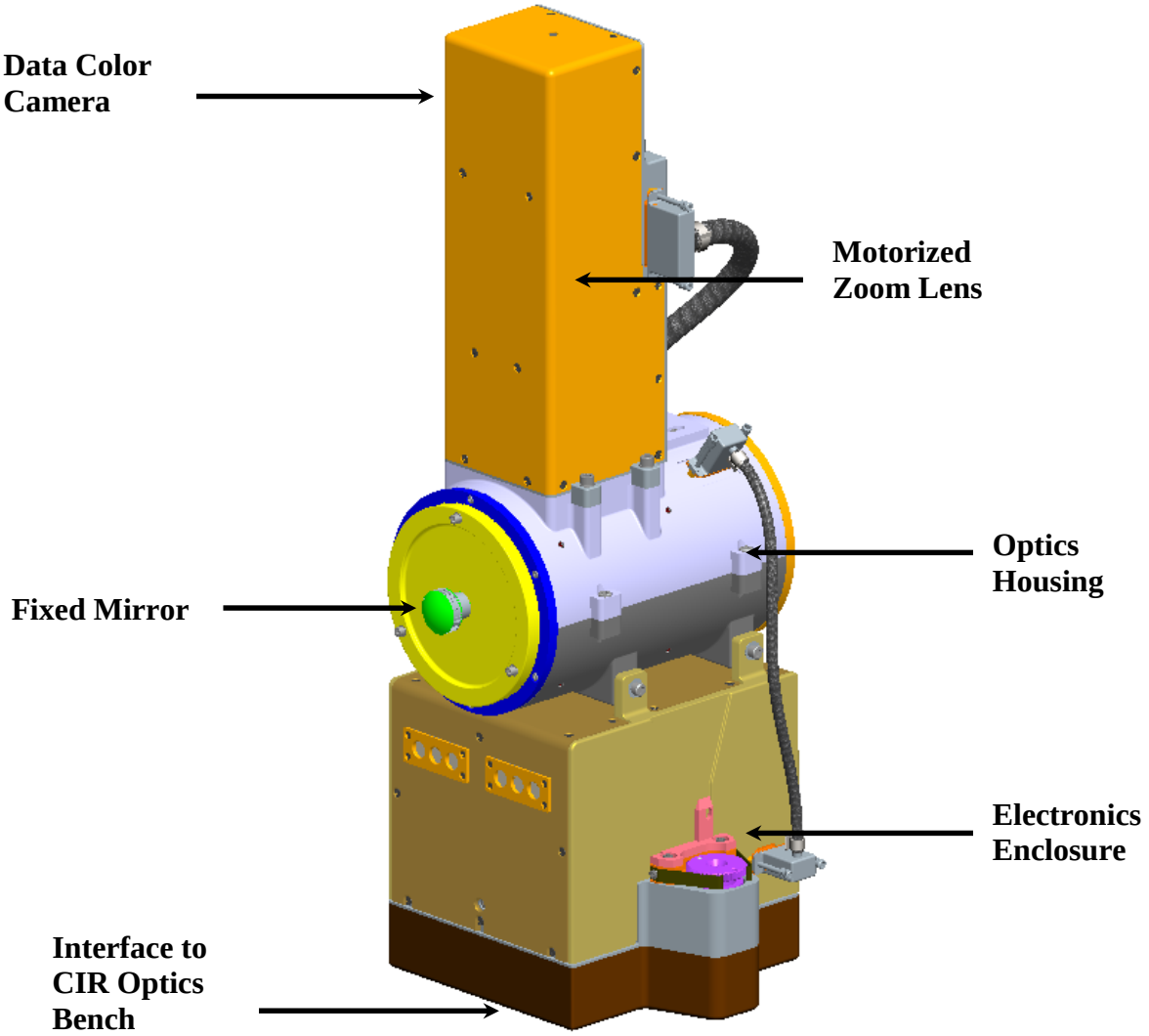


FIGURE 5. ACME Data Color Camera Package (DataCam) Design

6.0 INTERFACES

This section outlines the interfaces between the project's hardware as a system to other external systems, any launch configuration interfaces, and on-orbit external interfaces that are needed for successful operation of the ACME System. Internal interfaces are also addressed.

6.1 Internal Interfaces

The internal interfaces between ACME and CIR are listed below and the requirements for these interfaces will be defined in *ACME Interface Control Document for the Fluids and Combustion Facility*, FCF-ICD-CIR-ACME.

- Structural/Mechanical and Stowage
- Electrical Power
- Command and Data Handling
- Payload and Diagnostics
- Thermal Control
- Exhaust Vent
- FOMA Gas Delivery
- Environmental

6.2 External Interfaces

6.2.1 Launch Configuration Interfaces

ACME provided hardware is non-operational in whatever launch vehicle is chosen to get it on-orbit. ACME hardware will be stowed in standard containers provided by ACME and will only have physical and structural interfaces with the containers.

6.2.2 On-orbit External Interfaces

ACME has no direct external interfaces on-orbit. All interfaces with the FCF, US LAB, ISS and the ground are provided indirectly through CIR.

7.0 ACME ENGINEERING REQUIREMENTS

Engineering requirements are ACME's build-to requirement response to science requirements levied by NASA. This section is drawn from the information filled into the ZIN Requirements Development Matrix (T04003) using the Requirements Management Procedure (P04005.) ACME's requirements are organized according to the project's PBS. This helps to keep the reader and reviewer thinking in terms of the whole system.

The SSC_ID column is for useful organization and sorting of the system requirements in the Requirements Tracking Matrix (RTM) into categories by system, subsystem, or other traits for measurements and metrics that may be of interest. The SSC (System/Subsystem Classification) is a three digit number which relates to the Product Breakdown Structure (PBS) which in turn is a subset of the Work Breakdown Structure (WBS). The first digit of the SSC relates the requirement to the system assignment. The second digit of the SSC relates the requirement to the subsystem assignment. The third digit of the SSC is used to relate the requirement to the product classification or type: flight hardware product (0), flight software product (1), or ground software product (2), in which they affect. The SSC is followed by an underscore as a delineator and then a unique four digit identification number. The underscore, versus the dash, is used to keep the SSC and ID married as one object in the database. Requirements with like qualities of interest may be grouped together by ID number for ease of sorting.

Table 7-1—Engineering Requirements

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
	SPACE110	1.1	**General Experiment Configuration**
BURNER	530_0010	1.1.1	**Burners – Installation** ACME shall install a single burner one at a time within the CIR chamber.
BURNER	530_0020	1.1.2	**Burners – Exchangeable** ACME shall provide that all burners are exchangeable during spaceflight.
BURNER	530_0030	1.1.2	**Burners – ORU** ACME shall provide all burners as ORUs.
FLOW	520_0010	1.1.4.1	**Flow System H/W Compatibility - Hydrogen** ACME shall provide a gas delivery system that is compatible with the gaseous fuel hydrogen.
BURNER	530_0050	1.1.4.1	**Burner H/W Compatibility - Hydrogen** ACME shall provide all burners to be compatible with the gaseous fuel hydrogen.
FLOW	520_0020	1.1.4.1	**Flow H/W Compatibility - Methane** ACME shall provide a gas delivery system that is compatible with the gaseous fuel methane.
BURNER	530_0060	1.1.4.1	**Burner H/W Compatibility - Methane** ACME shall provide all burners to be compatible with the gaseous fuel methane.
FLOW	520_0030	1.1.4.1	**Flow H/W Compatibility - Ethane** ACME shall provide a gas delivery system that is compatible with the gaseous fuel ethane.
BURNER	530_0070	1.1.4.1	**Burner H/W Compatibility - Ethane** ACME shall provide all burners to be compatible with the gaseous fuel ethane.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
FLOW	520_0040	1.1.4.1	**Flow H/W Compatibility - Propane** ACME shall provide a gas delivery system that is compatible with the gaseous fuel propane.
BURNER	530_0080	1.1.4.1	**Burner H/W Compatibility - Propane** ACME shall provide all burners to be compatible with the gaseous fuel Propane.
FLOW	520_0050	1.1.4.1	**Flow H/W Compatibility - Ethylene** ACME shall provide a gas delivery system that is compatible with the gaseous fuel ethylene.
BURNER	530_0090	1.1.4.1	**Burner H/W Compatibility - Ethylene** ACME shall provide all burners to be compatible with the gaseous fuel Ethylene.
FLOW	520_0060	1.1.4.1	**Flow H/W Compatibility - Propylene** ACME shall provide a gas delivery system that is compatible with the gaseous fuel propylene.
BURNER	530_0100	1.1.4.1	**Burner H/W Compatibility - Propylene** ACME shall provide all burners to be compatible with the gaseous fuel Propylene.
FLOW	520_0070	1.1.4.1	**Flow H/W Compatibility - Fuel Mixtures** ACME shall provide a gas delivery system that is compatible with mixtures of any of the following gaseous fuels: 1) hydrogen 2) methane 3) ethane 4) propane 5) ethylene 6) propylene
BURNER	530_0110	1.1.4.1	**Burner H/W Compatibility - Fuel Mixtures** ACME shall provide all burners to be compatible with gaseous fuel mixtures of any of the following gaseous fuels: 1) hydrogen 2) methane 3) ethane 4) propane 5) ethylene 6) propylene
FLOW	520_0080	1.1.4.2	**Flow H/W Compatibility - Nitrogen** ACME shall provide a gas delivery system that is compatible with the inert gas nitrogen.
BURNER	530_0120	1.1.4.2	**Burner H/W Compatibility - Nitrogen** ACME shall provide all burners to be compatible with the inert gas nitrogen.
FLOW	520_0090	1.1.4.2	**Flow H/W Compatibility - Carbon Dioxide** ACME shall provide a gas delivery system that is compatible with the inert gas carbon dioxide.
BURNER	530_0130	1.1.4.2	**Burner H/W Compatibility - Carbon Dioxide** ACME shall provide all burners to be compatible with the inert gas carbon dioxide.
FLOW	520_0100	1.1.4.2	**Flow H/W Compatibility - Helium** ACME shall provide a gas delivery system that is compatible with the inert gas helium.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_0140	1.1.4.2	**Burner H/W Compatibility - Helium** ACME shall provide all burners to be compatible with the inert gas helium.
FLOW	520_0110	1.1.4.2	**Flow H/W Compatibility - Neon** ACME shall provide a gas delivery system that is compatible with the inert gas neon.
BURNER	530_0150	1.1.4.2	**Burner H/W Compatibility - Neon** ACME shall provide all burners to be compatible with the inert gas neon.
FLOW	520_0120	1.1.4.2	**Flow H/W Compatibility - Argon** ACME shall provide a gas delivery system that is compatible with the inert gas argon.
BURNER	530_0160	1.1.4.2	**Burner H/W Compatibility - Argon** ACME shall provide all burners to be compatible with the inert gas argon.
FLOW	520_0130	1.1.4.2	**Flow H/W Compatibility - Krypton** ACME shall provide a gas delivery system that is compatible with the inert gas krypton.
BURNER	530_0170	1.1.4.2	**Burner H/W Compatibility - Krypton** ACME shall provide all burners to be compatible with the inert gas krypton.
FLOW	520_0140	1.1.4.2	**Flow H/W Compatibility - Xenon** ACME shall provide a gas delivery system that is compatible with the inert gas xenon.
BURNER	530_0180	1.1.4.2	**Burner H/W Compatibility - Xenon** ACME shall provide all burners to be compatible with the inert gas xenon.
FLOW	520_0150	1.1.4.2	**Flow H/W Compatibility - Inert Mixtures** ACME shall provide a gas delivery system that is compatible with mixtures of any of the following inert gases: 1) nitrogen 2) carbon dioxide 3) helium 4) neon 5) argon 6) krypton 7) xenon
BURNER	530_0190	1.1.4.2	**Burner H/W Compatibility - Inert Mixtures** ACME shall provide all burners to be compatible with mixtures of any of the following inerts gases: 1) nitrogen 2) carbon dioxide 3) helium 4) neon 5) argon 6) krypton 7) xenon

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
FLOW	520_0160	1.1.4.1 1.1.4.2 1.1.4.3	**Flow H/W Compatibility - Fuel-Inert Mixtures** ACME shall provide a gas delivery system that is compatible with mixtures of any of the following gaseous fuels and inert gases: 1) hydrogen 2) methane 3) ethane 4) propane 5) ethylene 6) propylene 7) nitrogen 8) carbon dioxide 9) helium 10) neon 11) argon 12) krypton 13) xenon
BURNER	530_0200	1.1.4.1 1.1.4.2 1.1.4.3	**Burner H/W Compatibility - Fuel-Inert Mixtures** ACME shall provide all burners to be compatible with mixtures of any of the following gaseous fuels and inert gases: 1) hydrogen 2) methane 3) ethane 4) propane 5) ethylene 6) propylene 7) nitrogen 8) carbon dioxide 9) helium 10) neon 11) argon 12) krypton 13) xenon
FLOW	520_0170	1.1.4.4	**Flow H/W Compatibility - O_2 - Fuel Mixtures** ACME shall provide a gas delivery system that is compatible with mixtures of oxygen and one or more gaseous fuels (hydrogen, methane, ethane, propane, ethylene, and propylene) for oxygen concentrations up to 85% on a volume (molar) basis.
BURNER	530_0210	1.1.4.4	**Burner H/W Compatibility - O_2 - Fuel Mixtures** ACME shall provide all burners to be compatible with mixtures of oxygen and one or more gaseous fuels (hydrogen, methane, ethane, propane, ethylene, and propylene) for oxygen concentrations up to 85% on a volume (molar) basis.
FLOW	520_0180	1.1.4.4	**Flow H/W Compatibility - O_2 - Inert Mixtures** ACME shall provide a gas delivery system that is compatible with mixtures of oxygen and one or more inert gases (nitrogen, carbon dioxide, helium, neon, argon, krypton, and xenon) for oxygen concentrations up to 85% on a volume (molar) basis.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_0220	1.1.4.4	**Burner H/W Compatibility - O_2 - Inert Mixtures** ACME shall provide all burners to be compatible with mixtures of oxygen and one or more inert gases (nitrogen, carbon dioxide, helium, neon, argon, krypton, and xenon) for oxygen concentrations up to 85% on a volume (molar) basis.
FLOW	520_0190	1.1.4.4	**Flow H/W Compatibility - O_2 - Fuel & Inert Mixtures** ACME shall provide a gas delivery system that is compatible with mixtures of oxygen and one or more gaseous fuels (hydrogen, methane, ethane, propane, ethylene, and propylene) and one or more inert gases (nitrogen, carbon dioxide, helium, neon, argon, krypton, and xenon) for oxygen concentrations up to 85% on a volume (molar) basis.
BURNER	530_0230	1.1.4.4	**Burner H/W Compatibility - O_2 - Fuel & Inert Mixtures** ACME shall provide all burners to be compatible with mixtures of oxygen and one or more gaseous fuels (hydrogen, methane, ethane, propane, ethylene, and propylene) and one or more inert gases (nitrogen, carbon dioxide, helium, neon, argon, krypton, and xenon) for oxygen concentrations up to 85% on a volume (molar) basis.
EMS	510_0010	1.1.5	**Hardware Compatibility – CIR Chamber** The CIA hardware shall be compatible with concentrations of oxygen up to 40% on a volume (molar) basis.
		1.8	**Source Gas Supply**
FLOW	520_0200	1.8.1 1.8.1.1	**Experimental Fuels – H2** ACME shall provide 100% Hydrogen on a volume (molar) basis.
FLOW	520_0210	1.8.1 1.8.1.1	**Experimental Fuels – CH4** ACME shall provide 100% Methane on a volume (molar) basis.
FLOW	520_0220	1.8.1 1.8.1.1	**Experimental Fuels – C2H4** ACME shall provide 100% Ethylene on a volume (molar) basis.
FLOW	520_0230	1.8.1 1.8.1.1	**Experimental Fuels – Hydrocarbon Mixture 1** ACME shall provide a fuel mixture of 44.4/55.6 H2/CH4 on a volume (molar) basis.
FLOW	520_0240	1.8.1 1.8.1.1	**Experimental Fuels – Hydrocarbon Mixture 2** ACME shall provide a fuel mixture of 55.6/44.4 H2/CH4 on a volume (molar) basis.
FLOW	520_0250	1.8.1 1.8.1.1	**Experimental Fuels – Hydrocarbon Mixture 3** ACME shall provide a fuel mixture of 66.7/33.3 H2/CH4 on a volume (molar) basis.
FLOW	520_0260	1.8.1 1.8.1.2	**Oxidizer Mixture 1** ACME shall provide an oxidizer mixture of 85/15 O2/N2 on a volume (molar) basis.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
FLOW	520_0270	1.8.1 1.8.1.2	**Oxidizer Mixture 2** ACME shall provide an oxidizer mixture of 50/50 O2/N2 on a volume (molar) basis.
FLOW	520_0280	1.8.1 1.8.1.2	**Oxidizer Mixture 3** ACME shall provide an oxidizer mixture of 40/60 O2/N2 on a volume (molar) basis.
FLOW	520_0290	1.8.1 1.8.1.2	**Oxidizer Mixture 4** ACME HWCI shall provide an oxidizer mixture of 21/79 O2/N2 on a volume (molar) basis.
FLOW	520_0300	1.8.1 1.8.1.2	**Oxidizer Mixture 5** ACME shall provide an oxidizer mixture of 40/60 O2/He on a volume (molar) basis.
FLOW	520_0310	1.8.1 1.8.1.3	**Inert – N2** ACME shall provide 100% Nitrogen on a volume (molar) basis.
FLOW	520_0320	1.8.1 1.8.1.3	**Inert – He** ACME shall provide 100% Helium on a volume (molar) basis.
FLOW	520_0330	1.8.2 1.8.2.2	**Oxidizer Mixture 6** ACME shall provide an oxidizer mixture of 85/15 O2/CO2 on a volume (molar) basis.
FLOW	520_0340	1.8.2 1.8.2.2	**Oxidizer Mixture 7** ACME shall provide an oxidizer mixture of 40/60 O2/CO2 on a volume (molar) basis.
FLOW	520_0350	1.8.2 1.8.2.3	**Inert – CO2** ACME shall provide 100% Carbon Dioxide on a volume (molar) basis.
		1.1	**Chamber Atmosphere**
CIA	500_0081	1.11.4	**Chamber Atmosphere - Initial Pressure Precision** The ACME flow system shall provide the capability to vary the initial pretest chamber with a precision of ±3%.
CIA	500_0110	1.11.6	**Chamber Atmosphere – Free Volume** ACME shall provide a chamber free volume of at least 80 liters.
CIA	500_0120	1.11.7	**Chamber Atmosphere – Free** ACME should distribute hardware components uniformly in the chamber free volume relative to the chamber axis and the intersection of the CIR chamber window axes.
CIA	500_0130	1.11.8	**Chamber Atmosphere – ZOE Spherical Burner** ACME shall provide a free area of at least 200mm in diameter centered on the spherical burner and be kept clear of solid objects except the burner, tube, igniter, probes, TFP fibers and diagnostics for the spherical burner tests.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
CIA	500_0140	1.11.9	**Chamber Atmosphere – ZOE 1 Gas-jet, Coflow, & BRE** ACME shall provide a cylindrical free area for the gas-jet burner, coflow, and BRE burner tests that is at least 200mm in diameter, extends at least 100mm downstream and at least 50mm upstream from the burner outlet.
CIA	500_0150	1.11.9	**Chamber Atmosphere – ZOE 2 Gas-jet, Coflow, & BRE** ACME shall provide a cylindrical free area for the gas-jet burner and coflow burner tests that is at least 150mm in diameter, extends at least 200mm downstream from the burner outlet.
CIA	500_0180	1.11.12	**Chamber Atmosphere – Flame Only FOV** ACME shall provide chamber hardware with a non-reflective background such that there is nothing other than the flame that emits light within each optical detector's field of view and spectral range of sensitivity.
		1.9	**Burner Gas Delivery**
FLOW	520_0380	1.9.1	**Gas Supply - BRE, Spherical & Gas-Jet Burner** The ACME flow system shall simultaneously supply the BRE gas-jet or spherical burner with any combination of fuel, inert, or oxidizer source gas.
FLOW	520_0410	1.9.2	**Gas Supply - Normal Coflow Burner** The ACME flow system shall simultaneously deliver fuel, oxidizer and inert to either the inner tube or the outer tube of the coflow burner.
FLOW	520_0440	1.9.3	**Gas Supply Mixture – Control** The ACME flow system shall provide the capability for the composition of any mixture to be controlled to within ± 0.01 mole fraction.
FLOW	520_0450	1.9.3	**Gas Supply Mixture – MFC Response Time** ACME flow system shall know the gas mixture concentrations ejected from the burner as a function of time on a 1 second basis.
FLOW	520_0500	1.9.4	**Gas Flow Rate – Shutoff** ACME shall provide a flow system which has a gas flow shutoff capability for all gases.
FLOW	520_0530	1.9.4 1.9.5 1.9.5.1 1.9.5.1.1	**CLD Flame – Coflow Fuel Flow** ACME shall provide a flow system which is capable of providing a source fuel flow range of 0 to 0.2 slpm, on a N2 basis, within ±5% of the set point for the coflow burner.
FLOW	520_0540	1.9.4 1.9.5 1.9.5.1 1.9.5.1.2	**CLD Flame – Coflow Oxidizer Flow** ACME shall provide a flow system which is capable of providing a source oxidizer flow range of 0 to 15 slpm, on a N2 basis, within ± 5% of the set point for the coflow burner.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
FLOW	520_0550	1.9.4 1.9.5 1.9.5.1 1.9.5.1.3	**CLD Flame – Coflow Inert Flow** ACME shall provide a flow system which is capable of providing a source inert flow range of 0 to 0.1 slpm, on a N2 basis, within ± 5% of the set point for the coflow burner.
FLOW	520_0560	1.9.4 1.9.5 1.9.5.2 1.9.5.2.1	**E-FIELD Flame – Gas-Jet Fuel Flow** ACME shall provide a flow system which is capable of providing a source fuel flow range of 0 to 0.1 slpm, on a N2 basis, within ± 5% of the set point for the gas-jet burner.
FLOW	520_0570	1.9.4 1.9.5 1.9.5.2 1.9.5.2.3	**E-FIELD Flame – Gas-Jet Inert Flow** ACME shall provide a flow system which is capable of providing a source inert flow range of 0 to 0.05 slpm, on a N2 basis, within ± 5% of the set point for the gas-jet burner.
FLOW	520_0580	1.9.4 1.9.5 1.9.5.3 1.9.5.3.1	**E-FIELD Flame – Coflow Fuel Flow** ACME shall provide a flow system which is capable of providing a source fuel flow range of 0 to 0.1 slpm, on a N2 basis, within ± 5% of the set point for the coflow burner.
FLOW	520_0590	1.9.4 1.9.5 1.9.5.3 1.9.5.3.2	**E-FIELD Flame – Coflow Oxidizer Flow** ACME shall provide a flow system which is capable of providing a source oxidizer flow range of 0 to 2.0 slpm, on a N2 basis, within ± 5% of the set point for the coflow burner.
FLOW	520_0600	1.9.4 1.9.5 1.9.5.3 1.9.5.3.3	**E-FIELD Flame – Coflow Inert Flow** ACME shall provide a flow system which is capable of providing a source inert flow range of 0 to 0.05 slpm, on a N2 basis, within ± 5% of the set point for the coflow burner.
FLOW	520_0610	1.9.4 1.9.5 1.9.5.4 1.9.5.4.1	**Flame Design – Spherical Fuel Flow** ACME shall provide a flow system which is capable of providing a source fuel flow range of 0 to 0.3 slpm, on a N2 basis, within ± 5% of the set point for the spherical burner.
FLOW	520_0620	1.9.4 1.9.5 1.9.5.4 1.9.5.4.2	**Flame Design – Spherical Oxidizer Flow** ACME shall provide a flow system which is capable of providing a source oxidizer flow range of 0 to 3.0 slpm, on a N2 basis, within ± 5% of the set point for the spherical burner.
FLOW	520_0630	1.9.4 1.9.5 1.9.5.4 1.9.5.4.3	**Flame Design – Spherical Inert Flow** ACME shall provide a flow system which is capable of providing a source inert flow range of 0 to 10 slpm, on a N2 basis, within ± 5% of the set point for the spherical burner.
FLOW	520_0700	1.9.4 1.9.5 1.9.5.7 1.9.5.7.1	**s-Flame – Spherical Fuel Flow** ACME shall provide a flow system which is capable of providing a source fuel flow range of 0 to 0.5 slpm, on a N2 basis, within ± 5% of the set point for the spherical burner.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
FLOW	520_0710	1.9.4 1.9.5 1.9.5.7 1.9.5.7.3	**s-Flame – Spherical Inert Flow** ACME shall provide a flow system which is capable of providing a source inert flow range of 0 to 1.0 slpm, on a N2 basis, within ± 5% of the set point for the spherical burner.
FLOW	520_0711	1.9.4 1.9.5 1.9.5.8 1.9.5.8.1	**BRE- BRE Fuel Flow** ACME shall provide a flow system capable of providing a source fuel flow range of 0 to 2.0 slpm on a N2 basis, within ± 5% of the set point for the BRE Burner.
FLOW	520_0712	1.9.4 1.9.5 1.9.5.8 1.9.5.8.3	**BRE- BRE Inert Flow** ACME shall provide a flow system capable of providing a source inert flow range of 1.0 slpm on a N2 basis, within ± 5% of the set point for the BRE Burner.
FLOW	520_0720	1.9.6	**Gas Flow – Capacity** ACME shall provide a flow system which is capable of providing gases at the maximum flow rates (20 slpm for oxidizer and inert and 2.0 slpm for fuel) for at least 60 seconds.
FLOW	520_0730	1.9.7	**Burner Gas – Temperature** ACME shall design a flow system that causes no more than a 300±10K variance in burner gas temperature.
FLOW	520_0740	1.9.8	**Burner Gas – Dry Gases** ACME shall provide dry burner gas.
FLOW	520_0750	1.9.8	**Burner Gas – Trace Contaminants** ACME shall provide the burner gas such that the net amount of trace contaminants is less than 0.001 mole fraction.
		1.2	**Spherical Burner**
BURNER	530_0240	1.2.1	**Spherical Burner Tip – Porous Sphere** ACME shall provide each spherical burner with a porous sphere at the outlet tip.
BURNER	530_0250	1.2.1	**Spherical Burner Tip – 6.4 OD** ACME shall provide a spherical burner with a fixed outer diameter of 6.4mm for the spherical tip with a dimensional tolerance within ± 0.3mm and a measurement precision of ± 0.1mm.
BURNER	530_0260	1.2.1	**Spherical Burner Tip – 9.5 OD** ACME shall provide a spherical burner with a fixed outer diameter of 9.5mm for the spherical tip with a measured tolerance within ± 0.3mm and an accuracy of ± 0.1mm.
BURNER	530_0270	1.2.1	**Spherical Burner Tip – 12.7 OD** ACME shall provide a spherical burner with a fixed outer diameter of 12.7mm for the spherical tip with a measured tolerance within ± 0.3mm and an accuracy of ± 0.1mm.
BURNER	530_0280	1.2.2 1.2.2.3	**Spherical Burner – 6.4 OD Single 100 Supply** ACME shall provide a single supply tube of at least 100mm length from the sphere's center for the spherical burner with a 6.4 OD tip.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_0290	1.2.2 1.2.2.3	**Spherical Burner – 9.5 OD Single 100 Supply** ACME shall provide a single supply tube of at least 100mm length from the sphere's center for the spherical burner with a 9.5 OD tip.
BURNER	530_0300	1.2.2 1.2.2.3	**Spherical Burner – 12.7 OD Single 100 Supply** ACME shall provide a single supply tube of at least 100mm length from the sphere's center for the spherical burner with a 12.7 OD tip.
BURNER	530_0310	1.2.2 1.2.2.1	**Spherical Burner – 6.4 OD Differential Pressure** ACME shall provide a pressure drop, in the spherical burner supply tube, which is at most 20% of the pressure drop across the 6.4 OD burner sphere with a cold flow rate of 5 slpm.
BURNER	530_0320	1.2.2 1.2.2.1	**Spherical Burner – 9.5 OD Differential Pressure** ACME shall provide a pressure drop, in the spherical burner supply tube, which is at most 20% of the pressure drop across the 9.5 OD burner sphere with a cold flow rate of 5 slpm.
BURNER	530_0330	1.2.2 1.2.2.1	**Spherical Burner – 12.7 OD Differential Pressure** ACME shall provide a pressure drop, in the spherical burner supply tube, which is at most 20% of the pressure drop across the 12.7 OD burner sphere with a cold flow rate of 5 slpm.
BURNER	530_0340	1.2.2 1.2.2.2	**Spherical Burner – 6.4 OD Tip versus Supply OD** ACME shall provide an outer diameter, for each spherical burner supply tube, of no more than 25% of the 6.4 OD porous sphere.
BURNER	530_0350	1.2.2 1.2.2.2	**Spherical Burner – 9.5 OD Tip versus Supply OD** ACME shall provide an outer diameter, for each spherical burner supply tube, of no more than 25% of the 9.5 OD porous sphere.
BURNER	530_0360	1.2.2 1.2.2.2	**Spherical Burner – 12.7 OD Tip versus Supply OD** ACME shall provide an outer diameter, for each spherical burner supply tube, of no more than 25% of the 12.7 OD porous sphere.
BURNER	530-0370	1.2.3	**Spherical Burner – Flame Spherical Test** ACME shall produce a flame in microgravity conditions with rmin/rmax exceeding 0.8.
BURNER	530_0380	1.2.4, 1.2.4.1	**Spherical Burner – Supply Tube Axis Coincident** ACME shall position the spherical burner such that supply tube axis is coincident with the CIR chamber axis within ± 2mm along its length.
BURNER	530_0390	1.2.4, 1.2.4.1	**Spherical Burner – Supply Tube Axis Orthogonal** ACME shall position the spherical burner such that supply tube axis is **orthogonal** to the CIR chamber window axes.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_0400	1.2.4, 1.2.4.2	**Spherical Burner – Position 0** ACME shall position the spherical burner such that the sphere's center is optionally positioned within ± 2mm at the intersection of the CIR chamber window axes.
BURNER	530_0410	1.2.4, 1.2.4.2	**Spherical Burner – Position 30** ACME shall position the spherical burner such that the sphere's center is optionally positioned within ± 2mm, 30mm upstream from the intersection of the CIR window axes.
BURNER	530_0420	1.2.5	**Spherical Burner – Pressure Transducer** ACME shall position the differential pressure transducer downstream of the last solenoid valve.
BURNER	530_0430	1.2.5	**Spherical Burner – Volume** ACME shall provide an internal fluid volume of no more than 10 cc from the solenoid valve to the porous sphere, including the differential pressure transducer, for the spherical burners.
BURNER	530_0440	1.2.6	**Spherical Burner – Material** ACME shall provide spherical burners fabricated using stainless steel tubing.
BURNER	530_0450	1.2.7	**Spherical Burner – Temperature** All materials located within 75mm of the spherical burner tip center shall be capable of withstanding temperatures of at least 0 to 450 degrees Celsius.
		1.3	**Gas-Jet Burner**
BURNER	530_0460	1.3.1, 1.3.2	**Gas-Jet Burner – Dimensions** ACME shall provide each gas-jet burner consisting of a straight tube of a constant internal cross-section over a length of at least 40 times greater than the inner diameter starting from the tip.
BURNER	530_0470	1.3.3	**Gas-Jet Burner – Nozzle Plane** ACME shall provide Gas-Jet burners such that the plane defined by the nozzle tip is orthogonal to the burner tube axis ± 5 degrees.
BURNER	530_0480	1.3.4.1, 1.3.5, 1.3.6	**Gas-Jet Burner – Size A** ACME shall provide the gas-jet burner with a **0.4 I.D., 0.8 O.D** circular cross-sectional size, in mm, to a tolerance of ± 0.1mm with an accuracy of ± 0.05mm, and with the O.D. maintained up to 10mm from the tip.
BURNER	530_0490	1.3.4.2 1.3.5, 1.3.6	**Gas-Jet Burner – Size B** ACME shall provide the gas-jet burner with a **0.8 I.D., 1.2 O.D** circular cross-sectional size, in mm, to a tolerance of ± 0.1mm with an accuracy of ± 0.05mm, and with the O.D. maintained up to 10mm from the tip.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_0500	1.3.4.3 1.3.5, 1.3.6	**Gas-Jet Burner – Size C** ACME shall provide the gas-jet burner with a **1.3 I.D., 1.6 O.D** circular cross-sectional size, in mm, to a tolerance of ± 0.1mm with an accuracy of ± 0.05mm, and with the O.D. maintained up to 10mm from the tip.
BURNER	530_0510	1.3.4.4 1.3.5, 1.3.6	**Gas-Jet Burner – Size D** ACME shall provide the gas-jet burner with a **1.6 I.D., 2.0 O.D** circular cross-sectional size, in mm, to a tolerance of ± 0.1mm with an accuracy of ± 0.05mm, and with the O.D. maintained up to 10mm from the tip. .
BURNER	530_0520	1.3.4.5 1.3.5, 1.3.6	**Gas-Jet Burner – Size E** ACME shall provide the gas-jet burner with a **2.1 I.D., 2.4 O.D** circular cross-sectional size, in mm, to a tolerance of ± 0.1mm with an accuracy of ± 0.05mm, and with the O.D. maintained up to 10mm from the tip. .
BURNER	530_0530	1.3.4.6 1.3.5, 1.3.6	**Gas-Jet Burner – Size F** ACME shall provide the gas-jet burner with a **2.7 I.D., 3.1 O.D** circular cross-sectional size, in mm, to a tolerance of ± 0.1mm with an accuracy of ± 0.05mm, and with the O.D. maintained up to 10mm from the tip. .
BURNER	530_0540	1.3.4.7 1.3.5, 1.3.6	**Gas-Jet Burner – Size G** ACME shall provide the gas-jet burner with a **3.2 I.D., 3.6 O.D** circular cross-sectional size, in mm, to a tolerance of ± 0.1mm with an accuracy of ± 0.05mm, and with the O.D. maintained up to 10mm from the tip. .
BURNER	530_0550	1.3.7	**Gas-Jet Burner – Nozzle Quality** ACME shall provide an internal and external surface finish average roughness of no more than 100 μm for each Gas-Jet burner nozzle.
BURNER	530_0560	1.3.7	**Gas-Jet Burner – Tip Quality** ACME shall provide the gas-jet tip free of burrs, sharp edges, and smoothed to an average roughness of no more than 20 μm.
BURNER	530_0570	1.3.8, 1.3.8.1	**Gas-Jet Burner – Supply tube axis Coincident** ACME shall position the gas-jet burner such that supply tube axis is coincident with the CIR chamber axis within ± 2mm, along its length.
BURNER	530_0571	1.3.8, 1.3.8.1	**Gas -Jet Burner – Supply Tube Axis Orthogonal** ACME shall position the gas-jet burner such that supply tube axis is **orthogonal** to the CIR chamber window axes.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_0580	1.3.8, 1.3.8.2	**Gas-Jet Burner – Position Tip 1** ACME shall position each gas-jet burner such that the tip is within ± 2mm, 10mm upstream from the intersection of the CIR window axes.
BURNER	530_0590	1.3.8, 1.3.8.2	**Gas-Jet Burner – Position Tip 2** ACME shall position each gas-jet burner such that the tip is optionally positioned within ± 2mm, 30mm upstream from the intersection of the CIR window axes.
BURNER	530_0600	1.3.9	**Gas-Jet Burner – Material** ACME shall provide each gas-jet burner fabricated from stainless steel tubing.
BURNER	530_0610	1.3.10	**Gas-Jet Burner – Temperature** All materials located within 50mm of the Gas - Jet burner tip center shall be capable of withstanding temperatures of at least 0 to 1000 degrees Celsius.
		1.4	**Coflow Burner**
BURNER	530_0620	1.4.1	**Coflow Burner – Axisymmetric** ACME shall provide an axisymmetric coflow burner consisting of a small inner tube within a larger outer tube.
BURNER	530_0630	1.4.2	**Coflow Burner – Concentricity** ACME shall provide the coflow burner such that the inner and outer tubes are concentric within ± 0.3mm along their common length.
BURNER	530_0650	1.4.3	**Coflow Burner Outlet – Mesh** ACME shall provide a coflow burner annulus with a gas permeable outlet surface across the annulus and 3± 0.1mm upstream of the inner tube's outlet.
BURNER	530_0660	1.4.3	**Coflow Burner Outlet – Outer Tube Mesh Orthogonal** ACME shall provide coflow burners such that the annular outlet surface across the inner tube (outlet) is orthogonal to the burner axis within ± 5 degrees.
BURNER	530_0670	1.4.3	**Coflow Burner Outlet – Outer Tube Mesh Flat** ACME shall provide coflow burners such that the annular outlet surface across the inner tube (outlet) is flat to within ±0.1mm.
BURNER	530_0680	1.4.4	**Coflow Burner Outlet – Outer Tube** ACME shall provide coflow burners such that the outlet of the outer tube is 0.0 to 0.5mm downstream of the burner's outlet surface.
BURNER	530_0690	1.4.5	**Coflow Burner Inner Tube – ID & OD Dimensions** ACME shall provide coflow burners such that the circular inner tube has an inner diameter (ID) and an outer diameter (OD) within ± 0.1mm of 2.1 and 2.4mm respectively with an accuracy of ± 0.05mm.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_0700	1.4.5	**Coflow Burner Inner Tube – ID versus Length** ACME shall provide coflow burners such that the inner tube's inner diameter (ID) is constant for a length of 40 times the inner diameter.
BURNER	530_0710	1.4.6	**Coflow Burner – Inner Tube Surface Quality** ACME shall provide coflow burners such that each inner tube has an internal and external surface finish with an average roughness of no more than 100 μm.
BURNER	530_0720	1.4.6	**Coflow Burner – Tip Quality** ACME shall provide coflow burners such that the tip has a surface finish with an average roughness of no more than 20 μm.
BURNER	530_0730	1.4.7	**Coflow Burner – Inner Tube Volume** ACME shall provide coflow burner's inner tube such that the internal volume of the inner tube from the last solenoid valve to the tip, including the pressure transducer, is less than 10 cc.
BURNER	530_0740	1.4.8	**Coflow Burner – Outer Tube ID** ACME shall provide a flow profile of the coflow burners such that at the burner outlet, the outer tube has an inner diameter of 25 ± 0.5mm with an accuracy of ± 0.1mm.
BURNER	530_0750	1.4.9.a. 1.4.11 1.4.11.1	**Coflow Flow Profile – Axisymmetric – 20 cm/s** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at increments no greater than 5mm thereafter; for a **matched velocity** in the inner and outer tubes of **20 cm/s.**
BURNER	530_0760	1.4.9.a. 1.4.11 1.4.11.1	**Coflow Flow Profile – Axisymmetric – 30 cm/s** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at increments no greater than 5mm thereafter; for a **matched velocity** in the inner and outer tubes of **30** cm/s.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_0770	1.4.9.a. 1.4.11 1.4.11.1	**Coflow Flow Profile – Axisymmetric – 40 cm/s** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at increments no greater than 5mm thereafter; for a **matched velocity** in the inner and outer tubes of **40** cm/s.
BURNER	530_0780	1.4.9.a. 1.4.11 1.4.11.1	**Coflow Flow Profile – Axisymmetric – 50 cm/s** ACME shall provide a flow profile of the coflow burner, that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at increments no greater than 5mm thereafter; for a **matched velocity** in the inner and outer tubes of **50** cm/s.
BURNER	530_0790	1.4.9.a. 1.4.11 1.4.11.2	**Coflow Flow Profile – Axisymmetric 10 cm/s – 0 Degrees** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with a angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter at the **zero-degree** position orthogonal to the burner axis that is within 0.5mm downstream of the burner outlet; at increments of no greater than 1mmr within 5mm of the burner axis and at no greater than 5mm thereafter; for a **matched velocity** in the inner and outer tubes of **10** cm/s.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_0800	1.4.9.a. 1.4.11 1.4.11.2	**Coflow Flow Profile – Axisymmetric 10 cm/s – 45 Degrees** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with a angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter at the **45-degree** position orthogonal to the burner axis that is within 0.5mm downstream of the burner outlet; at increments of no greater than 1mm within 5mm of the burner axis and at no greater than 5mm thereafter; for a matched velocity in the inner and outer tubes of **10** cm/s.
BURNER	530_0810	1.4.9.a. 1.4.11 1.4.11.2	**Coflow Flow Profile – Axisymmetric 10 cm/s – 90 Degrees** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with a angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter at the **90-degree** position orthogonal to the burner axis that is within 0.5mm downstream of the burner outlet; at increments of no greater than 1mm within 5mm of the burner axis and at no greater than 5mm thereafter; for a matched velocity in the inner and outer tubes of **10** cm/s.
BURNER	530_0820	1.4.9.a. 1.4.11 1.4.11.2	**Coflow Flow Profile – Axisymmetric 10 cm/s – 135 Degrees** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with a angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter at the **135-degree** position orthogonal to the burner axis that is within 0.5mm downstream of the burner outlet; at increments of no greater than 1mm within 5mm of the burner axis and at no greater than 5mm thereafter; for a matched velocity in the inner and outer tubes of **10** cm/s.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_0830	1.4.9.a. 1.4.11 1.4.11.3	**Coflow Flow Profile – Axisymmetric Velocities Not Matched 0:10** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at no greater than 5mm thereafter; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:10** respectively.
BURNER	530_0840	1.4.9.a. 1.4.11 1.4.11.3	**Coflow Flow Profile – Axisymmetric Velocities Not Matched 0:20** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at no greater than 5mm thereafter; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:20** respectively.
BURNER	530_0850	1.4.9.a. 1.4.11 1.4.11.3	**Coflow Flow Profile – Axisymmetric Velocities Not Matched 0:30** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at no greater than 5mm thereafter; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:30** respectively.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_0860	1.4.9.a. 1.4.11 1.4.11.3	**Coflow Flow Profile – Axisymmetric Velocities Not Matched 0:40** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at no greater than 5mm thereafter; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:40** respectively.
BURNER	530_0870	1.4.9.a. 1.4.11 1.4.11.3	**Coflow Flow Profile – Axisymmetric Velocities Not Matched 0:50** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at no greater than 5mm thereafter; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:50** respectively.
BURNER	530_0880	1.4.9.a. 1.4.11 1.4.11.3	**Coflow Flow Profile – Axisymmetric Velocities Not Matched 10:30** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at no greater than 5mm thereafter; for velocities, in cm/s, **not matched** in the inner and outer tubes at **10:30** respectively.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_0890	1.4.9.a. 1.4.11 1.4.11.3	**Coflow Flow Profile – Axisymmetric Velocities Not Matched 10:50** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at no greater than 5mm thereafter; for velocities, in cm/s, **not matched** in the inner and outer tubes at **10:50** respectively.
BURNER	530_0900	1.4.9.a. 1.4.11 1.4.11.3	**Coflow Flow Profile – Axisymmetric Velocities Not Matched 30:10** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at no greater than 5mm thereafter; for velocities, in cm/s, **not matched** in the inner and outer tubes at **30:10** respectively.
BURNER	530_0910	1.4.9.a. 1.4.11 1.4.11.3	**Coflow Flow Profile – Axisymmetric Velocities Not Matched 30:50** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm; across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at no greater than 5mm thereafter; for velocities, in cm/s, **not matched** in the inner and outer tubes at **30:50** respectively.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_0920	1.4.9.a. 1.4.11 1.4.11.3	**Coflow Flow Profile – Axisymmetric Velocities Not Matched 50:10** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at no greater than 5mm thereafter; for velocities, in cm/s, **not matched** in the inner and outer tubes at **50:10** respectively.
BURNER	530_0930	1.4.9.a. 1.4.11 1.4.11.3	**Coflow Flow Profile – Axisymmetric Velocities Not Matched 50:30** ACME shall provide a flow profile of the coflow burner that have a uniform, axisymmetric flow profile at 1g using a cold flow, with no flame, such that the variation in axial velocity with angular position is less than 10% of the mean velocity for select radial positions from 0 to 10mm across one 30mm diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; at increments of no greater than 1mm within 5mm of the burner axis and at no greater than 5mm thereafter; for velocities, in cm/s, **not matched** in the inner and outer tubes at **50:30** respectively.
BURNER	530_0940	1.4.9.b. 1.4.11 1.4.11.1	**Coflow Flow Profile – Parallel Ejection, 20 cm/s** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for a **matched** velocity in the inner and outer tubes of **20** cm/s.
BURNER	530_0950	1.4.9.b. 1.4.11 1.4.11.1	**Coflow Flow Profile – Parallel Ejection, 30 cm/s** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for a **matched** velocity in the inner and outer tubes of **30** cm/s.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_0960	1.4.9.b. 1.4.11 1.4.11.1	**Coflow Flow Profile – Parallel Ejection, 40 cm/s** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for a **matched** velocity in the inner and outer tubes of **40** cm/s.
BURNER	530_0970	1.4.9.b. 1.4.11 1.4.11.1	**Coflow Flow Profile – Parallel Ejection, 50 cm/s** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for a **matched** velocity in the inner and outer tubes of **50** cm/s.
BURNER	530_0980	1.4.9.b. 1.4.11 1.4.11.2	**Coflow Flow Profile – Parallel Ejection 10 cm/s – 0 Degrees** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter, with a **zero-degree** angular position, 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for a **matched** velocity in the inner and outer tubes of **10** cm/s.
BURNER	530_0990	1.4.9.b. 1.4.11 1.4.11.2	**Coflow Flow Profile – Parallel Ejection 10 cm/s – 45 Degrees** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter, with a **45-degree** angular position, 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for a **matched** velocity in the inner and outer tubes of **10** cm/s.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1000	1.4.9.b. 1.4.11 1.4.11.2	**Coflow Flow Profile – Parallel Ejection 10 cm/s – 90 Degrees** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter, with a **90-degree** angular position, 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for a **matched** velocity in the inner and outer tubes of **10** cm/s.
BURNER	530_1010	1.4.9.b. 1.4.11 1.4.11.2	**Coflow Flow Profile – Parallel Ejection 10 cm/s – 135 Degrees** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter, with a **135-degree** angular position, 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for a **matched** velocity in the inner and outer tubes of **10** cm/s.
BURNER	530_1020	1.4.9.b. 1.4.11 1.4.11.3	**Coflow Flow Profile – Parallel Ejection 0:10** ACME shall provide a flow profile of the coflow burner, such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:10** respectively**.**
BURNER	530_1030	1.4.9.b. 1.4.11 1.4.11.3	**Coflow Flow Profile – Parallel Ejection 0:20** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:20** respectively.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1040	1.4.9.b. 1.4.11 1.4.11.3	**Coflow Flow Profile – Parallel Ejection 0:30** ACME shall provide a coflow burner with a flow profile such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:30** respectively.
BURNER	530_1050	1.4.9.b. 1.4.11 1.4.11.3	**Coflow Flow Profile – Parallel Ejection 0:40** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:40** respectively.
BURNER	530_1060	1.4.9.b. 1.4.11 1.4.11.3	**Coflow Flow Profile – Parallel Ejection 0:50** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:50** respectively.
BURNER	530_1070	1.4.9.b. 1.4.11 1.4.11.3	**Coflow Flow Profile – Parallel Ejection 10:30** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for velocities, in cm/s, **not matched** in the inner and outer tubes at **10:30** respectively.
BURNER	530_1080	1.4.9.b. 1.4.11 1.4.11.3	**Coflow Flow Profile – Parallel Ejection 10:50** ACME shall provide a coflow burner with a flow profile such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for velocities, in cm/s, **not matched** in the inner and outer tubes at **10:50** respectively.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1090	1.4.9.b. 1.4.11 1.4.11.3	**Coflow Flow Profile – Parallel Ejection 30:10** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for velocities, in cm/s, **not matched** in the inner and outer tubes at **30:10** respectively.
BURNER	530_1100	1.4.9.b. 1.4.11 1.4.11.3	**Coflow Flow Profile – Parallel Ejection 30:50** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for velocities, in cm/s, **not matched** in the inner and outer tubes at **30:50** respectively.
BURNER	530_1110	1.4.9.b. 1.4.11 1.4.11.3	**Coflow Flow Profile – Parallel Ejection 50:10** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for velocities, in cm/s, **not matched** in the inner and outer tubes at **50:10** respectively.
BURNER	530_1120	1.4.9.b. 1.4.11 1.4.11.3	**Coflow Flow Profile – Parallel Ejection 50:30** ACME shall provide a flow profile of the coflow burner such that the burner gas is ejected parallel to the chamber axis, at 1g using a cold flow – with no flame, without either swirl – helically rotating flow, or turbulence – eddies, where the axial velocity at any position across one diameter 0.5mm downstream of the inner tube outlet is constant with time within 3% of the time-averaged value; for velocities, in cm/s, **not matched** in the inner and outer tubes at **50:30** respectively.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1130	1.4.10, 1.4.10.1 1.4.11 1.4.11.1	**Coflow Flow Profile – Inner Boundary Layer – 20 cm/s** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an inner tube boundary layer thickness of less than 0.75mm outside the inner tube wall, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for a **matched** velocity in the inner and outer tubes of **20 cm/**s.
BURNER	530_1140	1.4.10, 1.4.10.1 1.4.11 1.4.11.1	**Coflow Flow Profile – Inner Boundary Layer – 30 cm/s** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm outside the inner tube wall, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for a matched velocity in the inner and outer tubes of **30 cm/s**.
BURNER	530_1150	1.4.10, 1.4.10.1 1.4.11 1.4.11.1	**Coflow Flow Profile – Inner Boundary Layer – 40 cm/s** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm outside the inner tube wall, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for a matched velocity in the inner and outer tubes of **40 cm/s.**
BURNER	530_1160	1.4.10, 1.4.10.1 1.4.11 1.4.11.1	**Coflow Flow Profile – Inner Boundary Layer – 50 cm/s** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm outside the inner tube wall, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for a matched velocity in the inner and outer tubes of **50** cm/s.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1170	1.4.10, 1.4.10.1 1.4.11 1.4.11.2	**Coflow Flow Profile – Inner Boundary Layer 10 cm/s – 0 Degrees** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm; outside the inner tube, into the annulus, to the position where the axial velocity reaches 90 % of the plug flow velocity; across one diameter orthogonal to the burner axis, with a **zero-degree** angular position, that is within 0.5-mm downstream of the inner tube outlet; for a matched velocity in the inner and outer tubes of **10 cm/s.**
BURNER	530_1180	1.4.10, 1.4.10.1 1.4.11 1.4.11.2	**Coflow Flow Profile – Inner Boundary Layer 10 cm/s – 45 Degrees** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm; outside the inner tube, into the annulus, to the position where the axial velocity reaches 90 % of the plug flow velocity; across one diameter orthogonal to the burner axis, with a **45-degree** angular position, that is within 0.5-mm downstream of the inner tube outlet; for a **matched velocity** in the inner and outer tubes of **10** cm/s
BURNER	530_1190	1.4.10, 1.4.10.1 1.4.11 1.4.11.2	**Coflow Flow Profile – Inner Boundary Layer 10 cm/s – 90 Degrees** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis, with a **90-degree** angular position, that is within 0.5mm downstream of the inner tube outlet; for a **matched velocity** in the inner and outer tubes of **10 cm/s.**
BURNER	530_1200	1.4.10, 1.4.10.1 1.4.11 1.4.11.2	**Coflow Flow Profile – Inner Boundary Layer 10 cm/s – 135 Degrees** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm; outside the inner tube, into the annulus, to the position where the axial velocity reaches 90 % of the plug flow velocity; across one diameter orthogonal to the burner axis, with a **135-degree** angular position, that is within 0.5-mm downstream of the inner tube outlet; for a matched velocity in the inner and outer tubes of **10** cm/s.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1210	1.4.10, 1.4.10.1 1.4.11 1.4.11.3	**Coflow Flow Profile – Inner Boundary Layer 0:10** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm; outside the inner tube, into the annulus, to the position where the axial velocity reaches 90 % of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5-mm downstream of the inner tube outlet; for velocities, in cm/s, not matched in the inner and outer tubes at **0:10** respectively.
BURNER	530_1220	1.4.10, 1.4.10.1 1.4.11 1.4.11.3	**Coflow Flow Profile – Inner Boundary Layer 0:20** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, not matched in the inner and outer tubes at **0:20** respectively.
BURNER	530_1230	1.4.10, 1.4.10.1 1.4.11 1.4.11.3	**Coflow Flow Profile – Inner Boundary Layer 0:30** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, not matched in the inner and outer tubes at **0:30** respectively.
BURNER	530_1240	1.4.10, 1.4.10.1 1.4.11 1.4.11.3	**Coflow Flow Profile – Inner Boundary Layer 0:40** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, not matched in the inner and outer tubes at **0:40** respectively.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1250	1.4.10, 1.4.10.1 1.4.11 1.4.11.3	**Coflow Flow Profile – Inner Boundary Layer 0:50** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, not matched in the inner and outer tubes at **0:50** respectively.
BURNER	530_1260	1.4.10, 1.4.10.1 1.4.11 1.4.11.3	**Coflow Flow Profile – Inner Boundary Layer 10:30** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, not matched in the inner and outer tubes at **10:30** respectively.
BURNER	530_1270	1.4.10, 1.4.10.1 1.4.11 1.4.11.3	**Coflow Flow Profile – Inner Boundary Layer 10:50** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, not matched in the inner and outer tubes at **10:50** respectively.
BURNER	530_1280	1.4.10, 1.4.10.1 1.4.11 1.4.11.3	**Coflow Flow Profile – Inner Boundary Layer 30:10** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, not matched in the inner and outer tubes at **30:10** respectively.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1290	1.4.10, 1.4.10.1 1.4.11 1.4.11.3	**Coflow Flow Profile – Inner Boundary Layer 30:50** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, not matched in the inner and outer tubes at **30:50** respectively.
BURNER	530_1300	1.4.10, 1.4.10.1 1.4.11 1.4.11.3	**Coflow Flow Profile – Inner Boundary Layer 50:10** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, not matched in the inner and outer tubes at **50:10** respectively.
BURNER	530_1310	1.4.10, 1.4.10.1 1.4.11 1.4.11.3	**Coflow Flow Profile – Inner Boundary Layer 50:30** ACME shall provide a flow profile of the coflow burner at 1g using a cold flow, with no flame, that has an **inner tube boundary layer** thickness of less than 0.75mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, **not matched** in the inner and outer tubes at **50:30** respectively.
BURNER	530_1320	1.4.10, 1.4.10.2 1.4.11 1.4.11.1	**Coflow Flow Profile – Outer Boundary Layer – 20 cm/s** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than 3.0mm outside the inner tube wall, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for a **matched** velocity in the inner and outer tubes of **20** cm/s.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1330	1.4.10, 1.4.10.2 1.4.11 1.4.11.1	**Coflow Flow Profile – Outer Boundary Layer – 30 cm/s** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an outer **tube boundary layer** thickness of less than **3.0mm** outside the inner tube wall, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for a matched velocity in the inner and outer tubes of **30** cm/s.
BURNER	530_1340	1.4.10, 1.4.10.2 1.4.11 1.4.11.1	**Coflow Flow Profile – Outer Boundary Layer – 40 cm/s** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than **3.0mm** outside the inner tube wall, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for a matched velocity in the inner and outer tubes of **40** cm/s.
BURNER	530_1350	1.4.10, 1.4.10.2 1.4.11 1.4.11.1	**Coflow Flow Profile – Outer Boundary Layer – 50 cm/s** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than **3.0mm** outside the inner tube wall, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for a matched velocity in the inner and outer tubes of **50** cm/s.
BURNER	530_1360	1.4.10, 1.4.10.2 1.4.11 1.4.11.2	**Coflow Flow Profile – Outer Boundary Layer 10 cm/s – 0 Degrees** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an outer **tube boundary layer** thickness of less than **3.0mm** outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis, with a **zero-degree** angular position, that is within 0.5mm downstream of the inner tube outlet; for a matched velocity in the inner and outer tubes of **10 cm/s**.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1370	1.4.10, 1.4.10.2 1.4.11 1.4.11.2	**Coflow Flow Profile – Outer Boundary Layer 10 cm/s – 45 Degrees** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an outer **tube boundary layer** thickness of less than **3.0mm** outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis, with a **45-degree** angular position, that is within 0.5mm downstream of the inner tube outlet; for a matched velocity in the inner and outer tubes of **10 cm/s**.
BURNER	530_1380	1.4.10, 1.4.10.2 1.4.11 1.4.11.2	**Coflow Flow Profile – Outer Boundary Layer 10 cm/s – 90 Degrees** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an outer **tube boundary layer** thickness of less than **3.0mm** outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis, with a **90-degre**e angular position, that is within 0.5mm downstream of the inner tube outlet; for a matched velocity in the inner and outer tubes of **10 cm/s**.
BURNER	530_1390	1.4.10, 1.4.10.2 1.4.11 1.4.11.2	**Coflow Flow Profile – Outer Boundary Layer 10 cm/s – 135 Degrees** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than **3.0mm** outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis, with a **135-degree** angular position, that is within 0.5mm downstream of the inner tube outlet; for a **matched** velocity in the inner and outer tubes of **10** cm/s.
BURNER	530_1400	1.4.10, 1.4.10.2 1.4.11 1.4.11.3	**Coflow Flow Profile – Outer Boundary Layer 0:10** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than 3.0mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:10** respectively.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1410	1.4.10, 1.4.10.2 1.4.11 1.4.11.3	**Coflow Flow Profile – Outer Boundary Layer 0:20** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than 3.0mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:20** respectively.
BURNER	530_1420	1.4.10, 1.4.10.2 1.4.11 1.4.11.3	**Coflow Flow Profile – Outer Boundary Layer 0:30** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than 3.0mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:30** respectively.
BURNER	530_1430	1.4.10, 1.4.10.2 1.4.11 1.4.11.3	**Coflow Flow Profile – Outer Boundary Layer 0:40** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than 3.0mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:40** respectively.
BURNER	530_1440	1.4.10, 1.4.10.2 1.4.11 1.4.11.3	**Coflow Flow Profile – Outer Boundary Layer 0:50** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than 3.0mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, **not matched** in the inner and outer tubes at **0:50** respectively.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1450	1.4.10, 1.4.10.2 1.4.11 1.4.11.3	**Coflow Flow Profile – Outer Boundary Layer 10:30** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than 3.0mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, **not matched** in the inner and outer tubes at **10:30** respectively.
BURNER	530_1460	1.4.10, 1.4.10.2 1.4.11 1.4.11.3	**Coflow Flow Profile – Outer Boundary Layer 10:50** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than 3.0mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, **not matched** in the inner and outer tubes at **10:50** respectively.
BURNER	530_1470	1.4.10, 1.4.10.2 1.4.11 1.4.11.3	**Coflow Flow Profile – Outer Boundary Layer 30:10** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than 3.0mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, **not matched** in the inner and outer tubes at **30:10** respectively.
BURNER	530_1480	1.4.10, 1.4.10.2 1.4.11 1.4.11.3	**Coflow Flow Profile – Outer Boundary Layer 30:50** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than 3.0mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, **not matched** in the inner and outer tubes at **30:50** respectively.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1490	1.4.10, 1.4.10.2 1.4.11 1.4.11.3	**Coflow Flow Profile – Outer Boundary Layer 50:10** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than 3.0mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, not matched in the inner and outer tubes at **50:10** respectively.
BURNER	530_1500	1.4.10, 1.4.10.2 1.4.11 1.4.11.3	**Coflow Flow Profile – Outer Boundary Layer 50:30** ACME shall provide a coflow burner at 1g using a cold flow, with no flame, that has an **outer tube boundary layer** thickness of less than 3.0mm outside the inner tube, into the annulus, to the position where the axial velocity reaches 90% of the plug flow velocity; across one diameter orthogonal to the burner axis that is within 0.5mm downstream of the inner tube outlet; for velocities, in cm/s, not matched in the inner and outer tubes at **50:30** respectively.
BURNER	530_1510	1.4.12.a	**Coflow Flame Profile – Edge Axisymmetric – 10 cm/s** ACME shall provide an axisymmetric, 50/50, CH4/N2 flame profile of the coflow burner at 1g such that the variation in radius of the visible **flame edge** with angular position is less than 10% of the mean radius for all radial positions from the flame base to tip; for matched velocities in the inner and outer tubes of **10 cm/s**.
BURNER	530_1520	1.4.12.a	**Coflow Flame Profile – Edge Axisymmetric – 20 cm/s** ACME shall provide an axisymmetric, 50/50, CH4/N2 flame profile of the coflow burner at 1g, such that the variation in radius of the visible **flame edge** with angular position is less than 10% of the mean radius for all radial positions from the flame base to tip; for matched velocities in the inner and outer tubes of **20 cm/s**.
BURNER	530_1530	1.4.12.a	**Coflow Flame Profile – Edge Axisymmetric – 30 cm/s** ACME shall provide an axisymmetric, 50/50, CH4/N2 flame profile of the coflow burner at 1g, such that the variation in radius of the visible **flame edge** with angular position is less than 10% of the mean radius for all radial positions from the flame base to tip; for matched velocities in the inner and outer tubes of **30 cm/s**.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1540	1.4.12.a	**Coflow Flame Profile – Edge Axisymmetric – 40 cm/s** ACME shall provide an axisymmetric, 50/50, CH4/N2 flame profile of the coflow burner at 1g, such that the variation in radius of the visible **flame edge** with angular position is less than 10% of the mean radius for all radial positions from the flame base to tip; for matched velocities in the inner and outer tubes of **40 cm/s**.
BURNER	530_1550	1.4.12.a	**Coflow Flame Profile – Edge Axisymmetric – 50 cm/s** ACME shall provide an axisymmetric, 50/50, CH4/N2 flame profile of the coflow burner at 1g, such that the variation in radius of the visible **flame edge** with angular position is less than 10% of the mean radius for all radial positions from the flame base to tip; for matched velocities in the inner and outer tubes of **50 cm/s**.
BURNER	530_1560	1.4.12.b	**Coflow Flame Profile – Base Axisymmetric – 10 cm/s** ACME shall provide an axisymmetric, 50/50, CH4/N2 flame profile of the coflow burner at 1g, such that the plane of the **flame's base** is orthogonal to the burner axis where the variation in axial distance from the visible flame base to burner outlet, along the circumference of the base, is no more than 1mm for matched velocities in the inner and outer tubes of **10 cm/s**.
BURNER	530_1570	1.4.12.b	**Coflow Flame Profile – Base Axisymmetric – 20 cm/s** ACME shall provide an axisymmetric, 50/50, CH4/N2 flame profile of the coflow burner at 1g, such that the plane of the **flame's base** is orthogonal to the burner axis where the variation in axial distance from the visible flame base to burner outlet, along the circumference of the base, is no more than 1mm for matched velocities in the inner and outer tubes of **20 cm/s**.
BURNER	530_1580	1.4.12.b	**Coflow Flame Profile – Base Axisymmetric – 30 cm/s** ACME shall provide an axisymmetric, 50/50, CH4/N2 flame profile of the coflow burner at 1g, with a temperature sensor installed, such that the plane of the **flame's base** is orthogonal to the burner axis where the variation in axial distance from the visible flame base to burner outlet, along the circumference of the base, is no more than 1mm for matched velocities in the inner and outer tubes of **30 cm/s.**

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1590	1.4.12.b	**Coflow Flame Profile – Base Axisymmetric – 40 cm/s** ACME shall provide an axisymmetric, 50/50, CH4/N2 flame profile of the coflow burner at 1g, such that the plane of the **flame's base** is orthogonal to the burner axis where the variation in axial distance from the visible flame base to burner outlet, along the circumference of the base, is no more than 1mm for matched velocities in the inner and outer tubes of **40 cm/s.**
BURNER	530_1600	1.4.12.b	**Coflow Flame Profile – Base Axisymmetric – 50 cm/s** ACME shall provide an axisymmetric, 50/50, CH4/N2 flame profile of the coflow burner at 1g, with a temperature sensor installed, such that the plane of the **flame's base** is orthogonal to the burner axis where the variation in axial distance from the visible flame base to burner outlet, along the circumference of the base, is no more than 1mm for matched velocities in the inner and outer tubes of **50 cm/s.**
BURNER	530_1610	1.4.12	**Coflow Burner Flame – Axisymmetric Drop Test Matched Velocity 10:50** ACME shall provide coflow burners such that the axisymmetric of a 50/50 CH4/N2 flame shall be demonstrated in drop tests for a limited set of the following conditions: a range of flow velocities in 10 cm/s increments from 10 to 50 cm/s where the inner and outer tube velocities are matched.
BURNER	530_1620	1.4.13, 1.4.13.1	**Coflow Burner Coincident** ACME shall position co-flow burners such that the supply tube axis is coincident with the CIR chamber axis within ± 2mm, along its length, and orthogonal to the CIR chamber window axes.
BURNER	530_1630	1.4.13, 1.4.13.2	**Coflow Burner Position** ACME shall position the coflow burners such that the tip is within ± 2mm, 12mm upstream from the intersection of the CIR window axes.
BURNER	530_1640	1.4.14	**Coflow Burner – Inner Tube Material** ACME shall provide the coflow burner inner tube fabricated from stainless steel tubing.
BURNER	530_1650	1.4.15	**Coflow Burner Temperature** ACME shall provide the coflow burner capable of withstanding temperatures of 0-1000 degrees Celsius within a 50mm radius of the burner outlet center.
		1.5	**Burner as Electrode**
BURNER	530_1660	1.5.1	**Gas-Jet Burner as Electrode** ACME shall provide each gas-jet burner with a resistance of less than 1 Ohm from the tip to the insulator and across its entire outlet surface.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1670	1.5.1	**Coflow Burner as Electrode** ACME shall provide each coflow burner with a resistance of less than 1 Ohm from the tip to the insulator and across its entire outlet surface.
BURNER	530_1680	1.5.2 1.7.5	**Gas-Jet Burner – Electrically Insulated** The ACME gas-jet burners shall have a resistance of greater than 10 mega Ohms from the burner tip to the burner support with the exception of the return path and the thermocouple.
BURNER	530_1690	1.5.2 1.7.5	**Coflow Burner – Electrically Insulated** ACME shall provide coflow burners that are electrically insulated from any mechanism that supports it within the chamber with the exception of wiring associated with the burner temperature measurement.
		1.1	**BRE Burner (BRE)**
BURNER	530_1700	1.13.1	**BRE - Outlet Plate** ACME shall provide each BRE burner with a circular gas-permeable outlet plate.
BURNER	530_1710	1.13.1	**BRE - 25mm Burner** ACME shall provide a circular BRE burner with a diameter of 25mm ± 1.0mm and is known within ± 0.1mm.
BURNER	530_1860	1.13.1	**BRE - 25mm Burner Inner wall** ACME shall provide a 25mm BRE burner with a housing inner wall diameter of 25mm ± 1mm.
BURNER	530_1870	1.13.1	**BRE - 50mm Burner** ACME shall provide a circular BRE burner with a diameter of 50mm ± 1.0mm and is known within ± 0.1mm.
BURNER	530_1880	1.13.1	**BRE - 50mm Burner Inner wall** ACME shall provide a 50mm BRE burner with a housing inner wall diameter of 50mm ± 1mm.
BURNER	530_1890	1.13.2	**BRE - Outlet Plate Bevel** ACME shall provide each BRE burner with an outlet plate that expands in a bevel from the outlet surface to the outer edge of the burner's cylindrical wall.
BURNER	530_1720	1.13.3	**BRE - Outlet Plate Orthogonally** ACME shall provide each BRE burner with an outlet plate orthogonal to the burner axis ± 0.1mm.
BURNER	530_1900	1.13.3	**BRE - Outlet Plate Flatness** ACME shall provide each BRE burner with an outlet plate flat to within ± 0.1mm.
BURNER	530_1730	1.13.4 3.4.1 3.10	**BRE - Outlet Plate Thickness** ACME shall provide each BRE burner with an outlet plate that is 5.3 ± 0.1mm thick.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1910	1.13.4	**BRE - Outlet Holes** ACME shall provide a BRE burner with an outlet plate with at least 60 holes across the entire outlet surface, with uniform diameters of no more than 1/15th of the burner diameter.
BURNER	530_1920	1.13.4	**BRE- Outlet Hole Rings** ACME shall provide a BRE burner with holes in the outlet plate in a uniform circular arrangement of concentric rings with a constant ring-to-ring spacing and a constant hole-to-hole spacing where the ratio of those two spacing's shall be between 0.5 and 2.
BURNER	530_1930	1.13.4	**BRE Outlet Hole Size** ACME shall provide a BRE burner with holes in the outlet surface where the open area fraction of the holes to the outlet surface is between 0.2 and 0.4.
BURNER	530_1940	1.13.4	**BRE Inner Hole Pattern Distance** ACME shall provide a BRE burner with holes in the outlet where the distance between the burner center and the innermost ring of holes shall be equal to the ring-to-ring spacing.
BURNER	530_1950	1.13.4	**BRE Outer Hole Pattern Distance** ACME shall provide a BRE burner with holes in the outlet where the distance between the outermost ring of holes and the edge of the outlet plate's surface shall be less than the ring-to-ring spacing and one half that distance.
BURNER	530_1960	1.13.4	**BRE Center Heat Flux Sensor Hole** ACME shall provide a BRE burner with a hole sized such that gas can flow through an annulus around the center heat flux sensor, where the cross-sectional area of that annulus equals the cross-sectional area of other holes without sensors.
BURNER	530_1970	1.13.4	**BRE Offset Heat Flux Sensor Hole** ACME shall provide a BRE burner with a hole sized such that if the offset heat flux sensor position coincides with an outlet hole the gas can flow through an annulus around the sensor, where the cross-sectional area of that annulus equals the cross-sectional area of the other holes without sensors.
BURNER	530_1740	1.13.5	**BRE - Outlet Plate Emissivity** ACME shall provide a BRE burner with an outlet plate that has an emissivity of 0.75 for 0.6-15 µm.
BURNER	530_1750	1.13.6	**BRE -Burner Wall Thickness** ACME shall provide each BRE burner with a cylindrical wall that is no more than 1mm thick within 25mm of the outlet plate.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
BURNER	530_1760	1.13.7	**BRE-Honeycomb Cylinder** ACME shall provide each BRE burner with a ceramic honeycomb cylinder that is 1 to 2mm upstream of the outlet plate which has open cells that are parallel with the burner axis and fully span the burner cross section.
BURNER	530_1980	1.13.7	**BRE- Honeycomb Cell Size** ACME shall provide each BRE burner with a honeycomb cylinder that has a cross sectional area of each cell that is no more than 2% of the surface area of a hemisphere with a radius equal to the cylinder's.
BURNER	530_1990	1.13.8	**BRE - Housing Insulation** ACME shall provide a BRE burner that has a thermally insulated outer cylindrical wall from the outlet plate to at least 50mm upstream, with a 6 ± 2mm of insulation with a thermal conductivity of no more than 0.06 W/mK at 260 degreesC (550 degrees F)
BURNER	530_1800	1.13.12 1.13.12.1	**BRE - Coincidence** ACME shall position each BRE burner such that its axis is coincident with the CIR chamber axis ± 2mm.
BURNER	530_1810	1.13.12 1.13.12.1	**BRE- Orthogonal** ACME shall position each BRE burner orthogonal to the CIR chamber window axes.
BURNER	530_1820	1.13.12 1.13.12.2	**BRE - Position** ACME shall position each BRE burner such that the plane of the CIR chamber window centers are 10 ± 2mm downstream of the inner tube outlet.
BURNER	530_1830	1.13.13	**BRE - Burner Outlet Material** ACME shall provide BRE burners with an outlet plate fabricated from copper.
BURNER	530_2000	1.13.13	**BRE - Burner Wall Material** ACME shall provide BRE burners with a cylindrical wall fabricated from stainless.
BURNER	530_1840	1.13.14	**BRE - Burner Temperature** ACME shall provide BRE burners that are capable of withstanding temperatures of 0-400 degrees C within 10mm of the burner outlet with the exception of the heat flux sensors.
		1.6	**Electrode Mesh**
E-FIELD	580_0010	1.6.1	**Electrode Mesh – Material Dimensions** ACME shall provide an electrode mesh that is composed of copper mesh with 16 to 22 wires per 25.4mm (1 inch) with a wire diameter of 0.38 to 0.64mm (0.015 to 0.025 inches).
E-FIELD	580_0020	1.6.2	**Electrode Mesh – Attached Electrically** ACME shall provide the mesh electrode electrically attached to a ring.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
E-FIELD	580_0030	1.6.2	**Electrode Mesh – Structural Description** ACME shall provide the mesh electrode copper ring with a 2 to 3mm cross-sectional diameter with an accuracy of ± 0.2mm.
E-FIELD	580_0040	1.6.2	**Electrode Mesh – Material Quality** ACME shall provide the mesh electrode free of electrical discharge at the maximum system potential.
E-FIELD	580_0050	1.6.3	**Electrode Mesh – OD** ACME shall provide the mesh electrode ring forming the edge with a maximized outer diameter of at least 80 ± 1mm.
E-FIELD	580_0060	1.6.3	**Electrode Mesh – Burner Distance** ACME shall provide the mesh electrode such that the axial distance from the burner tip to the electrode mesh is shorter than the distance between the electrode mesh and any other conductive surface.
E-FIELD	580_0070	1.6.4	**Electrode Mesh – Axis Centered** ACME shall provide the electrode mesh positioned so that its center is on the CIR chamber axis within ± 2mm.
E-FIELD	580_0080	1.6.4	**Electrode Mesh – Orthogonal Plane** ACME shall provide the electrode mesh positioned so that its center is orthogonal to the CIR chamber axis within ± 3 degrees.
E-FIELD	580_0090	1.6.5	**Electrode Mesh – CIA Orientation Default** ACME shall provide the electrode mesh positioned within ± 1mm at a default 25mm downstream of the chamber window axes.
E-FIELD	580_0100	1.6.5	**Electrode Mesh – CIA Orientation 2** ACME shall provide the electrode mesh positioned within ± 1mm 40mm downstream of the chamber window axes.
E-FIELD	580_0110	1.6.6	**Electrode Mesh – ORU** ACME shall provide the electrode mesh as an ORU and removable from the ACME Chamber Insert Assembly (CIA) during spaceflight.
		1.7	**Electric Field**
E-FIELD	580_0120	1.7.1 2.5.1 2.5.4	**Electric Field – DC Range** ACME shall provide an electric field generator capable of an output of ± 10 kV DC at no load with an equivalent DC output impedance of 50 Mega-ohms.
E-FIELD	580_0130	1.7.1	**Electric Field – DC Range Accuracy** ACME shall provide an electric field generator such that the voltage monitoring signal of the output is within an accuracy of ± 0.12% with a temperature coefficient of± 200 ppm per degree Celsius.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
E-FIELD	580_0140	1.7.2	**Electric Field – Ramping Stepped Voltage 100V** ACME shall provide an electric field with the capability to step the voltage potential of up to 100V.
E-FIELD	580_0141	1.7.2	**Electric Field – Ramping Stepped Voltage 10kV** ACME shall provide an electric field with the capability to step the voltage potential of up to10 kV.
E-FIELD	580_0150	1.7.3	**Electric Field – Frequency** ACME shall provide an electric field with the capability to select the stepped voltage frequency between 0-1 Hertz for the specified voltage range and accuracy.
E-FIELD	580_0151	1.7.3	**Electric Field- Step Magnitude** ACME shall provide an electric field with step magnitudes of no more than 10 kV.
E-FIELD	580_0320	1.7.6	**Electric Field – Hardware Zone of Exclusion (ZoE)** ACME shall provide electric field with a cylindrical zone of exclusion, kept free of any electrically conductive objects during operation of the electric field, which has a 30-mm greater diameter than the electrode mesh and extends from the burner to 20mm downstream of the mesh.
E-FIELD	580_0330	1.7.7	**Electric Field – Default Current Leakage** ACME shall provide an electric field such that the current leakage between the electrode mesh and the burner, measured in air at 1 bar absolute, is less than 100 nanoamps; with operational accuracy of ± 1 kV at both -10 kV and +10 kV and with the 35mm default spacing, accurate to ± 2mm.
E-FIELD	580_0340	1.7.8	**Electric Field – ORU Power Supply 1** ACME shall provide the high voltage power supply 1 as an ORU and removable from the ACME CIA during spaceflight.
E-FIELD	580_0350	1.7.8	**Electric Field – ORU Power Supply 2** ACME shall provide the high voltage power supply 2 as an ORU and removable from the ACME CIA during spaceflight.
E-FIELD	580_0360	1.7.8	**Electric Field – ORU HV Supply I/F Circuitry** ACME shall provide the high voltage supply interface circuitry as an ORU and removable from the ACME CIA during spaceflight.
E-FIELD	580_0370	1.7.8	**Electric Field – ORU HV Passive Network** ACME shall provide the high voltage passive network as an ORU and removable from the ACME CIA during spaceflight.
		1.1	**Ignition**

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
IGN	540_0010	1.10.1	**Ignition Reliability – Spherical 6.4 OD, Position 0** ACME shall provide an igniter that ignites all initial flow conditions in 1g at the limiting conditions in the test matrices within 1 second of igniter activation for the spherical burner with a 6.4mm O.D. tip at position zero, the intersection of the CIR chamber and window axes ± 2mm high and low.
IGN	540_0020	1.10.1	**Ignition Reliability – Spherical 6.4 OD, Position 30** ACME shall provide an igniter that ignites all initial flow conditions in 1g at the limiting conditions in the test matrices within 1 second of igniter activation for the spherical burner with a 6.4mm O.D. tip at position 30, 30mm ± 2mm upstream of the CIR window axes.
IGN	540_0030	1.10.1	**Ignition Reliability – Spherical 9.5 OD, Position 0** ACME shall provide an igniter that ignites all initial flow conditions in 1g at the limiting conditions in the test matrices within 1 second of igniter activation for the spherical burner with a 9.5mm O.D. tip at position zero, the intersection of the CIR chamber and window axes ± 2mm.
IGN	540_0040	1.10.1	**Ignition Reliability – Spherical 9.5 OD, Position 30** ACME shall provide an igniter that ignites all initial flow conditions in 1g at the limiting conditions in the test matrices within 1 second of igniter activation for the spherical burner with a 9.5mm O.D. tip at position 30, 30mm ± 2mm upstream of the CIR window axes.
IGN	540_0050	1.10.1	**Ignition Reliability – Spherical 12.7 OD, Position 0** ACME shall provide an igniter that ignites all initial flow conditions in 1g at the limiting conditions in the test matrices within 1 second of igniter activation for the spherical burner with a 12.7mm O.D. tip at position zero, the intersection of the CIR chamber and window axes ± 2mm
IGN	540_0060	1.10.1	**Ignition Reliability – Spherical 12.7 OD, Position 30** ACME shall provide an igniter that ignites all initial flow conditions in 1g at the limiting conditions in the test matrices within 1 second of igniter activation for the spherical burner with a 12.7mm O.D. tip at position 30, 30mm ± 2mm upstream of the CIR window axes.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
IGN	540_0070	1.10.1	**Ignition Reliability – Gas-Jet, Position 10** ACME shall provide an igniter that ignites all initial flow conditions in 1g at the limiting conditions in the test matrices within 1 second of igniter activation for the gas-jet burner at position 10, 10mm ± 2mm upstream from the intersection of the CIR chamber window axes.
IGN	540_0080	1.10.1	**Ignition Reliability – Gas-Jet, Position 30** ACME shall provide an igniter that ignites all initial flow conditions in 1g at the limiting conditions in the test matrices within 1 second of igniter activation for the gas-jet burner at position 30, 30mm ± 2mm upstream from the intersection of the CIR chamber window axes.
IGN	540_0090	1.10.1	**Ignition Reliability – Coflow, Position 0** ACME shall provide an igniter that ignites all initial flow conditions in 1g at the limiting conditions in the test matrices within 1 second of igniter activation for the coflow burner at position zero, the intersection of the CIR chamber and window axes ± 2mm.
IGN	540_0100	1.10.1	**Ignition Reliability – Coflow, Position 12** ACME shall provide an igniter that ignites all initial flow conditions in 1g at the limiting conditions in the test matrices within 1 second of igniter activation for the coflow burner at position 12, 12mm ± 2mm upstream from the intersection of the CIR chamber window axes.
IGN	540_0101	1.10.1	**Ignition Reliability - BRE** ACME shall provide an igniter that ignites all initial flow conditions within 1 second of igniter activation for the BRE burner.
		1.12	**Data Synchronization and Recording**
CDS	620_0010	1.12.2	**Data Recording – Storage for Non-Imaging Duration Default** ACME AVP CSCI shall provide data recording storage capacity for 12 mins of simultaneous non-imaging data, such that all non-imaging measurements begin at an uplinked time that is at most 5 minutes prior to ignition and continues to an uplinked time at most 5 minutes after extinction detection unless otherwise noted.
CDS	620_0020	1.12.3	**Data Recording – Storage for Imaging Duration Default** ACME AVP CSCI shall provide up to 4 mins of simultaneous image data recording storage capacity such that all imaging measurements begin at an uplinked time that is at most 1 minute prior to ignition and continues to an uplinked time at most 1 minute after extinction detection.
		2.1	**Chamber Pressure**

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
MEAS	570_0010	2.1.1	**Chamber Pressure – Accuracy** ACME shall ensure the absolute chamber pressure is measured with an accuracy of ± 1% of reading.
MEAS	570_0020	2.1.2	**Chamber Pressure – Range** ACME shall ensure the pressure measurement range is 0.1 to 4 bar absolute.
MEAS	570_0030	2.1.3	**Chamber Pressure – Sampling Frequency** ACME shall ensure the chamber pressure is measured at 1 sample per second or faster with a response time of 0.5 second or less.
		2.3	**Gas Flow Rate**
FLOW	520_0760	2.3.1	**Gas Flow Rate – Accuracy** ACME shall provide each gas flow rate measurement with an accuracy of ± 5% of the reading or better.
		2.4	**Burner Pressure Differential**
MEAS	570_0060	2.4.1	**Burner Pressure Differential Accuracy** ACME shall provide the pressure differential between the BRE and spherical burner supply tube inlet and the chamber with accuracy of ± 2 % of the reading.
MEAS	570_0070	2.4.1	**Burner Pressure Differential – Orientation** ACME shall provide the pressure differential for the BRE and spherical burner measured downstream of any valves.
MEAS	570_0080	2.4.2	**Burner Pressure Differential – Range** ACME shall provide a pressure differential measurement range for the spherical burner that is from 0.1 to 1.5 times the pressure drop across the spherical burner at a flow rate of 5 slpm, on a N2 basis, in cold flow testing (i.e., with no flame).
		2.5	**Electric Potential**
E-FIELD	580_0380	2.5.1, 2.5.4	**Electric Potential – Accuracy** ACME shall provide electric potential measurement between the mesh electrode and the high voltage (HV) ground terminal to an accuracy of within the larger of 20 VDC or ± 1% of the reading.
E-FIELD	580_0390	2.5.2	**Electric Potential – Range** ACME shall provide an electric potential measurement range of ± 10 kilovolts.
		3.1	**ACME Analog Camera**
OpsCam	550_0010	3.1.1	**Analog Camera – Wavelength Response** ACME shall provide a wavelength response of 400 to 700 nanometers.
OpsCam	550_0020	3.1.2	**Analog Camera – Focusing** ACME shall focus the analog camera on the CIR chamber axis where the Field of View (FOV) plane includes the CIR chamber axis.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
OpsCam	550_0030	3.1.3	**Analog Camera – FOV** ACME shall provide an analog camera FOV of at least a 90mm diameter circle centered on the intersection of the CIR window axes.
OpsCam	550_0040	3.1.4	**Analog Camera – Adjustable Settings** ACME shall provide the analog camera with the ability to adjust at least the gain (or iris) setting.
OpsCam	550_0070	3.1.5	**Analog Camera – Frame Rate** ACME shall provide the analog camera with a minimum of 25 frames per second.
OpsCam	550_0080	3.1.6	**Analog Camera – Near-Real Time Downlink** ACME shall provide the analog camera with near real-time downlink.
OpsCam	550_0110	3.1.7	**Analog Camera– ORU** ACME shall provide the analog camera system as an ORU.
OpsCam	550_0120	3.1.8	**Analog Camera – Illumination** ACME shall provide analog camera system with a white light source for illuminating the imaging views before, during, or after tests.
		3.2	**ACME Data Camera**
DataCam	310_0010	3.2.1	**Data Camera – Spectral Range** ACME shall provide a data camera with a wavelength response of 400 to 700 nanometers.
DataCam	310_0020	3.2.2	**Data Camera – Sensitivity** ACME shall provide a data camera with a sensitivity, especially to dim blue light at 431 nanometers, matching or exceeding cameras equipped with a Sony ICX285 2/3” Progressive scan CCD.
DataCam	310_0030	3.2.3	**Data Camera – f-number** ACME shall provide a data camera lens with an f-number of no more than f2.8 and as small as possible.
DataCam	310_0040	3.2.4 3.2.4.1	**Data Camera – Spatial Resolution** ACME shall provide a data camera with a spatial resolution at the focal plane is 0.2% or less of the corresponding FOV dimension; when simultaneously imaging at its maximum framing rate with full sensor.
DataCam	310_0050	3.2.4 3.2.4.2	**Data Camera – Bit Depth** ACME shall provide a data camera with a bit depth of at least 3x12; when simultaneously imaging at its maximum framing rate with full sensor.
DataCam	310_0060	3.2.5	**Data Camera – Focusing** ACME shall provide data camera that is focused on the CIR chamber axis.
DataCam	310_0070	3.2.6 3.2.10	**Data Camera – FOV Centered** ACME shall provide a data camera with a FOV for each imaging system centered on and perpendicular to a CIR window axis.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
DataCam	310_0080	3.2.7 3.2.10	**Data Camera – FOV Spherical Flames** ACME shall provide a data camera with a FOV of at least 90mm in diameter for the spherical and BRE flames.
DataCam	310_0090	3.2.8 3.2.10	**Data Camera – FOV Gas-Jet & Coflow Flames** ACME shall provide a data camera with a FOV of 35mm along the chamber axis and at least 20mm orthogonal to the CIR chamber axis for the gas-jet and coflow flames.
DataCam	310_100	3.2.9 3.2.10	**Data Camera FOV – Rectangular FOV** ACME shall provide a data camera with the long dimension of the FOV aligned with the chamber axis, if the FOV is rectangular.
DataCam	310_0120	3.2.11	**Data Camera – Filter Option** ACME shall provide the capability to utilize an optical filter with data color imaging system.
DataCam	310_0140	3.2.11	**Data Camera – BG7 Filter** ACME shall provide a BG7 Schott glass filter with a transmittance from 400-700 nm matching within ± 5% of 1mm thickness for the data color camera.
DataCam	310_0141	3.2.11	**Data Camera - Band Pass Filter 1** ACME shall provide a 430 nm band pass filter with transmission curves that match within ± 5% of transmittance data for a 10nm full width at half maximum transmission.
DataCam	310_0142	3.2.11	**Data Camera - Band Pass Filter 2** ACME shall provide a 450 nm band pass filter with transmission curves that match within ± 5% of Thorlab transmittance data for a 10nm full width at half maximum transmission (FWHM.)
DataCam	310_0143	3.2.11	**Data Camera - Filter Position** ACME shall provide a filter barrel with filter positions 0-3 as empty, BG7, 430nm, and 450nm.
DataCam	310_0150	3.2.12	**Data Camera – Frame Rate** ACME data camera shall provide 20 frames/s or faster when imaging the full sensor at a bit depth of at least 3x12.
DataCam	310_0160	3.2.13	**Data Camera – Adjustable Settings Bit Depth** ACME data camera shall provide the ability to adjust the bit depth settings of the data camera between tests prior to ignition.
DataCam	310_0190	3.2.13	**Data Camera – Adjustable Settings, Gain** ACME data camera shall provide the ability to adjust the gain settings between tests prior to ignition via uplink control.
DataCam	310_0220	3.2.13	**Data Camera – Adjustable Settings Iris** ACME data camera shall provide the ability to adjust the iris settings between tests prior to ignition via uplink control.
DataCam	310_0250	3.2.13	**Data Camera – Adjustable Settings, Exposure Time** ACME data camera shall provide the ability to adjust the exposure time settings between tests prior to ignition via uplink control.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
DataCam	310_0280	3.2.13	**Data Camera – Adjustable Settings, Framing Rate** ACME data camera shall provide the ability to adjust the framing rate settings between tests prior to ignition via uplink control.
DataCam	310_0310	3.2.13	**Data Camera – Adjustable Settings, Binning** ACME data camera shall provide the ability to adjust the binning settings between tests prior to ignition via uplink control.
DataCam	310_0340	3.2.13	**Data Camera – Adjustable Settings, Region Of Interest (ROI)** ACME data camera shall provide the ability to adjust the Region Of Interest (ROI) settings between tests prior to ignition via uplink control.
DataCam	310_0470	3.2.15	**Data Color Imaging – ORU** ACME shall provide the data color imaging system(s) as an ORU and removable from the Optics Bench UML during spaceflight.
		3.4	**Burner Temperature Measurement**
MEAS	570_0120	3.4.1.1	**Spherical Temperature – Location** ACME shall provide **spherical** burner surface temperature measurement at a location between 20 degrees and 90 degrees of the burner tube.
MEAS	570_0130	3.4.1.1	**Spherical Temperature – Probe Non-Visible** The ACME burner design shall ensure the probe is not visible during mission science in at least one view of the data color imaging.
MEAS	570_0150	3.4.1.2	**Gas-Jet Surface Temperature – Location** ACME shall provide a **gas-jet** burner surface temperature measurements on the exterior surface of the burner as close as possible to its outlet tip where the measurement device does not disturb the axisymmetric of the flow and flame.
MEAS	570_0170	3.4.1.4	**Burner Temperature – Probe Size** ACME shall provide temperature measurements of all burner types such that, if a probe used, it is smaller than 250 microns in thickness.
MEAS	570_0180	3.4.1.5	**Burner Temperature – Accuracy** ACME shall provide temperature measurements, of all burner types, with a precision of ± 2 degrees Celsius and an accuracy of ± 5 degrees Celsius.
MEAS	570_0190	3.4.1.6	**Gas-Jet Temperature – Range** ACME shall provide a temperature measurement with a range of 0 to 300 degrees Celsius for the **gas-jet** burner.
MEAS	570_0200	3.4.1.7	**Spherical Temperature – Range** ACME shall provide a **spherical** burner surface temperature measurement range of 0 to 700 degrees Celsius.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
MEAS	570_0201	3.4.1.9	**BRE Temperature Measurements - Heat Flux Location** ACME shall provide a **BRE** burner outlet surface temperature measurements at both heat flux measurement locations.
MEAS	570_0202	3.4.1.9	**BRE Temperature Measurements - Additional Locations - 25mm Burner** ACME shall provide an additional BRE burner outlet surface temperature measurements not in an outlet hole and at 180 degrees from the offset heat flux sensor at radial positions of 5 ± 0.5mm and 10 ± 0.5mm for the 25mm burner.
MEAS	570_0203	3.4.1.9	**BRE Temperature Measurements - Additional Locations - 50mm Burner** ACME shall provide an additional BRE burner outlet surface temperature measurements not in an outlet hole and at 180 degrees from the offset heat flux sensor at radial positions of 5 ± 0.5mm and 15 ± 0.5mm for the 50mm burner.
MEAS	570_0204	3.4.1.10	**BRE Temperature Measurements - Probe** ACME shall provide temperature measurement probes that are .3mm in diameter or less that are thermally bonded to the burner outlet plate.
MEAS	570_0205	3.4.1.11	**BRE - Temperature Range** ACME shall provide a **BRE** burner surface temperature measurement range of 0 to 500 degrees Celsius.
		3.4.3	**Hot Soot-Free Region Temperature**
DIAG	560_0010	3.4.3.2	**Hot Soot-Free Region Temperature** ACME shall provide hot soot-free region temperature measurements made simultaneously on 5 or more lines spaced 5mm ± 0.5mm apart from neighboring lines.
DIAG	560_0020	3.4.3.3	**Hot Soot-Free Region Temperature – TFP Reposition Between tests** ACME shall provide hot soot-free region temperature measurements with the ability to reposition the measurement lines or fibers **between** tests to uplinked positions.
DIAG	560_0030	3.4.3.3	**Hot Soot-Free Region Temperature – TFP Reposition During Tests** ACME shall provide hot soot-free region temperature measurements with the ability to reposition the measurement lines, or fibers, **during** tests to uplinked positions.
DIAG	560_0060	3.4.3.3	**Hot Soot-Free Region Temperature – Axial Position** ACME shall provide the ability to move the axial position along its length of a TFP fiber relative to the burner as a function of time within ± 1mm.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
DIAG	560_0090	3.4.3.4	**Hot Soot-Free Region Temperature – Fiber Retractable** ACME shall provide hot soot-free region temperature measurements where the fibers, if used, shall be retractable with minimal disturbance to beyond 100mm from the burner.
DIAG	560_0110	3.4.3.4	**Hot Soot-Free Region Temperature – Fiber Removable** ACME shall provide the TFP fiber array fully removable for replacement.
DIAG	560_0130	3.4.3.9	**Hot Soot-Free Region Temperature – Fiber Orientation 1** ACME shall provide fiber orientation such that they intersect with and are orthogonal to the CIR chamber axis.
DIAG	560_0150	3.4.3.9	**Hot Soot-Free Region Temperature – Fiber Orientation 2** ACME shall provide fiber orientation such that they are orthogonal, with a tolerance of ± 25 degrees, to the line of sight of the corresponding imaging system.
DIAG	560_0160	3.4.3.13	**Hot Soot-Free Region Temperature – TFP Fibers as ORU** ACME shall provide the fibers mounted within ORU(s) and removable from the ACME CIA during spaceflight, if the hot soot-free region temperature measurement is made by TFP.
		3.4.4	**Far-Field Temperature Measurement**
MEAS	570_0230	3.4.4.1, 3.4.4.2	**Far-Field Temperature – Avoidance Region** ACME shall provide far-field temperature measurements along unobstructed lines passing through the intersection of the CIR chamber window axes that avoid the region within 20 degrees of the burner supply tube.
MEAS	570_0240	3.4.4.3	**Far-Field Temperature – Near Radial Array** ACME shall provide far-field temperature measurements from at least the following radial locations measured in mm from the intersection of the CIR chamber window axes: 13, 25, 50, 100, 200, and 200 in a direction orthogonal to the other 200mm position (when the intersection of the CIR window axes is considered the origin of the two rays.)

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
MEAS	570_0250	3.4.4.4	**Far-Field Temperature – Distant Radial Array** ACME shall provide far-field temperature measurements from at least the following radial locations measured in mm from the intersection of the CIR chamber window axes: 45, 60, 100, 100 (in a direction orthogonal to the other like-distance position, when the intersection of the CIR window axes is considered the origin of the two rays), 200, 200 (in a direction orthogonal to the other like-distance position, when the intersection of the CIR window axes is considered the origin of the two rays.)
MEAS	570_0260	3.4.4.5	**Far-Field Temperature – Positioning Accuracy** ACME shall provide far-field temperature measurements with each position known in three dimensions to within ± 3mm or less.
MEAS	570_0270	3.4.4.6	**Far-Field Temperature – Range Inside 40-mm** ACME shall provide far-field temperature measurements in the range of 700-1700 degrees Celsius for the "near" radial array for positions inside 40mm from the intersection of the CIR chamber window axes: 13 and 25mm.
MEAS	570_0280	3.4.4.6	**Far-Field Temperature – Range Outside 40-mm** ACME shall provide far-field temperature measurements in the range of 0-1250 degrees Celsius for each radial array for positions outside of 40mm from the intersection of the CIR chamber window axes: 45, 60, 100, and 200mm.
MEAS	570_0290	3.4.4.7	**Far-Field Temperature – Sensor Precision Inside 40-mm** ACME shall provide far-field temperature measurements such that the temperature sensors have precisions for each radial array of ± 1.2C for the positions inside 40mm from the intersection of the CIR chamber window axes: 13 and 25mm.
MEAS	570_0300	3.4.4.7	**Far-Field Temperature – Sensor Precision Outside 40-mm** ACME shall provide far-field temperature measurements such that the temperature sensors have precisions for each radial array of ± 1.2C for the positions outside 40mm from the intersection of the CIR chamber window axes: 45, 50, 60, 100, and 200mm.
MEAS	570_0330	3.4.4.9	**Far-Field Temperature Measurement – ZOE** ACME shall provide far-field temperature measurements with a cylindrical zone of exclusion with a diameter of 50mm and maximized extending from the intersection of the CIR chamber window axes to 10mm beyond each temperature measurement position.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
MEAS	570_0350	3.4.4.10	**Far-Field Temperature Measurement – TC Probe Parallel** ACME shall provide far-field temperature measurements such that if thermocouples are used, each probe (including the junction) shall be within 10 degrees of parallel to a ray from the intersection of the CIR chamber window axes to the temperature measurement position to a distance of at least 100mm from the intersection of the CIR chamber windows.
MEAS	570_0360	3.4.4.11	**Far-Field Temperature Measurement – TC Probe OD** ACME shall provide far-field temperature measurements such that each probe within a radial distance of 100mm shall have an outer diameter limit of the following or smaller: 0.2mm within 5mm of the measurement position; 2mm within 20mm of the measurement position; 3.5mm within 50mm of the measurement position.
MEAS	570_0370	3.4.4.12	**Far-Field Temperature Measurement – ORU** ACME shall provide far-field temperature measurements such that the sensors within 100mm of the intersection of the CIR chamber window axes that are ORU's
		3.6	**Radiant Emission Measurement**
DIAG	560_0180	3.6.2	**Radiant Emission – Minimum FOV** ACME shall provide flame radiant emissions measurements such that the minimum region for each measurement is at least a 120-mm diameter FOV at the intersection of the CIR chamber window axes.
DIAG	560_0190	3.6.3	**Radiant Emission – Orthogonal FOV Array** ACME shall provide flame radiant emissions measurements with a linear array of five measurements made where the view from each detector is equidistant from and orthogonal to the CIR chamber axis within ± 3 degrees.
DIAG	560_0200	3.6.3	**Radiant Emission – Orthogonal Centered FOV** ACME shall provide flame radiant emissions measurements with one of five's FOV is centered on the intersection of the CIR chamber window axes.
DIAG	560_0210	3.6.3	**Radiant Emission – Orthogonal Uniform FOVs** ACME shall provide flame radiant emissions measurements where four of five FOVs are positioned uniformly, with equal spacing of no more than 30mm ± 1mm, across a span of at least 60mm ± 1mm in each direction.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
DIAG	560_0220	3.6.4	**Radiant Emission – Non-Orthogonal, Additional FOV** ACME shall provide flame radiant emissions measurements where one additional FOV is also made at angle(s) other than orthogonal to the CIR chamber axis for a FOV centered on the intersection of the CIR chamber window axes.
DIAG	560_0230	3.6.4	**Radiant Emission – Non-Orthogonal, Additional FOV Downstream** ACME shall provide flame radiant emissions measurements where the additional measurement, non-orthogonal to the chamber axis, are uniformly spaced in an angle relative to the orthogonal direction, and cover a span from orthogonal to at least 60 degrees ± 2 degrees downstream, of the burners.
DIAG	560_0240	3.6.5	**Radiant Emission – Spectral Range** ACME shall provide flame radiant emissions measurements where the detector has a spectral range of 0.2 to 11 microns and an irradiance range of 0.0001-0.1 W/cm2.
DIAG	560_0241	3.6.8	**Radiant Emissions - BRE** ACME shall provide one primary and two or more secondary radiant flame emissions for BRE burner tests.
DIAG	560_0242	3.6.8	**Radiant Emission - BRE FOV Size** ACME shall provide a field of view for the BRE radiant emission measurements that fully includes a cylindrical zone downstream of the burner surface that is 50mm in diameter by 50mm in length.
DIAG	560_0243	3.6.8	**Radiant Emission - BRE FOV Position** ACME shall provide a cylindrical field of view for BRE radiant emission measurements that is aligned with the chamber and BRE burner axis.
DIAG	560_0244	3.6.8	**Radiant Emission - BRE FOV Shape** ACME shall provide a conical field of view for each radiant emission measurement that have the same angle between the field of view edges and a plane passing through each measurements center axis.
DIAG	560_0510	3.6.8	**Radiant Emission -BRE Burner Exclusion From Primary** ACME shall provide a primary radiant emission measurement that blocks the BRE burner from the FOV with a blocked thickness of the cylindrical zone of less than 5mm when measured parallel with the burner axis.
DIAG	560_0520	3.6.8	**Radiant Emission - BRE Burner Exclusion of Lense** ACME shall not use lenses on the radiant emission measurements to form the field of view.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
DIAG	560_0530	3.6.8	**Radiant Emission - BRE Burner Identical** ACME shall use identical sensors for all radiant emission measurements.
		3.7	**Chemiluminescent Emission Measurement**
DIAG	560_0290	3.7.2	**Chemiluminescent Emission – Simultaneous** ACME shall provide simultaneous chemiluminescent emissions measurements at 310 and 431 nanometers with a 10 nanometer or larger FWHM, and broadband from 300 to 600 nanometers.
DIAG	560_0300	3.7.3	**Chemiluminescent Emission – Minimum FOV** ACME shall provide chemiluminescent emissions measurements with a 120mm diameter FOV, centered on the intersection of the CIR chamber window axes.
DIAG	560_0310	3.7.4	**Chemiluminescent Emission – Sensitivity** ACME shall provide chemiluminescent emissions measurements with a typical Radiant Sensitivity of at least 40 mA/W (as measured at the peak sensitivity wavelength.)
DIAG	560_0340	3.7.6	**Chemiluminescent Emission – ORU** ACME shall provide chemiluminescent emissions sensors as ORU(s) and removable from the ACME CIA during spaceflight.
		3.8	**Ion Current Measurement**
E-FIELD	580_0430	3.8.1, 3.8.2, 3.8.3	**Ion Current Range** ACME shall provide measurements of ion current produced between the burner and mesh electrode where it's amplitude is over a range of at least 0 to 25 micro amps with an error of no more than the larger of ± 5% of the reading or ± 0.05 micro amps.
MEAS		3.1	**Heat Flux (BRE burner tests only)**
MEAS	570_1050	3.10.1 1.13.12	**Heat Flux - Sensor** ACME shall provide heat flux measurements at the outlet of the BRE burner using a Schmidt-Boelter thermopile type heat flux transducer.
MEAS	570_1060	3.10.2 1.13.12	**Heat Flux - 25mm Burner Center** ACME shall take a heat flux measurement at the center ± 0.1mm of the 25mm BRE burner.
MEAS	570_1070	3.10.2 1.13.12	**Heat Flux - 50mm Burner Center** ACME shall take a heat flux measurement at the center ± 0.1mm of the 50mm BRE burner.
MEAS	570_1080	3.10.2 1.13.12	**Heat Flux 25mm Burner Offset** ACME shall take a heat flux measurement at a position with an offset radius of 10mm ± 0.1mm for the 25mm BRE burner.
MEAS	570_1090	3.10.2 1.13.12	**Heat Flux 50mm Burner Offset** ACME shall take a heat flux measurement at a position with an offset radius of 15mm ± 0.1mm for the 50mm BRE burner.

System	SSC_ID	ISRD Trace	ERD REQ Title/Description
MEAS	570_1200	3.10.3	**Heat Flux - Relative Size** ACME shall provide heat flux sensors that have a diameter of less than 10% of the BRE burner's outlet surface.
MEAS	570_1210	3.10.4	**Heat Flux - Orientation** ACME shall provide downstream facing heat flux sensors that are flush with the BRE burner outlet surface ± 0.05mm.
MEAS	570_1220	3.10.4	**Heat Flux - Flush Mount** ACME shall provide heat flux sensors that are flush with the BRE burner outlet surface ± 0.05mm.
MEAS	570_1230	3.10.5	**Heat Flux - FOV** ACME shall provide heat flux sensors that have no restrictions to their fields of view, geometrically by the burner.
MEAS	570_1240	3.10.6	**Heat Flux - Emissivity** ACME shall provide heat flux sensors whose surface has an emissivity of .93 for a range of 0.6-15 microns.
MEAS	570_1250	3.10.7	**Heat Flux - Range** ACME shall provide a heat flux measurement range of 0-50 kW/m2.
MEAS	570_1260	3.10.8	**Heat Flux - Precision** ACME shall provide heat flux measurements with a precision of ± 1%.
MEAS	570_1270	3.10.8	**Heat Flux - Accuracy** ACME shall provide heat flux measurements with an accuracy of ± 5% of full scale.
MEAS	570_1340	3.10.9 3.10.7	**Heat Flux - Measurement** ACME shall provide a BRE burner where the heat flux measurements remain within 0-50 kW/m2.

8.0 ACME OPERATIONAL REQUIREMENTS

This section specifies operational requirements needed to support the training, testing, operations, design and development of ACME. This section specifies requirements needed to support the training, testing, operations, design and development of ACME.

Table 8-1—Operational Requirements

System	SSC_ID	ISRD Trace	OPS Trace	ERD REQ Title/Description
			1.0	[CIR & ACME] **General OPS**
OPS	603_0010		1.1	**OPS - Flexible Control** ACME shall provide flexible experiment control based on explicit functions which can be selected, repeated as needed, and ordered to create a variety of overall experiment sequences.
OPS	603_0020		1.2	**OPS - Console Methods** ACME shall be capable of operations by all of the following methods: 1) automated software control, 2) operator control 3) mix of automated and operator control.
OPS	603_0030		1.3	**OPS - Varied Duration S/W Sequencing** ACME shall provide for automated software control through sequences of commands that are each either instant, or timed with a variable duration that is a parameter of that function.
OPS	603_0040		1.4 1.5	**OPS - Extensible Interface** ACME shall be capable of software upgrades.
OPS	613_0140	1.12.7	1.6	**OPS - Independent Functions and Status Monitoring** ACME shall be capable of handling new commands while simultaneously: (1) acquiring imaging and non-imaging data. (2) flowing gas. (3) illuminating the chamber. (4) projecting light from the CIR illumination package. (5) energizing the electric field. (6) sensing for flame extinction.
OPS	603_0060		1.7	**Command Time Log** ACME shall provide a detailed time log of commands issued for each test conducted.
			2.0	[CIR & ACME] **Pause Functions**
OPS	603_0070		2.1	**OPS - Automated Pausing** ACME shall be capable of pausing for a duration of 0.01 to 999.0s with a precision of 0.01 s or less between functions.
OPS	603_0080		2.2	**OPS - Manual Pausing** ACME shall be capable of pausing for a specified duration between functions, where it shall be possible to specify a duration in the range of at least 10 s to 99 minutes, with an uncertainty of no more than the larger of 1) 5 s 2)10% of the duration.
			3.0	[CIR & ACME] **Hardware Set Up Functions**

System	SSC_ID	ISRD Trace	OPS Trace	ERD REQ Title/Description
OPS	503_0190		3.1	**OPS Crew - External ORU Swap** ACME shall be capable of installing/removing the following hardware at specified positions external to the CIR chamber: 1) ACME avionics package 2) GCIP 3) CIR and ACME imaging packages 4) IPSUs 5) CIR illumination package 6) gas bottles 7) filter cartridge.
OPS	503_0200		3.2	**OPS Crew - Internal ORU Swap** ACME shall be capable of installing/removing the following hardware at specified positions on the ACME chamber insert assembly: 1) burner 2) igniter tip 3) electric field components 4) mass flow controllers 5) Thin Filament Pyrometry (TFP) fiber array 6) far-field thermocouple array 7) Photomultiplier Tube (PMT) array.
OPS	503_0210		3.3	**OPS Crew- Soot Clean Off** ACME shall be capable of cleaning the soot off of gas-jet burners.
			4.0	[CIR] **Chamber Ops Functions**
OPS	523_0850	1.11.1	4.1	**OPS - Venting to Specified Pressure** ACME shall be capable of venting the CIR chamber atmosphere to a specified pressure within a range of 0.01 to 3 bar, with an uncertainty of no more than the greater of 0.01 bar or 1% of the specified pressure.
OPS	503_0220	1.11.1	4.2	**OPS - Scrubbing CIR Atmosphere** ACME shall be capable of activating/deactivating scrubbing for the CIR chamber atmosphere.
OPS	573_0390		4.3	**OPS - CIR Chamber Sealed** ACME shall be capable of verifying that the CIR chamber is sealed.
OPS	523_0860	1.11.1 1.11.4	4.4	**OPS - Filling Chamber** ACME shall be capable of adding a specified gas to the CIR chamber to a specific pressure with a precision that meets the composition and pressure requirements documented in ISRD 1.11.1 and 1.11.4.
OPS	503_0230	1.11.2 1.11.3	4.5	**OPS - Chamber Fan** ACME shall be capable of activating and deactivating the CIR chamber fan.
OPS	573_0400	3.9.1	4.6	**OPS - GCIP** ACME shall be capable of providing species concentration measurements of the chamber atmosphere through operating the GCIP.
			5.0	[CIR & ACME] **Initiation & Shut Down Functions**

System	SSC_ID	ISRD Trace	OPS Trace	ERD REQ Title/Description
OPS	523_0870	1.9.3 1.9.4 2.3.1	5.1	**Powering Ops** ACME OPS shall be capable of independently powering-up, in advance of operation.
OPS	313_0480	3.2.11	5.2	**OPS - Camera Filter** ACME shall be capable of inserting and removing optical filters from the ACME data analysis camera.
OPS	553_0150	3.00	5.3	**OPS - Camera Setting - Analog Camera Gain** ACME shall be capable of configuring the CIR camera settings for Gain.
OPS	603_0090	1.12.2 1.12.7	5.4	**OPS - Non-Imaging Recording** ACME shall be capable of starting and stopping non-imaging data acquisition 5 minutes in advance of ignition and after extinction for: (1) Chamber pressure (2) Gas flow rates (3) Burner pressure differential – only required for spherical flame tests (4) Electric potential – for E-FIELD Flames tests only (5) Acceleration – whenever SAMS is available (6) Temperature: burner (7) Temperature: hot soot-containing regions AND/OR hot soot-free regions (8) Temperature: far-field – except for E-FIELD Flames tests (9) Soot volume fraction (10) Radiant emission (11) Chemiluminescent emission (12) Ion current – for E-FIELD Flames only. (13) Heat Flux - Only for BRE Burner tests
OPS	603_0100	1.12.3 1.12.7	5.5	**OPS - Imaging Record Start** ACME shall be capable of starting and stopping imaging data acquisition 1 minute prior to ignition and after extinction for: (1) Color imaging for operations (2) Color imaging for data analysis (3) Ultraviolet (UV) imaging,
			6.0	[ACME] **Chamber Illumination Functions**
OPS	553_0160	3.1.8	6.1	**OPS - Illumination On/Off** ACME shall be capable of activating and deactivating the chamber illumination system.
OPS	553_0170	3.1.8	6.2	**OPS - Illumination Timing** ACME OPS shall be capable of activating any bank of chamber lights for a duration of 0.03 to 999.99s with a precision of 0.01 s and then deactivating those same bank of chamber lights.
OPS	553_0180	3.1.8	6.3	**OPS - Illumination Flashing** ACME OPS shall be capable of activating any bank of chamber lights for a duration of 0.03 to 999.99s with a precision of 0.01 s and then deactivating the same bank of chamber lights for a duration of 0.03 to 999.99s with a precision of 0.01 s for 1 to 9,999 cycles.

System	SSC_ID	ISRD Trace	OPS Trace	ERD REQ Title/Description
			7.0	[CIR] **Illumination Package Functions**
OPS	323_0040	3.5.4 3.5.6	7.1	**OPS - CIR Illumination On** ACME shall be capable of activating the CIR illumination package at a specified intensity.
OPS	323_0050	3.5.4 3.5.6	7.2	**OPS - CIR Active Illumination Adjust** ACME shall be capable of adjusting the intensity of the illumination while the illumination package is activated.
OPS	323_0060	3.5.4 3.5.6	7.3	**OPS - Illumination Off** ACME shall be capable of deactivating the CIR illumination package.
			8.0	[ACME] **Ignition Functions**
OPS	543_0200	1.10.2	8.1	**OPS - Specifying Igniter Speed** ACME shall be capable of translating the igniter at a rate of 0.0 to 50.0 mm/s with an uncertainty of less than 0.1 mm/s.
OPS	503_0240	1.11.8 1.11.9	8.1	**OPS - Specifying Igniter Position** ACME OPS shall be capable of translating the igniter to a specified position with a precision of 1mm and over a range extending nominally from the burner tip to at least beyond the zone of exclusion.
OPS	543_0210	1.10.2 1.10.3 1.10.4	8.2	**OPS - Specifying Ignition Energizing Voltage** ACME OPS shall be capable of energizing the igniter at a voltage of 50 to 100% of max power with a precision of 2%.
OPS	543_0220		8.2	**OPS - Specifying Ignition Energizing Duration** ACME shall be capable of energizing the igniter for a duration of 0 to 10s with a precision of 0.1s.
			9.0	[CIR & ACME] **Flow Control Functions**
OPS	523_0880	1.9.3 1.9.4 2.3.1	9.1	**OPS - MFCs Zeroing** ACME shall be capable of zeroing all ACME MFCs.
OPS	523_0890		9.2	**OPS - Venting Gas Lines** ACME shall be capable of venting all or a subset of the gas delivery lines.
OPS	523_0900		9.3	**OPS - Purging Gas Delivery Lines** ACME shall be capable of purging all or a subset of the gas delivery lines with the gases with which those lines are currently connected.
OPS	523_0910	1.9.4	9.4	**OPS - Setting CIR MFCs** ACME shall be capable of setting all or a subset of CIR MFCs for specified steady flow rates within a range of 0 to 100% and a precision of 0.1% or less.
OPS	523_0920	1.9.4	9.5	**OPS - Setting ACME MFCs** ACME shall be capable of setting all or a subset of ACME MFCs for specified steady flow rates within a range of 0 to 100% and a precision of .01% or less.
OPS	523_0930	1.9.4	9.6	**OPS - ACME Flows - Step** ACME OPS shall be capable of setting all or a subset of ACME flows to: 1) step upward and/or downward by specified values and

System	SSC_ID	ISRD Trace	OPS Trace	ERD REQ Title/Description
				2) pausing to hold all flows steady for a specified wait duration with a precision of 0.1 s or less for a specified number of cycles ranging from at least 1 to 999.
OPS	523_0940	1.9.4	9.7	**OPS - ACME Flows - Linear Ramp** ACME shall be capable of setting all or a subset of ACME flows to: 1) linearly ramp upward/downward by values of 0 to 100% and a precision of 0.1% for a duration of 0.1 to 999.9s with a precision of 0.1s. 2) and pausing with all flows held steady for a wait duration with a precision of 0.1 s or less.
OPS	523_0950	1.9.4	9.8	**OPS - ACME Flows - Step - A &B Values** ACME shall be capable of setting all or a subset of the flows to: 1) step upward/downward by a value of 0 to 100% and a precision of 0.1% 2) pausing with all flows held steady for a duration of 0.1 to 999.9s with a precision of 0.1s. 3) stepping the flows downward/upward by a value of 0 to 100% and a precision of 0.1%. 4) pausing with all flows held steady for a duration with a precision of 0.1s.
OPS	523_0960	1.9.4	9.9	**OPS - ACME Flows - Ramp - A&B Values** ACME shall be capable of setting all or a subset of ACME flows to: 1) ramp upward/downward by a value of 0 to 100% and a precision of 0.1% duration of 0.1 to 999.9s with a precision of 0.1 s. 2) pausing with all flows held steady for a duration of 0.0 to 999.9s with a precision of 0.1 s. 3) ramping the flow downward/upward by a value of 0 to 100% and a precision of 0.1% for a duration of 0.1 to 999.9s with a precision of 0.1s. 4) pausing with all flows held steady for a duration of 0.0 to 999.9s with a precision of 0.1s for 1 to 999 cycles.
OPS	523_0970	1.9.4	9.10	**OPS - SV Control (CIR/ACME)** ACME OPS shall be capable of opening and closing the CIR and ACME solenoid valves.
OPS	523_0980	1.9.4	9.11	**OPS - SV Control (ACME)** ACME OPS shall be capable of opening and closing all ACME solenoid valves or any specified subset thereof.
			10.0	[ACME] **TFP Pyrometry Functions**
OPS	563_0280	1.11.8 1.11.9	10.1	**OPS - TFP Positioning - Operational Range** ACME shall be capable of translating the TFP fiber array at

System	SSC_ID	ISRD Trace	OPS Trace	ERD REQ Title/Description
		3.4.3.3 3.4.3.4		a specified rate of 100 mm/s or greater, with an uncertainty of 0.1 mm/s or less, to a position within its operational range.
OPS	563_0290	1.11.8 1.11.9 3.4.3.3	10.2	**OPS - TFP Positioning - Specified Rate** ACME shall be capable of 1) translating the TFP fiber array upward and/or downward at a specified rate to a position within its operational range. 2) pausing for a specified wait duration with a precision of 0.01 s or less and within a range of 0.00 to 999.99 s for a specified number of cycles ranging from 1 to 999.
OPS	563_0300	3.4.3.3	10.3	**OPS - TFP Positioning - Ramp up** ACME shall be capable of 1) translating the TFP fiber array upward at a specified ramp-up value. 2) pausing for a specified high-position duration with a precision of 0.01 s or less, and within a range of at least 0.00 to 99.9 s 3) ramping the TFP fiber array downward by a specified ramp-down values as high as 100 mm/s with an uncertainty of no more than 0.1 mm/s. 4) pausing for a specified low-position duration with a precision of 0.01 s or less, and within a range of at least 0.00 to 99.99 s for a specified number of cycles (1-999)
			11.0	[ACME] **Electric Field Functions**
OPS	583_0470	1.7.1 1.7.4	11.1	**OPS - E-Field - Activate** ACME shall be capable of activating the electric field at the specified polarity (positive or negative) and voltage with a precision and range at least meeting ISRD 1.7.1.
OPS	583_0480	1.7.1 1.7.4	11.2	**OPS - E-Field - Step Change** ACME shall be capable of setting the electric field potential, by a step change by adjusting the polarity and voltage of the electric field with a precision and range outlined in ISRD 1.7.1.
OPS	583_0490	1.7.2 1.7.4	11.3	**OPS - E-Field - Deactivate** ACME shall be capable of deactivating the electric field.
OPS	583_0500	1.7.1 1.7.2 1.7.4	11.4	**OPS - E-Field - Stepping Voltage** ACME shall be capable of 1) stepping the electric field voltage upward and/or downward by a specified value outlined in ISRD 1.7.1 2) pausing with the voltage held steady for a wait duration of 0.000 to 999.999s with a precision of 0.005 s or less for 1 to 9,999 cycles.
OPS	583_0510	1.7.1 1.7.2 1.7.4	11.5	**OPS - E-Field - Ramping Voltage** ACME shall be capable of 1) ramping the electric field voltage upward and/or

System	SSC_ID	ISRD Trace	OPS Trace	ERD REQ Title/Description
				downward at a specified step size outlined in ISRD 1.7.1 for a ramp duration of 0.005 to 999.999s with a precision of 0.01s. 2) pausing with the voltage held steady for a wait duration of 0.000 to 999.999s with a precision of 0.01s for 1 to 9,999 cycles.
OPS	583_0520	1.7.1 1.7.2 1.7.3 1.7.4	11.6	**OPS - E-Field - Rectangular Wave** ACME shall be capable of 1) stepping the electric field voltage upward by specified step-up values outlined in ISRD 1.7.1. 2) pausing with the voltage held steady for a duration of 0.005 to 999.999s and a precision of 0.005 s 3) stepping the voltage downward by a specified step-down value outlined in ISRD 1.7.1. 4) pausing with the voltage held steady for a duration of 0.005 to 999.999s and a precision of 0.005 for 1 to 9,999 cycles.
OPS	583_0530	1.7.1 1.7.3 1.7.4	11.7	**OPS - E-Field - Triangular Wave** ACME shall be capable of 1) ramping the electric field voltage upward by a ramp size outlined in ISRD 1.7.1 for a duration of 0.000 to 999.999s with a precision of 0.005s. 2) pausing with the voltage held steady for a duration of 0.000 to 999.999s with a precision of 0.005s 3) ramping the electric field voltage downward by a ramp-down size outlined in ISRD 1.7.1 for a duration of 0.000 to 999.999s with a precision of 0.005s. 4) pausing with the voltage held steady for a duration of 0.000 to 999.999s with a precision of 0.005 s for 1 to 9,999 cycles
			12.0	[ACME] **Extinction Functions**
OPS	563_0310	3.7.1	12.1	**OPS - Flame Extinction - Detection** ACME shall be capable of detecting flame extinction after ignition, by monitoring one or more non-imaging instruments and verifying that averaged measurements over a duration of 0.0 to 99.9s with a precision of 0.1s are below specified values
OPS	563_0540	3.7.1	12.1	**OPS - Flame Extinction - Detection Selection** ACME shall be capable of specifying whether 1) any specified measurement indicates extinction. 2) half of the specified measurements indicate extinction. 3) all specified measurements indicate extinction.

System	SSC_ID	ISRD Trace	OPS Trace	ERD REQ Title/Description
DIAG	560_0280	3.7.1		**Chemiluminescent Emission – Flame Extinction** ACME shall provide chemiluminescent emissions measurements where flame extinction is detected.
OPS	523_0990	1.9.4 3.7.1	12.2	**OPS - Flame Extinction- Gas Flow Shutoff** ACME shall be capable of setting all gas flows to zero in response to extinction detection after a duration of 0.0 to 99.99s with a precision of 0.1s.
OPS	523_1000	1.9.4 3.7.1	12.3	**OPS - Flame Extinction - Single Step Flow** ACME shall be capable of setting all or a subset of flows in response to extinction detection to step upward/downward 0 to 100% with a precision of .01%
OPS	523_1010	1.9.4 3.7.1	12.4	**OPS - Flame Extinction - Single Ramp Flow** ACME shall be capable of setting any flows within a range of 0 to 100% and a precision of 0.1% In response to extinction detection, for a ramp duration of 0.1 to 999.9s with a precision of 0.1s and holding the flows at their new values.
OPS	523_1020	1.9.4 3.7.1	12.5	**OPS - Flame Extinction - Stepping Flow** ACME shall be capable of setting all or a subset of flows to 1)step upward and/or downward In response to extinction detection by a range of 0 to 100% and a precision of 0.1%. 2) pausing with all flows held steady for duration of 0.0 to 99.9s with a precision of 0.1 s and then setting all gas flows to zero.
OPS	523_1030	1.9.4 3.7.1	12.6	**OPS - Flame Extinction - Ramping Flow** ACME shall be capable of setting all or a subset of flows to: 1) ramp upward/downward in response to extinction detection by 0 to 100% and a precision of 0.1% over a duration of 0 to 99.9s with a precision of 0.1 s. 2) pausing with all flows held steady for a duration of 0 to 99.9s with a precision of 0.1s and then setting all gas flows to zero.
OPS	563_1040	3.7.1	12.7	**OPS - Flame Extinction - Shutoff** ACME shall be capable of deactivating extinction detection.
		3.1		[ACME] **Heat Flux (BRE burner tests only)**
BURNER	530_2010	1.13.16		**BRE - Burner Down-Mass** ACME shall return both BRE burners to Earth at the end of spaceflight testing.
		1.1		[ACME] **General Experiment Configuration**
BURNER	530_0040	1.1.3		**Burners - Duplicate to PS** ACME shall provide at least one duplicate of each burner to the Project Scientist except where burners with different tube lengths are otherwise identical, then only one such burner is required for transfer to the Project Scientist.
EMS	510_0020	1.1.6		**Simultaneous Hardware Configuration** ACME shall apply all configuration requirements

System	SSC_ID	ISRD Trace	OPS Trace	ERD REQ Title/Description
				simultaneously unless otherwise specified such as those for burners, electrodes, e-field, gas supply and delivery, ignition, chamber atmosphere, data synchronization and recording, and imaging.
		1.1		[CIR & ACME] **Ignition**
IGN	540_0110	1.10.2		**Ignition – Hydrodynamic Disturbance** ACME shall provide ignition where the igniter is inserted at least 30 seconds prior to ignition, and removed from the free area within 0.5 second after igniter off.
		1.1		[CIR] **Chamber Atmosphere**
CIA	500_0010	1.11.1		**Chamber Atmosphere – Composition CIR Support** The ACME flow system shall provide the capability to vary the gas composition within the CIR chamber with an accuracy of ±0.005 mole fraction of each concentration.
CIA	500_0020	1.11.1		**Chamber Atmosphere – Dry Gases CIR Support** The ACME flow system shall provide dry chamber gas.
CIA	500_0021	1.11.1		**Chamber Atmosphere - Maintain Dry** The ACME flow system shall maintain a dry chamber atmosphere.
CIA	500_0030	1.11.1		**Chamber Atmosphere – Trace Contaminants CIR Support** The ACME flow system shall provide chamber gas containing less than a mole fraction of 0.01 of contaminants H2O and CO2.
CIA	500_0040	1.11.2		**Chamber Atmosphere – Mixed CIR Support** The ACME flow system shall provide a well mixed chamber atmosphere such that the chamber fan moves 100 times the gas volume in the chamber.
CIA	500_0050	1.11.3		**Chamber Atmosphere – Quiescence** ACME shall provide quiescent conditions of the chamber atmosphere achieved with a hold period of at least 300 seconds prior to ignition, following filling, venting or actuator motion.
CIA	500_0060	1.11.3		**Chamber Atmosphere – Convective Disturbances** ACME shall provide chamber atmospheric conditions such that convective disturbances during a test point are limited to those caused by burner gas flow, combustion, the igniter, and the motion of the mesh electrode and any diagnostics.
CIA	500_0070	1.11.3		**Chamber Atmosphere – Quiescence Data Sampling** ACME shall provide quiescent conditions extending through the entire duration of the data collection.
CIA	500_0080	1.11.4		**Chamber Atmosphere – Initial Pressure Range** ACME shall provide the capability to vary the initial, pre-test chamber pressure in the range 0.1 to 3 bar absolute.
CIA	500_0090	1.11.5		**Chamber Atmosphere – Initial Temperature Range** ACME shall provide the initial, pre-test ambient temperature of 290 to 310 degrees Kelvin.
CIA	500_0100	1.11.5		**Chamber Atmosphere – Initial Temperature Uniformity** ACME shall provide the initial, pre-test ambient temperature that is uniform within ± 5 degrees Kelvin.

System	SSC_ID	ISRD Trace	OPS Trace	ERD REQ Title/Description
CIA	500_0160	1.11.10		**Chamber Atmosphere – Ambient Disturbances** ACME shall provide a chamber atmosphere such that ambient disturbances are limited to those caused by burner flow, combustion, the igniter, the motion of any probes or the electrode mesh, and g-jitter during each test, including the entire data collection period.
CIA	500_0170	1.11.11		**Chamber Atmosphere – ISS Ops Disturbances** ACME shall provide a chamber atmosphere such that no test points are conducted when thrusters fire or spacecraft dock or undock from ISS.
		2.2		[CIR] **Chamber Oxygen Concentration**
MEAS	570_0040	2.2.1		**Chamber Oxygen Concentration - Filling/Scrubbing** ACME shall measure the oxygen concentration of the chamber atmosphere after procedures for filling and/or scrubbing are completed using the CIR's GCIP when available.
MEAS	570_0050	2.2.1		**Chamber Oxygen Concentration – Well-Mixed** ACME shall measure the oxygen concentration of the chamber atmosphere after that atmosphere has been well-mixed using the CIR's GCIP when available.
		2.3		[CIR] **Gas Flow Rate**
FLOW INTEG	520_0800	2.3.4		**Gas Flow Rate – Recording CIR Chamber** ACME shall ensure that gas flow in all supply lines to the **CIR chamber** is always measured and recorded during each test even if and when there is no gas flow in certain line(s).
		2.5		[ACME] **Electric Potential**
E-FIELD	580_0420	2.5.4		**Electric Potential – E-FIELD Flames tests only** ACME shall provide electric potential measurement between the mesh electrode and the high voltage (HV) ground terminal for the E-FIELD Flames tests.
		2.6		[CIR] **Chamber Acceleration**
MEAS INTEG	570_0110	2.6.1		**Chamber Acceleration Measurement** ACME shall provide chamber acceleration measurement(s) through SAMS, when available, with a sensor head at or near the CIR, unless waived by the Project Scientist.
		3.3		[CIR] **Ultraviolet (UV) Imaging**
UVI	320_0010	3.3.1		**UV Imaging – OH* Filter** ACME shall provide ultraviolet (UV) imaging using the CIR LLL-UV Package, where camera and all associated optics are optimized for UV imaging through an OH* filter (310 nanometers) with a 10 nanometers FWHM, i.e., Full Width at Half Maximum transmission.
UVI	320_0020	3.3.2		**UV Imaging FOV – Spherical & BRE** ACME shall provide ultraviolet imaging for 75mm diameter FOV, for the spherical and BRE flames, using the CIR LLL-UV Package.
UVI	320_0030	3.3.3		**UV Imaging FOV – Gas-Jet & Coflow** ACME shall provide ultraviolet imaging for 60mm ± 5mm along the chamber axis and at least 30mm wide (i.e. in the

System	SSC_ID	ISRD Trace	OPS Trace	ERD REQ Title/Description
				direction orthogonal to the CIR chamber axis), for the gas-jet and coflow flames, using the CIR LLL-UV Package.
		3.4.2		[CIR] **Hot Soot Region Temperature Measurement**
HiBMS2	320_2010	3.4.2.1 3.4.2.2		**Hot Soot Region Temperature – Spherical and BRE** ACME shall provide hot soot-containing region temperature measurements for spherical and BRE flames with a minimum measurement region of 85-mm diameter centered on the intersection of the CIR chamber window axes.
HiBMS2	320_2020	3.4.2.1 3.4.2.3		**Hot Soot Region Temperature – Gas-Jet & Coflow** ACME shall provide hot soot-containing region temperature measurements for the gas-jet and coflow flames such that the measurement region is 50 ± 5mm along the chamber axis and at least 20mm wide centered on the intersection of the CIR chamber window axes.
HiBMS2	320_2040	3.4.2.5		**Hot Soot Region Temperature – Sensor Gain** HiBMS2 shall provide hot soot-containing region temperature measurements such that the sensor has a variable gain via uplink prior to ignition.
HiBMS2	320_2050	3.4.2.6		**Hot Soot Region Temperature – Sampling** HiBMS2 AMP CSCI shall provide hot soot-containing region temperature measurements such that the measurements occur at least every 5 seconds with a response time of 5 seconds.
HiBMS2	320_2060	3.4.2.7		**Hot Soot Region Temperature Measurement – Control** HiBMS2 shall provide hot soot-containing region temperature measurements such that the measurements start and stop according to uplink prior to ignition.
		3.4.3		[CIR] **Hot Soot-Free Region Temperature**
HiBMS2	320_2070	3.4.3.1 3.4.3.5		**Hot Soot-Free Region Temperature – Spherical & BRE** HiBMS2 shall provide hot soot-free region temperature measurements for the spherical and BRE flames where the minimum measurement region is an 85mm diameter, in the direction orthogonal to the CIR chamber axis, centered on the intersection of the CIR chamber window axes.
HiBMS2	320_2080	3.4.3.6		**Hot Soot-Free Region Temperature – Gas-Jet & Coflow** HiBMS2 shall provide hot soot-free region temperature measurements for the gas-jet and coflow flames where the measurement region is 50mm ± 5mm along the chamber axis and at least 20mm wide, in the direction orthogonal to the CIR chamber axis, centered on the intersection of the CIR chamber window axes.
DIAG	560_0120	3.4.3.8		**Hot Soot-Free Region Temperature – Fiber Size** ACME shall provide 15-micron SiC fibers for hot soot-free region temperature measurement.
HiBMS2	320_2100	3.4.3.10		**Hot Soot-Free Region Temperature – Sensor Gain** HiBMS2 shall provide hot soot-free region temperature measurements such that the sensor allows variable gain via uplink prior to ignition.
HiBMS2	321_2110	3.4.3.11		**Hot Soot-Free Region Temperature – Sampling** The IMAG HiBMS2 HWCI shall provide hot soot-free

System	SSC_ID	ISRD Trace	OPS Trace	ERD REQ Title/Description
				region temperature measurements that occur at least every 5 seconds with a response time of 5 seconds.
HiBMS2	322_2120	3.4.3.12		**Hot Soot-Free Region Temperature – Control** The IMAG HiBMS2 HWCI shall provide hot soot-free region temperature measurements that start and stop according to uplink prior to ignition.
DIAG	560_0550	3.4.3.14		**Hot Soot-Free Region Temperature – Platinum Wire** ACME shall provide the option of using a TFP array where the furthest downstream fiber is replaced with a platinum wire that is 200 ± 50 microns in diameter where the wire diameter is known to ± 5 microns.
		3.5		[CIR] **Soot Volume Fraction Measurement**
HiBMS1	320_1010	3.5.1 3.5.2		**Soot Volume Fraction – Spherical and BRE FOV** HiBMS1 shall provide soot volume fraction measurement, for the spherical and BRE flames; such that the measurement region is at least a 45mm diameter centered on the intersection of the CIR chamber window axes.
HiBMS1	320_1020	3.5.1 3.5.3		**Soot Volume Fraction – Gas-Jet & Coflow FOV** HiBMS1 shall provide soot volume fraction measurement, for the spherical flames; such that the measurement region is at least 25mm along the CIR chamber axis and at least 25mm wide (i.e., in the direction orthogonal to the CIR chamber axis), centered on the intersection of the CIR chamber window axes.
HiBMS1	320_1030	3.5.4		**Soot Volume Fraction – Image Types** The CIR illumination package shall be available to provide soot volume fraction measurement such that there are three image types: illumination with no flame, illumination with the flame, and the flame without illumination.
HiBMS1	320_1040	3.5.5		**Soot Volume Fraction Measurement – Frequency** HiBMS1 shall provide soot volume fraction measurement such that the measurements are made at 1 sample per second or faster with a response time of 1 second.
HiBMS1	320_1050	3.5.6		**Soot Volume Fraction – Control** HiBMS1 shall provide soot volume fraction measurement such that the measurements start and stop according to uplink prior to ignition.
		3.9		[CIR] **Post-Test Gas Composition**
MEAS	570_0380	3.9.1, 3.9.2		**Post-Test Gas Composition** ACME shall be able to reference the CIR GCIP.
		1.12		[ACME] **Data Synchronization and Recording**
CDS	620_0110	1.12.5		**Data Downlink - Select Image Downlink** ACME shall provide the capability to downlink image data from only selected imaging systems.
CDS	620_0120	1.12.6		**Data Downlink - Select Data Downlink** ACME shall provide the capability to downlink a fraction of the acquired image data from selected time periods.
CDS	620_0130	1.12.6		**Data Downlink - Select Data Time Period** ACME shall provide the capability to select different time periods for each imaging system where a fraction of the data is downlinked.

9.0 CARRIER REQUIRMENTS

All ACME carrier requirements are contained in the *ACME Interface Control Document (ICD)*, FCF-ICD-CIR-ACME, which will be completed by the FCF CIR Sustaining Engineering Team.

10.0 DERIVED REQUIREMENTS

This section specifies derived requirements for the ACME system. Derived Requirements are those requirements not directly linked to a science or operational requirement as defined by the NASA project scientist.

Table 10-1—Derived Requirements

System	SSC_ID	Parent ID/ ISRD Trace	ERD REQ Title/Description
FLOW	520_1430	ACME Risk 021	**Spare - Mass Flow Controller** ACME shall provide 1 spare mass flow controller assembly for each oxidizer, inert and fuel flow rate.
CIA	500_0250	ACME Risk 010	**Spare - Data Cube** ACME shall provide 1 spare data cube assemblies.
AVP	600_0200		**Spare - Hard Drive Assembly** ACME shall provide 2 spare hard drive assemblies.
OpsCam	550_190		**Spare - OpsCam** ACME shall provide 1 spare operational camera assembly.
DataCam	310_610		**Spare - DataCam** ACME shall provide 1 spare digital color camera assemblies.
DIAG	560_0510		**Spare - TFP** ACME shall provide 10 spare thin filament pyrometry array assemblies.
MEAS	570_1330		**Spare - Far Field TC Array** ACME shall provide 10 spare far field thermocouple array assemblies for both the near and far positions.
MEAS	570_1330		**Spare - 200mm Thermocouples** ACME shall provide 5 spare 200mm thermocouple assemblies for both the relay and igniter positions.
IGN	540_0310		**Spare - Igniter Tips** ACME shall provide 20 spare igniter tips.
E-FIELD	580_0640		**Spare - E-FIELD Mesh Electrode** ACME shall provide 1 spare E-field electrode mesh.
E-FIELD	580_0650		**Spare - E-FIELD HV Power Supply** ACME shall provide 1 spare E-field high voltage power supply.
BURNER	530_2030		**Spare - Spherical Burners** ACME shall provide 5 spare spherical burners of each size.
BURNER	530_2040		**Spare - Gas-Jet Burners** ACME shall provide 2 spare gas jet burners of each size.
BURNER	530_2050		**Spare - BRE Burners** ACME shall provide 1 spare BRE burner of each size.
EMS	510_0110		**Spare - Fuel Hose** ACME shall provide 1 spare fuel hose assembly.
EMS	510_0120		**Spare - Oxygen Hose** ACME shall provide 1 spare oxygen hose assembly.
EMS	510_0130		**Spare - Inert Hose** ACME shall provide 1 spare Inert hose assembly.
EMS	510_0140		**Spare - Green Water Cooling Hose** ACME shall provide 1 spare green water cooling hose assembly.

System	SSC_ID	Parent ID/ ISRD Trace	ERD REQ Title/Description
EMS	510_0150		**Spare - Grey Water Cooling Hose** ACME shall provide 1 spare grey water cooling hose assembly.

11.0 SOFTWARE REQUIREMENTS

This section specifies software requirements for the ACME Main Processor (AMP), ACME Ground Software (AGS), and Computer Software Configuration Items (CSCI).

Table 11-1—Software Requirements

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
				General
CIR	611_0170	FCF-ICD-CIR-ACME 8.1.7.3.1		The AMP CSCI shall immediately enter Initialization state upon power on.
CIR	611_0180	FCF-ICD-CIR-ACME 8.1.7.3.2		The AMP CSCI shall transition to Operational state upon successful initialization.
CIR	611_0190	FCF-ICD-CIR-ACME 8.1.7.3.2.1		The AMP CSCI shall have an Idle state within the Operational state which denotes the experiment is suspended.
CIR	611_0200	FCF-ICD-CIR-ACME 8.1.7.3.3		The AMP CSCI shall transition into Off-Nominal state when a significant error occurs either during initialization or nominal operation.
CIR	611_0210	FCF-ICD-CIR-ACME 8.1.7.3.4		The AMP CSCI shall transition to Safed state upon command.
CIR	612_0220	FCF-ICD-CIR-ACME 8.2.1.2		The AGS CSCI shall obtain telemetry once a second upon connection to the AMP CSCI.
CIR	612_0230	FCF-ICD-CIR-ACME 8.2.1.2		The AGS CSCI shall display the current software state from the telemetry report.
CIR	611_0240	FCF-ICD-CIR-ACME 8.2.1.2		The AGS CSCI shall display health and status from the telemetry report.
CIR	611_0250	FCF-ICD-CIR-ACME 8.1.7.5		The AMP CSCI shall report telemetry upon command.
CIR	611_0260	FCF-ICD-CIR-ACME 8.1.7.5		The AMP CSCI shall report the current software state in the telemetry report.
CIR	611_0270	FCF-ICD-CIR-ACME 8.1.7.5		The AMP CSCI shall report health and status in the telemetry report.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
CIR	612_0280	FCF-ICD-CIR-ACME 8.1.1.1.5		The AGS CSCI shall communicate with the FCF IOP MP according to FCF-ICD-0076A, the FCF SW ICD.
CIR	612_0290	FCF-ICD-CIR-ACME 8.1.1.1.5		The AGS CSCI shall communicate with the AMP CSCI according to ACME-DOC-008, the ACME Command and Telemetry ICD.
CIR	611_0300	FCF-ICD-CIR-ACME 8.1.1.1.5		The AMP CSCI shall communicate with the FCF IOP MP according to FCF-ICD-0076A.
CIR	611_0310	FCF-ICD-CIR-ACME 8.1.1.1.5		The AMP CSCI shall communicate with the FCF IPSU MPs according to FCF-ICD-0076A.
CIR	611_0320	FCF-ICD-CIR-ACME 8.1.1.1.5		The AMP CSCI shall interface with the PMD Prodigy/CMD PC/104 Motion Control Board.
CIR	611_0330	FCF-ICD-CIR-ACME 8.1.1.1.5		The AMP CSCI shall interface with the AGS CSCI according to ACME-DOC-008, the ACME Command and Telemetry ICD.
CIR	612_0340	FCF-ICD-CIR-ACME 8.2.1.1		The AGS CSCI shall use the TReK library to communicate with the TSC.
MCS AMP	611_0010	1.12.1		**Data Synchronization** ACME shall have all data time stamped and synchronized to within +/- 0.03 sec and referenced within +/- 0.03 sec of GMT.
MCS AGS	612_0090	1.12.5		**Data Downlink – Image System - Commanded** The AGS CSCI shall provide a command to transfer data files to the IOP.
MCS AMP	611_0080	1.12.5		**Data Downlink – Image System - Configured** The AMP CSCI shall transfer data files to the IOP upon command.
MCS AGS	612_0110	1.12.6		**Data Downlink – Image Select - Commanded** The AGS CSCI shall provide a command to transfer subsets of the color imaging data to the IOP MP.
DataCam AMP	311_0560	3.2		**Data Color Imaging Interface** The AMP CSCI shall interface with the AVT GC1380CH digital color camera.
MCS AMP	611_0150	1.12		**Cube Interface** The AMP CSCI shall interface with the UEI PowerDNA Gigabit cube.
MCS AMP	611_0160	1.12		**Hard Drive Interface** The AMP CSCI shall store data to the SATA disk.
OPS	602_0150	4.0		**AGS Uplink - Commanded** The AGS CSCI shall provide a command to uplink a sequence to the AMP CSCI.
OPS	602_0160	4.0		**AGS Sequence Saved** The AGS CSCI shall allow a sequence to be saved to file.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
OPS	602_0170	4.0		**AGS Sequence Loaded** The AGS CSCI shall allow a sequence to be loaded from file.
OPS	601_0110	4.0		**Automated Sequence Uplink - Configured** The AMP CSCI shall allow an automated sequence to be uplinked.
OPS	601_0120	4.0		**Automated Sequence Execute** The AMP CSCI shall execute the uplinked automated sequence upon command.
OPS	601_0130	4.0		**Automated Sequence Accept** The AMP CSCI shall only accept uplinked sequences when in Operational state.
MCS AGS	612_0030	1.12.2 1.12.3 1.12.7	5.4 5.5	**Data Recording - Hard Drive - Commanded** The AGS CSCI shall provide and uplinkable command to configure the hard drive to record data up to 5 minutes before and after a test point.
MCS AMP	611_0020	1.12.2 1.12.3 1.12.7	5.4 5.5	**Data Recording - Hard Drive - Configured** The AMP CSCI shall configure upon command the hard drive to record data up to 5 minutes before and after a test point.
MCS AGS	612_0350			**Upstream Relay Board 1 - Enable/ Disable - Commanded** The AGS CSCI shall provide an uplinkable command to configure the upstream relay board 1 to enable/disable.
MCS AMP	611_0360			**Upstream Relay Board 1 - Enable/ Disable - Configured** The AMP CSCI shall configure upon command the upstream relay board 1 to enable/disable by applying a voltage of 0V or +5V through channel A012.
MCS AGS	612_0370			**Downstream Relay Board 2 - Enable/ Disable - Commanded** The AGS CSCI shall provide an uplinkable command to configure the downstream relay board 2 to enable/disable.
MCS AMP	611_0380			**Downstream Relay Board 2 - Enable/ Disable - Configured** The AMP CSCI shall configure upon command the downstream relay board 2 to enable/disable by applying a voltage of 0V or +5V through channel A030.
				Flow System

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
FLOW AGS	522_1070	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Mass Flow Controller A1-Commanded Enable/Disable** The AGS CSCI shall provide an uplinkable command to configure MFC A1 enable/disable.
FLOW AMP	521_1080	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Mass Flow Controller A1 Configured Enable/Disable** The AMP CSCI shall configure upon command MFC A1 to enable/disable by applying a voltage of 0 or +5V through channel AO9.
FLOW AGS	522_1090	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Mass Flow Controller A2-Commanded Enable/Disable** The AGS CSCI shall provide an uplinkable command to configure MFC A2 enable/disable.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
FLOW AMP	521_1100	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Mass Flow Controller A2 Configured Enable/Disable** The AMP CSCI shall configure upon command MFC A2 to enable/disable by applying a voltage of 0 or +5V through channel AO10.
FLOW AGS	522_1110	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Mass Flow Controller A3 Commanded Enable/Disable** The AGS CSCI shall provide an uplinkable command to configure MFC A3 enable/disable.
FLOW AMP	521_1120	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Mass Flow Controller A3 Configured Enable/Disable** The AMP CSCI shall configure upon command MFC A3 to enable/disable by applying a voltage of 0 or +5V through channel AO11.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
FLOW AGS	522_1130	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Mass Flow Controller A1 - Control Commanded** The AGS CSCI shall provide an uplinkable command to configure MFC A1.
FLOW AMP	521_1140	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Mass Flow Controller A1 - Control Configured** The AMP CSCI shall configure upon command MFC A1 by applying a voltage of 0 to +5V through channel AO0.
FLOW AGS	522_1150	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Mass Flow Controller A2 - Control Commanded** The AGS CSCI shall provide an uplinkable command to configure MFC A2.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
FLOW AMP	521_1160	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Mass Flow Controller A2 - Control Configured** The AMP CSCI shall configure upon command MFC A2 by applying a voltage of 0 to +5V through channel AO1.
FLOW AGS	522_1170	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Mass Flow Controller A3 - Control Commanded** The AGS CSCI shall provide an uplinkable command to configure MFC A3.
FLOW AMP	521_1180	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Mass Flow Controller A3 - Control Configured** The AMP CSCI shall configure upon command MFC A3 by applying a voltage of 0 to +5V through channel AO2.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
FLOW AGS	522_1190	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A1 - Enable/Disable Commanded** The AGS CSCI shall provide an uplinkable command to configure SV A1 enable/disable.
FLOW AMP	521_1200	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A1- Enable/Disable Configured** The AMP CSCI shall provide an uplinkable command to configure SV A1 by applying a voltage of 0V or +5V through channel AO15.
FLOW AGS	522_1210	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valves A2 - Enable/Disable Commanded** The AGS CSCI shall provide an uplinkable command to configure SV A2 enable/disable.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
FLOW AMP	521_1220	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A2 - Enable/Disable Configured** The AMP CSCI shall provide an uplinkable command to configure SV A2 by applying a voltage of 0V or +5V through channel AO16.
FLOW AGS	522_1230	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A3 - Enable/Disable Commanded** The AGS CSCI shall provide an uplinkable command to configure SV A3 enable/disable.
FLOW AMP	521_1240	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A3 - Enable/Disable Configured** The AMP CSCI shall provide an uplinkable command to configure SV A3 by applying a voltage of 0V or +5V through channel AO17.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
FLOW AGS	522_1250	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A4 - Enable/Disable Commanded** The AGS CSCI shall provide an uplinkable command to configure SV A4 enable/disable.
FLOW AMP	521_1260	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A4 - Enable/Disable Configured** The AMP CSCI shall provide an uplinkable command to configure SV A4 by applying a voltage of 0V or +5V through channel AO18.
FLOW AGS	522_1270	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A5 - Enable/Disable Commanded** The AGS CSCI shall provide an uplinkable command to configure SV A5 enable/disable.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
FLOW AMP	521_1280	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A5 - Enable/Disable Configured** The AMP CSCI shall provide an uplinkable command to configure SV A5 by applying a voltage of 0V or +5V through channel AO19.
FLOW AGS	522_1290	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A6 - Enable/Disable Commanded** The AGS CSCI shall provide an uplinkable command to configure SV A6 enable/disable.
FLOW AMP	521_1300	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A6 - Enable/Disable Configured** The AMP CSCI shall provide an uplinkable command to configure SV A6 by applying a voltage of 0V or +5V through channel AO20.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
FLOW AGS	522_1310	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A7 - Enable/Disable Commanded** The AGS CSCI shall provide an uplinkable command to configure SV A7 enable/disable.
FLOW AMP	521_1320	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A7 - Enable/Disable Configured** The AMP CSCI shall provide an uplinkable command to configure SV A7 by applying a voltage of 0V or +5V through channel AO21.
FLOW AGS	522_1330	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A8 - Enable/Disable Commanded** The AGS CSCI shall provide an uplinkable command to configure SV A8 enable/disable.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
FLOW AMP	521_1340	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A8 - Enable/Disable Configured** The AMP CSCI shall provide an uplinkable command to configure SV A8 by applying a voltage of 0V or +5V through channel AO22.
FLOW AGS	522_1350	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A9 - Enable/Disable Commanded** The AGS CSCI shall provide an uplinkable command to configure SV A9 enable/disable.
FLOW AMP	521_1360	1.9.1 1.9.2 1.9.4 1.11.1	4.1 4.2 4.3 4.4 4.6 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10 9.11	**Flow System - Solenoid Valve A9 - Enable/Disable Configured** The AMP CSCI shall provide an uplinkable command to configure SV A9 by applying a voltage of 0V or +5V through channel AO23.
FLOW AMP	521_0770	2.3.3	5.4	**Gas Flow Rate – Sampling Frequency** The AMP CSCI shall collect gas flow measurements for all mass flow controllers at 10Hz.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
FLOW AGS	522_1370	2.3.3 2.3.4	5.4	**Mass Flow Controller A1 - Telemetry Display** The AGS CSCI shall display gas flow measurements for MFC A1.
FLOW AMP	521_1380	2.3.3 2.3.4	5.4	**Mass Flow Controller A1 - Telemetry Report** The AMP CSCI shall report gas flow measurements for MFC A1 by reading a voltage of 0V to +5V from channel Ain-0 (207).
FLOW AGS	522_1390	2.3.3 2.3.4	5.4	**Mass Flow Controller A2 - Telemetry Display** The AGS CSCI shall display gas flow measurements for MFC A2.
FLOW AMP	521_1400	2.3.3 2.3.4	5.4	**Mass Flow Controller A2 - Telemetry Report** The AMP CSCI shall report gas flow measurements for MFC A2 by reading a voltage of 0V to +5V from channel Ain-1 (207.)
FLOW AGS	522_1410	2.3.3 2.3.4	5.4	**Mass Flow Controller A3 - Telemetry Display** The AGS CSCI shall display gas flow measurements for MFC A3.
FLOW AMP	521_1420	2.3.3 2.3.4	5.4	**Mass Flow Controller A3 - Telemetry Report** The AMP CSCI shall report gas flow measurements for MFC A3 by reading a voltage of 0V to +5V from channel Ain-2 (207.)
				Igniter
IGN AGS	542_0230	1.10.2 1.10.3 1.10.4	8.1 8.2	**Igniter - Enable/Disable - Commanded** The AGS CSCI shall provide an uplinkable command to configure the igniter enable/disable.
IGN AMP	541_0240	1.10.2 1.10.3 1.10.4	8.1 8.2	**Igniter - Enable/Disable - Configured** The AMP CSCI shall configure upon command the igniter enable/disable by applying a voltage of 0V or +5V through channel AO7.
IGN AGS	542_0250	1.10.2 1.10.3 1.10.4	8.1 8.2	**Igniter Stepper Reset - Commanded** The AGS CSCI shall provide an uplinkable command to configure the igniter stepper reset.
IGN AMP	541_0260	1.10.2 1.10.3 1.10.4	8.1 8.2	**Igniter Stepper Reset - Configured** The AMP CSCI shall configure upon command the igniter stepper reset by applying a voltage of 0V or +5V through channel AO14.
IGN AGS	542_0270	1.10.2 1.10.3 1.10.4	8.1 8.2	**Igniter Deployment - Commanded** The AGS CSCI shall provide an uplinkable command to configure the igniter position.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
IGN AMP	541_0280	1.10.2 1.10.3 1.10.4	8.1 8.2	**Igniter Deployment - Configured** The AMP CSCI shall configure upon command the igniter position.
IGN AGS	542_0290	1.10.3 1.10.4	8.2	**Igniter Output Voltage Control - Commanded** The AGS CSCI shall provide an uplinkable command to configure the igniter output voltage for a specified duration.
IGN AMP	541_0300	1.10.3 1.10.4	8.2	**Igniter Output Voltage Control - Configured** The AMP CSCI shall configure upon command the igniter output voltage for a specified duration by applying a voltage of 0V to +.4V through channel AO8.
				ACME Data Camera
MCS AMP	611_0040	1.12.3	5.5	**Data Recording – Imaging Duration Default Configured** The AMP CSCI shall configure all imaging measurements to begin at an uplinked time that is at most 1 minute prior to ignition and continues to an uplinked time at most 1 minute after extinction detection.
DataCam AGS	312_0570	3.2.7 3.2.8	5.3	**ACME Data Camera - Field of View (Focus) - Commanded** The AGS CSCI shall provide an uplinkable command to configure the data camera's focus.
DataCam AMP	311_0580	3.2.7 3.2.8	5.3	**Data Camera - Field of View (Focus) - Configured** The AMP CSCI shall configure upon command the data camera's focus.
DataCam AGS	312_0180	3.2.13	5.3	**Data Camera - Bit Depth - Commanded** The AGS CSCI shall provide an uplinkable command to configure the digital color camera's Bit Depth.
DataCam AMP	311_0170	3.2.13	5.3	**Data Camera - Bit Depth - Configured** The AMP CSCI shall configure upon command the digital color camera's Bit Depth.
DataCam AGS	312_0210	3.2.13	5.3	**Data Camera - Gain - Commanded** The AGS CSCI shall provide an uplinkable command to configure the digital color camera's Gain.
DataCam AMP	311_0200	3.2.13	5.3	**Data Camera - Gain - Configured** The AMP CSCI shall configure upon command the digital color camera's Gain.
DataCam AGS	312_0240	3.2.13	5.3	**Data Camera - Iris - Commanded** The AGS CSCI shall provide an uplinkable command to configure the digital color camera's iris.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
DataCam AMP	311_0230	3.2.13	5.3	**Data Camera - Iris - Configured** The AMP CSCI shall configure upon command the digital color camera's iris.
DataCam AGS	312_0270	3.2.13	5.3	**Data Camera - Exposure Time - Commanded** The AGS CSCI shall provide an uplinkable command to configure the digital color camera's exposure time.
DataCam AMP	311_0260	3.2.13	5.3	**Data Camera - Exposure Time - Configured** The AMP CSCI shall configure upon command the digital color camera's exposure time.
DataCam AGS	312_0300	3.2.13	5.3	**Data Camera - Frame Rate - Commanded** The AGS CSCI shall provide an uplinkable command to configure the digital color camera's frame rate.
DataCam AMP	311_0290	3.2.13	5.3	**Data Camera - Frame Rate - Configured** The AMP CSCI shall configure upon command the digital color camera's frame rate.
DataCam AGS	312_0330	3.2.13	5.3	**Data Camera - Binning - Commanded** The AGS CSCI shall provide an uplinkable command to configure the digital color camera's binning.
DataCam AMP	311_0320	3.2.13	5.3	**Data Camera - Binning - Configured** The AMP CSCI shall configure upon command the digital color camera's binning.
DataCam AGS	312_0360	3.2.13	5.3	**Data Camera - Region Of Interest (ROI) - Commanded** The AGS CSCI shall provide an uplinkable command to configure the digital color camera's region of interest (ROI).
DataCam AMP	311_0350	3.2.13	5.3	**Data Camera - Region Of Interest (ROI) - Configured** The AMP CSCI shall configure upon command the digital color camera's region of interest (ROI).
DataCam AGS	312_0590	3.2.11	5.2	**Data Camera - Filter Barrel - Commanded** The AGS CSCI shall provide an uplinkable command to configure the digital camera's filter barrel.
DataCam AMP	311_0600	3.2.11	5.2	**Data Camera - Filter Barrel - Configured** The AMP CSCI shall configure upon command the digital camera's filter barrel.
				ACME Analog Camera/ Illumination Package

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
OpsCam AGS	552_0050	3.1.4	5.3	**Analog Camera - Gain - Commanded** The AGS CSCI shall provide an uplinkable command to configure the analog operations camera's gain (iris.)
OpsCam AMP	551_0060	3.1.4	5.3	**Analog Camera - Gain -Configured** The AMP CSCI shall configure upon command the analog operations camera's gain (iris.)
OpsCam AGS	552_0090	3.1.8	6.1 6.2 6.3	**Illumination Package Bank 1 - Commanded** The AGS CSCI shall provide an uplinkable command to configure LED bank 1 of the illumination package.
OpsCam AMP	551_0100	3.1.8	6.1 6.2 6.3	**Illumination Package Bank 1 - Configured** The AMP CSCI shall configure upon command LED bank 1 of the illumination package by applying a voltage of 0V to +5V through channel AO28.
OpsCam AGS	552_0130	3.1.8	6.1 6.2 6.3	**Illumination Package Bank 2- Commanded** The AGS CSCI shall provide an uplinkable command to configure LED bank 2 of the illumination package.
OpsCam AMP	551_0140	3.1.8	6.1 6.2 6.3	**Illumination Package Bank 2 - Configured** The AMP CSCI shall configure upon command LED bank 2 of the illumination package by applying a voltage of 0V to +5V through channel AO29.
				Diagnostics
DIAG AMP	571_0800	3.6.7	5.4 12.1	**Radiant Emission – Sampling Frequency** The AMP CSCI shall collect radiant emission measurements from all thermopiles at 60 Hz.
DIAG AGS	572_0810	3.4.1.8 3.4.4.8	5.4 12.1	**Radiometer Offset Position Cold Junction - Telemetry Display** The AGS CSCI shall display temperature measurements for radiometer offset cold junction from the telemetry report.
DIAG AMP	571_0820	3.4.1.8 3.4.4.8	5.4 12.1	**Radiometer Offset Position Cold Junction - Telemetry Report** The AMP CSCI shall report temperature measurements for Radiometer cold junction by reading a voltage of 0V to +1.25V from channel Ain-18 (225.)

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
DIAG AGS	572_0830	3.6.8	5.4 12.1	**Radiometer +60 Position - Telemetry Display** The AGS CSCI shall display radiant emission measurements from the +60 position radiometer from the telemetry report.
DIAG AMP	571_0840	3.6.8	5.4 12.1	**Radiometer +60 Position - Telemetry Report** The AMP CSCI shall report radiant emission measurements for the +60 position radiometer by reading a voltage of 0V to +5V from channel Ain-7 (225.)
DIAG AGS	572_0850	3.6.8	5.4 12.1	**Radiometer +30 Position - Telemetry Display** The AGS CSCI shall display radiant emission measurements from radiometer +30 from the telemetry report.
DIAG AMP	571_0860	3.6.8	5.4 12.1	**Radiometer +30 Position - Telemetry Report** The AMP CSCI shall report radiant emission measurements for the +30 position radiometer by reading a voltage of 0V to +5V from channel Ain-9 (225.)
DIAG AGS	572_0870	3.6.8	5.4 12.1	**Radiometer Center Position - Telemetry Display** The AGS CSCI shall display radiant emission measurements from the center position radiometer from the telemetry report.
DIAG AMP	571_0880	3.6.8	5.4 12.1	**Radiometer Center Position - Telemetry Report** The AMP CSCI shall report radiant emission measurements for the center position radiometer by reading a voltage of 0V to +5V from channel Ain-11 (225.)
DIAG AGS	572_0890	3.6.8	5.4 12.1	**Radiometer -30 Position - Telemetry Display** The AGS CSCI shall display radiant emission measurements from the -30 position radiometer from the telemetry report.
DIAG AMP	571_0900	3.6.8	5.4 12.1	**Radiometer -30 Position - Telemetry Report** The AMP CSCI shall report radiant emission measurements for the -30 position radiometer by reading a voltage of 0V to +5V from channel Ain-13 (225.)
DIAG AGS	572_0910	3.6.8	5.4 12.1	**Radiometer -60 Position - Telemetry Display** The AGS CSCI shall display radiant emission measurements from the -60 position radiometer from the telemetry report.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
DIAG AMP	571_0920	3.6.8	5.4 12.1	**Radiometer -60 Position - Telemetry Report** The AMP CSCI shall report radiant emission measurements for the -60 position radiometer by reading a voltage of 0V to +5V from channel Ain-15 (225.)
DIAG AGS	572_0930	3.6.8	5.4 12.1	**Radiometer Offset Position - Telemetry Display** The AGS CSCI shall display radiant emission measurements from the offset position radiometer from the telemetry report.
DIAG AMP	571_0940	3.6.8	5.4 12.1	**Radiometer Offset Position - Telemetry Report** The AMP CSCI shall report radiant emission measurements for the offset position radiometer by reading a voltage of 0V to +5V from channel Ain-17 (225.)
DIAG AGS	572_0950	3.4.1.8 3.4.4.8	5.4 12.1	**Radiometer +60 Position Cold Junction - Telemetry Display** The AGS CSCI shall display temperature measurements for radiometer +60 position cold junction from the telemetry report.
DIAG AMP	571_0960	3.4.1.8 3.4.4.8	5.4 12.1	**Radiometer +60 Position Cold Junction - Telemetry Report** The AMP CSCI shall report temperature measurements for radiometer +60 position cold junction by reading a voltage of 0V to +1.25V from channel Ain-8 (225.)
DIAG AGS	572_0970	3.4.1.8 3.4.4.8	5.4 12.1	**Radiometer +30 Position Cold Junction - Telemetry Display** The AGS CSCI shall display temperature measurements for radiometer +30 position cold junction from the telemetry report.
DIAG AMP	571_0980	3.4.1.8 3.4.4.8	5.4 12.1	**Radiometer +30 Position Cold Junction - Telemetry Report** The AMP CSCI shall report temperature measurements for radiometer +30 position cold junction by reading a voltage of 0V to +1.25V from channel Ain-10 (225.)
DIAG AGS	572_0990	3.4.1.8 3.4.4.8	5.4 12.1	**Radiometer Center Position Cold Junction - Telemetry Display** The AGS CSCI shall display temperature measurements for radiometer center position cold junction from the telemetry report.
DIAG AMP	571_1000	3.4.1.8 3.4.4.8	5.4 12.1	**Radiometer Center Position Cold Junction - Telemetry Report** The AMP CSCI shall report temperature measurements for radiometer center position cold junction by reading a voltage of 0V to +1.25V from channel Ain-12 (225.)

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
DIAG AGS	572_1010	3.4.1.8 3.4.4.8	5.4 12.1	**Radiometer -30 Position Cold Junction - Telemetry Display** The AGS CSCI shall display temperature measurements for radiometer -30 cold junction from the telemetry report.
DIAG AMP	571_1020	3.4.1.8 3.4.4.8	5.4 12.1	**Radiometer -30 Position Cold Junction - Telemetry Report** The AMP CSCI shall report temperature measurements for radiometer -30 cold junction by reading a voltage of 0V to +1.25V from channel Ain-14 (225.)
DIAG AGS	572_1030	3.4.1.8 3.4.4.8	5.4 12.1	**Radiometer -60 Position Cold Junction - Telemetry Display** The AGS CSCI shall display temperature measurements for radiometer -60 cold junction from the telemetry report.
DIAG AMP	571_1040	3.4.1.8 3.4.4.8	5.4 12.1	**Radiometer -60 Position Cold Junction - Telemetry Report** The AMP CSCI shall report temperature measurements for radiometer -60 cold junction by reading a voltage of 0V to +1.25V from channel Ain-16 (225.)
DIAG AGS	562_0360	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 1 - Enable/ Disable - Commanded** The AGS CSCI shall provide an uplinkable command to configure PMT1 enable/disable.
DIAG AMP	561_0370	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 1 - Enable/ Disable - Configured** The AMP CSCI shall configure upon command PMT1 enable/disable by applying a voltage of 0V or +10V through channel A03.
DIAG AGS	562_0380	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 2 - Enable/ Disable - Commanded** The AGS CSCI shall provide an uplinkable command to configure PMT2 enable/disable.
DIAG AMP	561_0390	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 2 - Enable/ Disable - Configured** The AMP CSCI shall configure upon command PMT2 enable/disable by applying a voltage of 0V or +10V through channel A03.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
DIAG AGS	562_0400	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 3 - Enable/ Disable - Commanded** The AGS CSCI shall provide an uplinkable command to configure PMT3 enable/disable.
DIAG AMP	561_0410	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 3 - Enable/ Disable - Configured** The AMP CSCI shall configure upon command PMT3 enable/disable by applying a voltage of 0V or +10V through channel A03.
DIAG AGS	562_0420	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 1 - Gain Adjust - Commanded** The AGS CSCI shall provide an uplinkable command to configure PMT1 gain adjust.
DIAG AMP	561_0430	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 1 - Gain Adjust - Configured** The AMP CSCI shall configure upon command PMT1 gain adjust by applying a voltage of 0V to +5V through channel A025.
DIAG AGS	562_0440	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 2 - Gain Adjust - Commanded** The AGS CSCI shall provide an uplinkable command to configure PMT2 gain adjust.
DIAG AMP	561_0450	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 2 - Gain Adjust - Configured** The AMP CSCI shall configure upon command PMT2 gain adjust by applying a voltage of 0V to +5V through channel A026.
DIAG AGS	562_0460	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 3 - Gain Adjust - Commanded** The AGS CSCI shall provide an uplinkable command to configure PMT3 gain adjust.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
DIAG AMP	561_0470	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 3 - Gain Adjust - Configured** The AMP CSCI shall configure upon command PMT3 gain adjust by applying a voltage of 0V to +5V through channel A027.
DIAG AMP	561_500	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube - Sampling Rate** ACME shall sample chemiluminescent emission measurements at 120 samples/sec or faster with a temporal resolution of .005 s.
DIAG AGS	572_0450	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 1 - Telemetry Display** The AGS CSCI shall display chemiluminescent emission measurements from PMT1 from the telemetry report.
DIAG AMP	571_0460	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 1 - Telemetry Report** The AMP CSCI shall report chemiluminescent emission measurements from PMT1 by reading a voltage of 0V to +4V on channel Ain-8 (207).
DIAG AGS	572_0470	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 2 - Telemetry Display** The AGS CSCI shall display chemiluminescent emission measurements from PMT2 from the telemetry report.
DIAG AMP	571_0480	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 2 - Telemetry Report** The AMP CSCI shall report chemiluminescent emission measurements from PMT2 by reading a voltage of 0V to +4V on channel Ain-9 (207.)
DIAG AGS	572_0490	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 3 - Telemetry Display** The AGS CSCI shall display chemiluminescent emission measurements from PMT3 from the telemetry report.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
DIAG AMP	571_0500	3.7.1 3.7.5	12.1 12.2 12.3 12.4 12.5 12.6 12.7	**Photomultiplier Tube 3 - Telemetry Report** The AMP CSCI shall report chemiluminescent emission measurements from PMT3 by reading a voltage of 0V to +4V on channel Ain-10 (207.)
				Measurement System
MEAS AMP	571_0090	2.4.3	1.6 5.4	**Pressure Measurement – Sampling Frequency** The AMP CSCI shall collect pressure measurement from all pressure transducers at 10 Hz.
MEAS AGS	572_0510	4.0	1.6 5.4	**Pressure Transducer 1 - Telemetry Display** The AGS CSCI shall display pressure measurements for PT1 from the telemetry report.
MEAS AMP	571_0410	2.4.3	1.6 5.4	**Pressure Transducer 1 - Telemetry Report** The AMP CSCI shall report pressure measurements for PT1 by reading a voltage of 0V - +5V from channel Ain-4 (207.)
MEAS AGS	572_0520	4.0	1.6 5.4	**Pressure Transducer 2 - Telemetry Display** The AGS CSCI shall display pressure measurements for PT2 from the telemetry report.
MEAS AMP	571_0530	2.4.3	1.6 5.4	**Pressure Transducer 2 - Telemetry Report** The AMP CSCI shall report pressure measurements for PT2 by reading a voltage of 0V - +5V from channel Ain-5 (207.)
MEAS AGS	572_0540	4.0	1.6 5.4	**Pressure Transducer 3 - Telemetry Display** The AGS CSCI shall display pressure measurements for PT3 from the telemetry report.
MEAS AMP	571_0550	2.4.3	1.6 5.4	**Pressure Transducer 3 - Telemetry Report** The AMP CSCI shall report pressure measurements for PT3 by reading a voltage of 0V - +5V from channel Ain-6 (207.)
MEAS AMP	571_0310	3.4.4.8	1.6 5.4 12.1	**Temperature Measurement - Sampling Frequency** The AMP CSCI shall collect temperature measurements from all thermocouples at 1 Hz.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
MEAS AGS	572_0440	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type B Thermocouple Rake A - Telemetry Display** The AGS CSCI shall display temperature measurements for type B thermocouple rake A from the telemetry report.
MEAS AMP	571_0430	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type B Thermocouple Rake A - Telemetry Report** The AMP CSCI shall report temperature measurements for type B thermocouple rake A by reading a voltage of 0mV to +14mV from channel Ain-0 (225.)
MEAS AGS	572_0560	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type K Thermocouple Rake A - Telemetry Display** The AGS CSCI shall display temperature measurements for type K thermocouple rake A from the telemetry report.
MEAS AMP	571_0570	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type K Thermocouple Rake A - Telemetry Report** The AMP CSCI shall report temperature measurements for type K thermocouple rake A by reading a voltage of -6.4mV to -55mV from channel Ain-0 (225).
MEAS AGS	572_0580	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type B Thermocouple Rake B - Telemetry Display** The AGS CSCI shall display temperature measurements for type B thermocouple rake B from the telemetry report.
MEAS AMP	571_0590	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type B Thermocouple Rake B - Telemetry Report** The AMP CSCI shall report temperature measurements for type B thermocouple rake B by reading a voltage of 0mV +14mV from channel Ain-1 (225.)
MEAS AGS	572_0600	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type K Thermocouple Rake B - Telemetry Display** The AGS CSCI shall display temperature measurements for type K thermocouple rake B from the telemetry report.
MEAS AMP	571_0610	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type K Thermocouple Rake B - Telemetry Report** The AMP CSCI shall report temperature measurements for type K thermocouple rake B by reading a voltage of -6.4mV to -55mV from channel Ain-1 (225).
MEAS AGS	572_0640	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type K Thermocouple Rake C - Telemetry Display** The AGS CSCI shall display temperature measurements for type K thermocouple rake C from the telemetry report.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
MEAS AMP	571_0650	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type K Thermocouple Rake C - Telemetry Report** The AMP CSCI shall report temperature measurements for type K thermocouple rake C by reading a voltage of -6.4mV to -55mV from channel Ain-2 (225.)
MEAS AGS	572_0660	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type K Thermocouple Rake D - Telemetry Display** The AGS CSCI shall display temperature measurements for type K thermocouple rake D from the telemetry report.
MEAS AMP	571_0670	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type K Thermocouple Rake D - Telemetry Report** The AMP CSCI shall report temperature measurements for type K thermocouple rake D by reading a voltage of -6.4mV to -55mV from channel Ain-3 (225.)
MEAS AGS	572_0700	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type K Thermocouple 200A - Telemetry Display** The AGS CSCI shall display temperature measurements for type K thermocouple 200A from the telemetry report.
MEAS AMP	571_0710	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type K Thermocouple 200A - Telemetry Report** The AMP CSCI shall report temperature measurements for type K thermocouple 200A by reading a voltage of -6.4mV to -55mV from channel Ain-4 (225.)
MEAS AGS	572_0720	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type K Thermocouple 200B - Telemetry Display** The AGS CSCI shall display temperature measurements for type K thermocouple 200B from the telemetry report.
MEAS AMP	571_0730	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type K Thermocouple 200B - Telemetry Report** The AMP CSCI shall report temperature measurements for type K thermocouple 200B by reading a voltage of -6.4mV to -55mV from channel Ain-5 (225.)
MEAS AGS	572_0740	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type K Thermocouple Burner - Telemetry Display** The AGS CSCI shall display temperature measurements for type K thermocouple on burner from the telemetry report.
MEAS AMP	571_0750	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Type K Thermocouple Burner - Telemetry Report** The AMP CSCI shall report temperature measurements for type K thermocouple on burner by reading a voltage of -6.4mV to -55mV from channel Ain-6 (225.)

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
MEAS AGS	572_0760	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Thermocouple Cold Junction - Telemetry Display** The AGS CSCI shall display temperature measurements for type K thermocouple cold junction from the telemetry report.
MEAS AMP	571_0770	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Thermocouple Cold Junction - Telemetry Report** The AMP CSCI shall report temperature measurements for type K thermocouple cold junction by reading a voltage of .803V to + 1.1V from channel Ain-24 (225.)
MEAS AGS	572_0780	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Thermocouple Spare - Telemetry Display** The AGS CSCI shall display temperature measurements for the thermocouple spare from the telemetry report.
MEAS AMP	571_0790	3.4.1.8 3.4.4.8	1.6 5.4 12.1	**Thermocouple Spare - Telemetry Report** The AMP CSCI shall report temperature measurements for the thermocouple spare by reading a voltage of 0V to +1.25V from channel Ain-20 (225.)
MEAS AMP	581_0600	2.5.3	5.4	**Electric Potential – Sampling Frequency** The AMP CSCI shall collect E-Field Electric potential measurements at 120 sample/s with a temporal resolution of 1ms.
MEAS AGS	582_0610	2.5.3	5.4	**Electric Potential - Telemetry Display** The AGS CSCI shall display E-FIELD electric potential measurements from the telemetry report.
MEAS AMP	581_0540	2.5.3	5.4	**Electric Potential - Telemetry Report** The AMP CSCI shall report E-FIELD electric potential measurements by reading a voltage of -4V to +4V from channel Ain-7 (207.)
MEAS AMP	581_0450	3.8.4	5.4	**Ion Current – Sampling Frequency** The AMP CSCI shall collect E-Field ion current measurements at 120 Hz.
MEAS AGS	582_0620	3.8.4	5.4	**Ion Current - Telemetry Display** The AGS CSCI shall display E-Field Ion current measurements from the telemetry report.
MEAS AMP	581_0550	3.8.4	5.4	**Ion Current - Telemetry Report** The AMP CSCI shall report E-Field ion current measurements by reading a voltage of -1.5V to +1.5V from channel Ain-3 (207.)
MEAS AGS	572_1290	3.10.1 3.10.2 1.13.12		**BRE Heat Flux Measurement Center - Telemetry Display** The AGS CSCI shall display heat flux measurements for the center position sensor from the telemetry report.

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
MEAS AMP	571_1300	3.10.1 3.10.2 1.13.12		**BRE Heat Flux Measurement Center - Telemetry Report** The AMP CSCI shall report heat flux measurements for the center position sensor by reading a voltage of 0V to +1.25V from channel Ain-20 (225.)
MEAS AGS	572_1310	3.10.1 3.10.2 1.13.12		**BRE Heat Flux Measurement Offset - Telemetry Display** The AGS CSCI shall display heat flux measurements for the offset position sensor from the telemetry report.
MEAS AMP	571_1320	3.10.1 3.10.2 1.13.12		**BRE Heat Flux Measurement Offset - Telemetry Report** The AMP CSCI shall report heat flux measurements for the offset position sensor by reading a voltage of 0V to +1.25V from channel Ain-21 (225.)
				E-Field
E-FIELD AGS	582_0170	1.7.1 1.7.2 1.7.4	11.1 11.3	**Electric Field - HV1 Enable/Disable - Commanded** The AGS CSCI shall provide an uplinkable command to configure the E-field HV1 enable/disable.
E-FIELD AMP	581_0180	1.7.1 1.7.2 1.7.4	11.1 11.3	**Electric Field - HV1 Enable/Disable - Configured** The AMP CSCI shall configure upon command the E-field HV1 enable/disable by applying a voltage of 0V or +5V through channel AO5.
E-FIELD AGS	582_0560	1.7.1 1.7.2 1.7.4	11.1 11.3	**Electric Field - HV2 Enable/Disable - Commanded** The AGS CSCI shall provide an uplinkable command to configure the E-field HV2 enable/disable.
E-FIELD AMP	581_0570	1.7.1 1.7.2 1.7.4	11.1 11.3	**Electric Field - HV2 Enable/Disable - Configured** The AMP CSCI shall configure upon command the E-field HV2 enable/disable by applying a voltage of 0V or +5V through channel AO6.
E-FIELD AGS	582_0580	1.7.1 1.7.2 1.7.4	11.1 11.3	**Electric Field - HV Control - Commanded** The AGS CSCI shall provide an uplinkable command to configure the E-Field HV output voltage.
E-FIELD AMP	581_0590	1.7.1 1.7.2 1.7.4	11.1 11.3	**Electric Field - HV Control - Configured** The AMP CSCI shall configure upon command the E-Field HV control by applying a voltage of 0V to +5V through channel A04..
				Thin Filament Pyrometry Array

System	SSC_ID	Parent ID/ ISRD Trace	Ops Requirement Trace	ERD REQ Title/Description
DIAG AGS	562_0040	1.11.8 1.11.9 3.4.3.3 3.4.3.4	10.1 10.2 10.3	**Thin Filament Pyrometry (TFP)-Reposition - Commanded** The AGS CSCI shall provide an uplinkable command to change the TFP array position.
DIAG AMP	561_0050	1.11.8 1.11.9 3.4.3.3 3.4.3.4	10.1 10.2 10.3	**Thin Filament Pyrometry (TFP) - Reposition - Configured** The AMP CSCI shall configure upon command to change the TFP array position.
DIAG AGS	562_0480	1.11.8 1.11.9 3.4.3.3 3.4.3.4	10.1 10.2 10.3	**Thin Filament Pyrometry (TFP) - Stepper Reset - Commanded** The AGS CSCI shall provide an uplinkable command to configure the TFP stepper reset.
DIAG AMP	561_0490	1.11.8 1.11.9 3.4.3.3 3.4.3.4	10.1 10.2 10.3	**Thin Filament Pyrometry (TFP) - Stepper Reset - Configured** The AMP CSCI shall configure upon command the TFP stepper reset by applying a voltage of 0V or +5V through channel AO13.
				Heat Flux Measurement
MEAS AMP	571_01280	3.10.10		**Heat Flux - Sampling** ACME shall sample heat flux measurements at 120 samples/sec or faster with a response time of 0.5s or faster.

12.0 SAFETY REQUIREMENTS

All ACME Safety Requirements are outlined in the Phase 0/1 Flight Safety Data Package Rev A (ACME-SDP-001 Rev. A).

APPENDIX A ACRONYMS AND ABBREVIATIONS <APPENDIX>

A.1 Scope

This appendix lists the acronyms and abbreviations used in this document.

Table A-1—Acronyms

1g	1 time gravitational force
A	amp
ACME	Advanced Combustion via Microgravity Experiments
Atm	Atmosphere
BRE	Burning Rate Emulator
C	Celsius
CDR	Critical Design Review
CIA	Chamber Insert Assembly
CIR	Combustion Integration Rack
cm	Centimeter
ConOps	Concept of Operations Document
EMS	Experiment Mounting Structure
ERD	Engineering Requirements Document
FCF	Fluids and Combustion Facility
FOMA	Fuel/Oxidizer Management Assembly
FOV	Field of View
Fps	Frames per second
GRC	Glenn Research Center
HiBMS	High Bit-Depth Multi-Spectral
HV	High Voltage
Hz	Hertz
ICD	Interface Control Document
ID	Identification
IOP	Input/ Output Processor
IRR	Interface Resource Ring
ISS	International Space Station
K	Kelvin
kV	Kilovolt
MDCA	Multi-user Droplet Combustion Apparatus
MFC	Mass flow controller
Min	Minute
mm	millimeter
ms	millisecond

NASA	National Aeronautics and Space Administration
PBS	Product Breakdown Structure
PDR	Preliminary Design Review
PI	Principal Investigator
PIL	Principal Investigator Location
PMT	Photomultiplier tube
PS	Project Scientist
OD	Quick Disconnect
RDR	Requirement Definition Review
SDP	Safety Data Package
SEMP	Systems Engineering Management Plan
SiC	Silicon Carbide
SRD	Science Requirements Document
WBS	Work Breakdown Structure